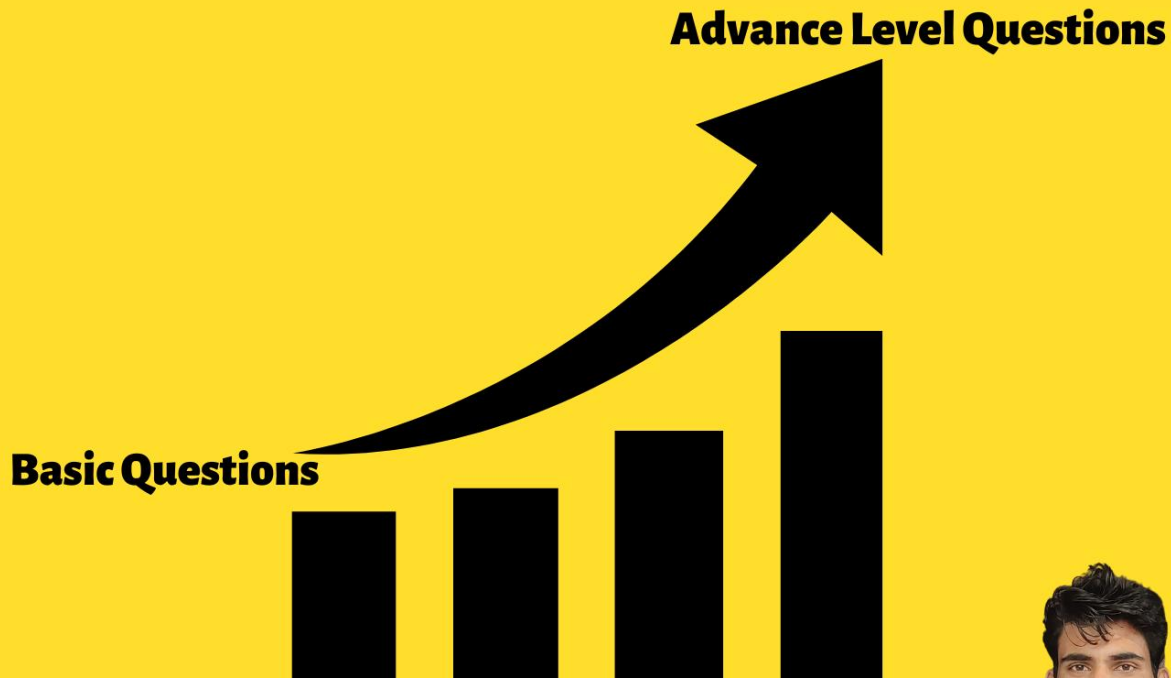


MATHS

MENSURATION

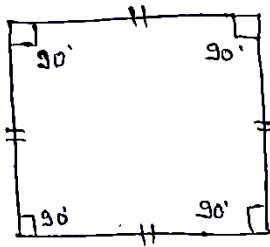
**ALL TYPE WISE
QUESTIONS**



Maths with Gajanand



Square



* Perimeter = $4 \times \text{side}$

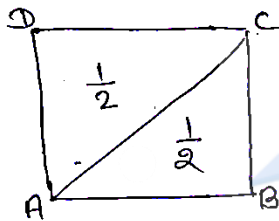
* Area = $(\text{side})^2$

* Diagonal_(d) = $\sqrt{2} \text{ side}$

* Area = $\frac{1}{2} \times d^2$

Important Points :-

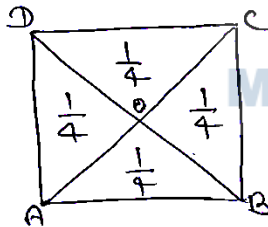
1



$\Rightarrow$ Diagonal intersect into two equal Part

Area($\triangle ACD$) = Area($\triangle ABC$)

2

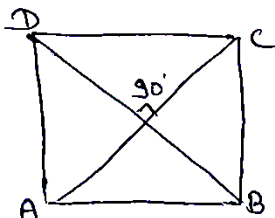


$\Rightarrow$ 2 Diagonal cut in four equal part

Area($\triangle DOC$) = Ar($\triangle BOC$) = Ar($\triangle AOB$) = Ar($\triangle AOD$)

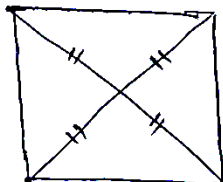
$\Rightarrow \frac{1}{4} \text{ Area}(ABCD)$

3



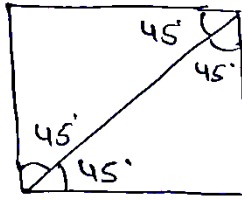
* Two diagonals are perpendicular to each others.

4



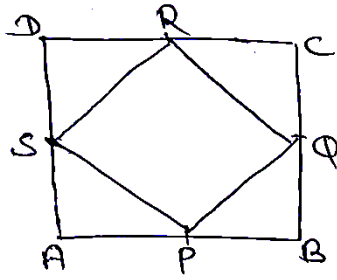
Diagonals intersect each other in equal parts.

5



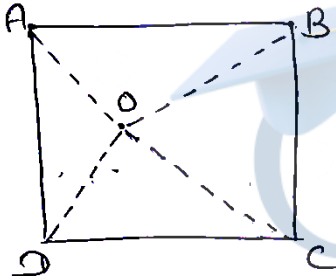
Diagonal is Angle Bisector.

6



$$\text{Area}(PQRS) = \frac{1}{2} \text{Area}(ABCD)$$

7

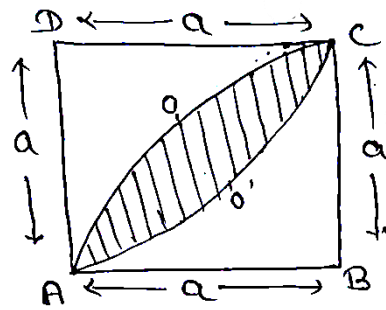


$$* AO^2 + OC^2 = BO^2 + DO^2 *$$

MathswithGajanand

Leaf Area

1



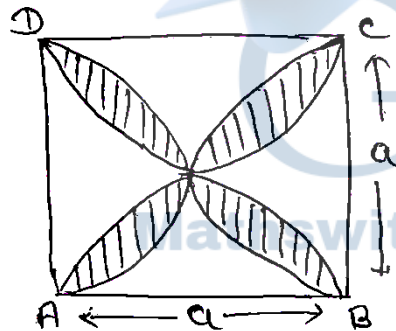
$$\text{Area} = \frac{4}{7} a^2$$

Prove :- $[\text{Area of } AOCB - \triangle ABC] \times 2$

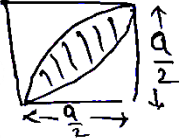
$$\text{Area} = \left[\frac{\pi a^2}{4} - \frac{1}{2} a^2 \right] \times 2$$

$$\text{Area} = \frac{4}{7} a^2$$

2



$$\text{Area} = \frac{4}{7} a^2$$

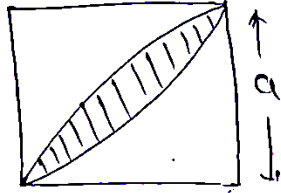
Prove:- $\text{Area} = 4 \times$ 

$$\text{Area} = 4 \times \left[\frac{4}{7} \times \left(\frac{a}{2} \right)^2 \right]$$

$$\text{Area} = 4 \times \frac{4}{7} \times \frac{a^2}{4}$$

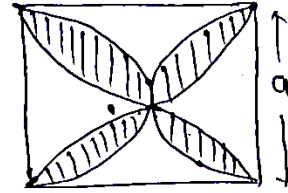
$$\text{Area} = \frac{4}{7} a^2$$

*



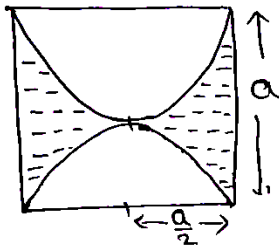
$$\text{Area} = \frac{4}{7}a^2$$

||



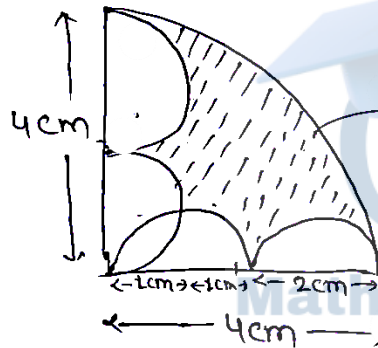
$$\text{Area} = \frac{4}{7}a^2$$

3



$$\text{Area} = \left[a^2 - \frac{\pi a^2}{4} \right]$$

SSC
Que. :-



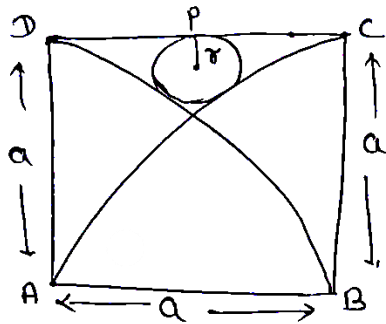
Area =

Soln :-

$$\begin{aligned} \text{Area} &= \text{Area of quarter circle with radius 4} - \left(\text{Area of two quarter circles with radius 2} + \text{Area of square} \right) \\ &= \frac{\pi a^2}{4} - \left(2 \times \frac{1}{4} \pi (2)^2 + (2)^2 \right) \\ &= \frac{\pi (4)^2}{4} - \left[\frac{3}{2} \pi (1)^2 + (1)^2 \right] \\ &= 4\pi - \left(\frac{3\pi}{2} + 1 \right) \\ &= \boxed{\frac{5\pi}{2} - 1} \quad \text{Ans} \end{aligned}$$

Circle radius in Square

1



$$* \quad r = \frac{a}{16} \quad *$$

Prove:-

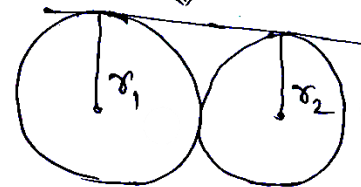
$$CD = PD + PC$$

$$a = 2\sqrt{ar} + 2\sqrt{ar}$$

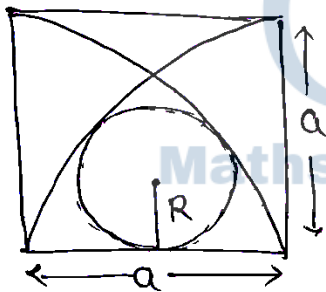
$$a^2 = 16ar$$

$$r = \frac{a}{16}$$

$$C.T = 2\sqrt{r_1 r_2}$$



2



$$* \quad R = \frac{3a}{8} \quad *$$

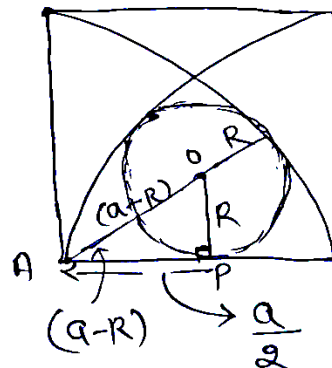
Prove:- In $\triangle AOP$

$$(a-R)^2 = R^2 + \frac{a^2}{4}$$

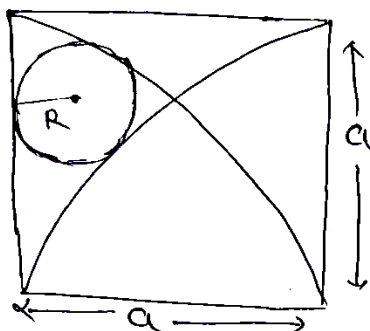
$$a^2 + R^2 - 2aR = R^2 + \frac{a^2}{4}$$

$$2aR = \frac{3a^2}{4}$$

$$R = \frac{3a}{8}$$



3



$$R = \frac{a}{6}$$

Prove:-

$$K^2 = (a-R)^2 - R^2$$

$$K^2 = (a+R)^2 - (a-R)^2$$

$$(a-R)^2 - R^2 = (a+R)^2 - (a-R)^2$$

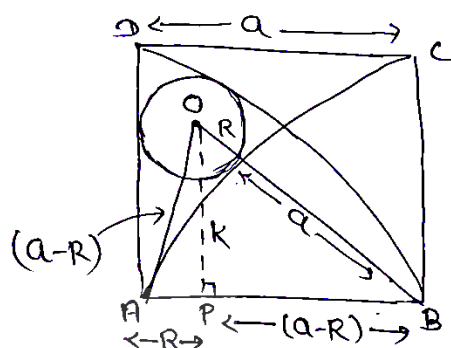
$$(a-R-R)(a-R+R) = 4aR$$

$$(a-2R)a = 4aR$$

$$a-2R = 4R$$

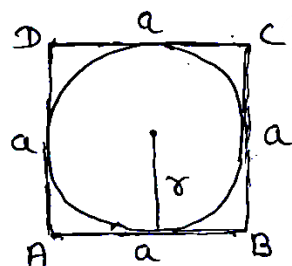
$$a = 6R$$

$$R = \frac{a}{6}$$



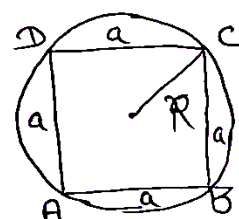
Points:- Circle Radius in Square:-

Case 1 $\rightarrow$ Inside



$$\text{Radius } (r) = \frac{AB}{2}$$

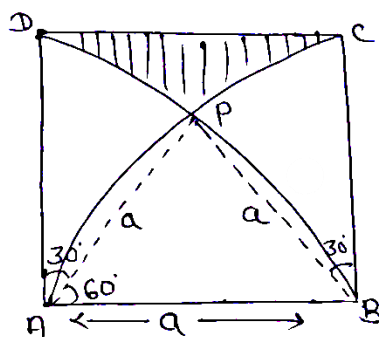
Case 2 $\rightarrow$ Outside



$$\text{Radius } (R) = \frac{a}{\sqrt{2}}$$

Area in Square

1



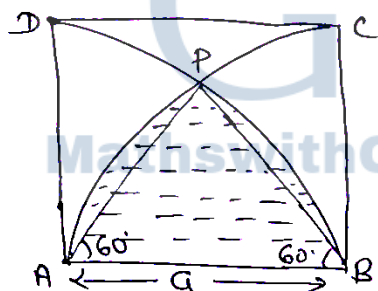
$$\text{Area of } PCD = \text{Ar}(ABCD) - \left[(\text{Area of } \triangle APB) + 2(\text{Sector of } 30^\circ) \right]$$

$$= a^2 - \left[\frac{\sqrt{3}}{4} a^2 + 2 \times \left(\frac{\pi a^2}{360} \times 30^\circ \right) \right]$$

$$= a^2 - \left[\frac{\sqrt{3}}{4} a^2 + \frac{\pi a^2}{6} \right]$$

$$\text{Area of } PCD = \frac{a^2}{12} [12 - 2\pi - 3\sqrt{3}]$$

2



$$\text{Area of } APB = (\text{60}^\circ \text{ sector Area}) + (\text{1 leaf Area})$$

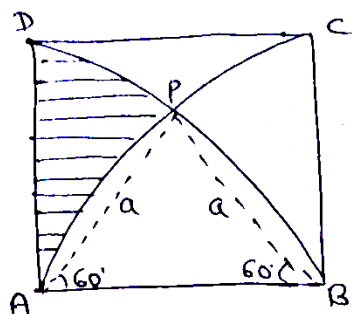


$$= \frac{\pi a^2}{360} \times 60^\circ + \left(\frac{\pi a^2}{360} \times 60^\circ - \frac{\sqrt{3}}{4} a^2 \right)$$

$$= \frac{\pi a^2}{3} - \frac{\sqrt{3}}{4} a^2$$

$$\text{Area} = \frac{a^2}{12} [4\pi - 3\sqrt{3}]$$

[3]



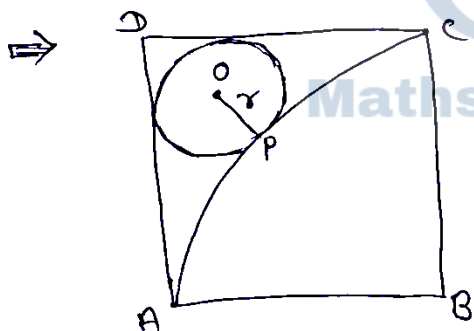
Area of APD = (Sector 30° Area) - (1 leaf Area)

$$= \frac{\pi a^2}{360} \times 30^\circ - \left(\frac{\pi a^2}{360} \times 60^\circ - \frac{\sqrt{3}}{4} a^2 \right)$$

$$= \frac{\pi a^2}{12} - \frac{\pi a^2}{6} + \frac{\sqrt{3}}{4} a^2$$

$$\text{Required Area} = \frac{\sqrt{3}}{4} a^2 - \frac{\pi a^2}{12}$$

$$\text{Required Area} = \frac{a^2}{12} [3\sqrt{3} - \pi]$$



$$\boxed{r = (\sqrt{2} - 1)^2 a}$$

Prove:- (OD) + r + (PB) = (BD)

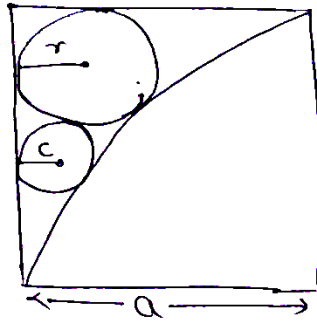
$$\sqrt{2} r + r + a = \sqrt{2} a$$

$$(\sqrt{2} + 1) r = (\sqrt{2} - 1) a$$

$$r = \frac{(\sqrt{2} - 1) a}{(\sqrt{2} + 1)}$$

$$\boxed{r = (\sqrt{2} - 1)^2 a} \text{ OR } (3 - 2\sqrt{2}) a$$

⇒



if
 $a = 1 \text{ cm}$

$c = ?$

Soln:-

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{r}} + \frac{1}{\sqrt{a}}$$

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{(\sqrt{2}-1)^2 a}} + \frac{1}{1}$$

$$\frac{1}{\sqrt{c}} = \frac{1}{(\sqrt{2}-1)} + 1$$

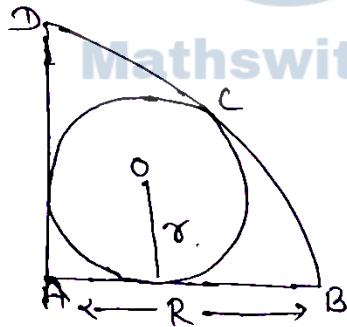
$$\frac{1}{\sqrt{c}} = \frac{\sqrt{2}}{\sqrt{2}-1}$$

$$c = \frac{(\sqrt{2}-1)^2}{2}$$

∴

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$$

⇒



$$r = (\sqrt{2}-1)R$$

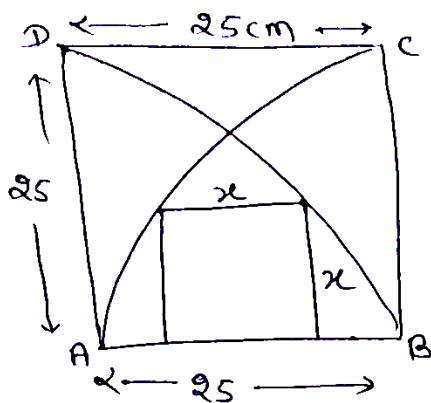
Prove:- $AC = AO + OC$

$$R = \sqrt{2}r + r$$

$$r = \frac{R}{(\sqrt{2}+1)}$$

$$r = (\sqrt{2}-1)R$$

⇒



$$x = ?$$

Soln: In ΔARQ

$$625 = x^2 + \left(\frac{25}{2} + \frac{x}{2}\right)^2$$

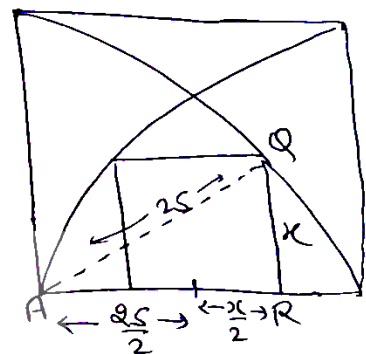
$$625 = x^2 + \frac{625 + x^2 + 50x}{4}$$

$$2500 = 5x^2 + 50x + 625$$

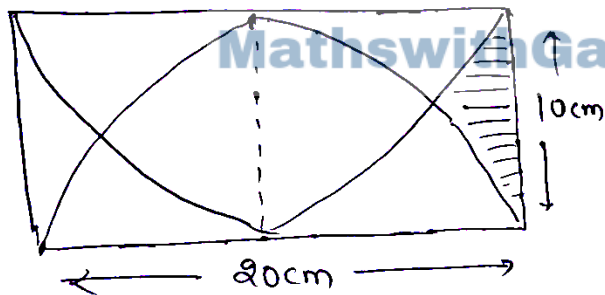
$$5x^2 + 50x - 1875 = 0$$

$$x = -25, 15$$

$$x = 15 \text{ Ans}$$



Que.

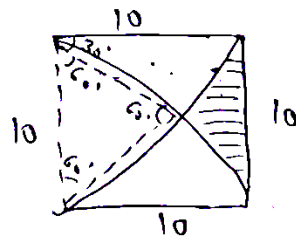


Area = ?

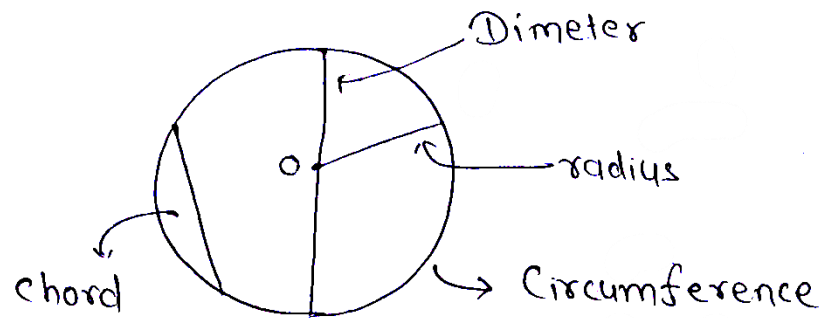
$$\text{Soln}:- \text{Area} = a^2 - \left(\frac{\sqrt{3}}{4} a^2 + 2 \times \frac{\pi a^2}{360} \times 30^\circ \right)$$

$$= 100 - \left(\frac{\sqrt{3}}{4} \times 100 + 2 \times \frac{\pi}{12} \times 100 \right)$$

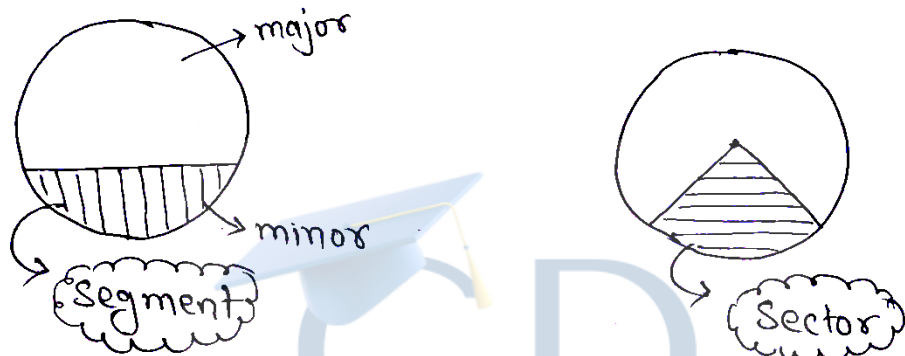
$$= 100 \left[1 - \frac{\sqrt{3}}{4} - \frac{\pi}{6} \right] \text{ Ans}$$



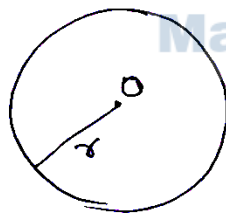
Circle



⇒



⇒

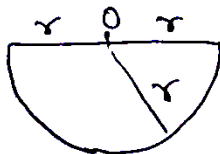


Circle

$$\text{Circumference} = 2\pi r$$

$$\text{Area} = \pi r^2$$

⇒



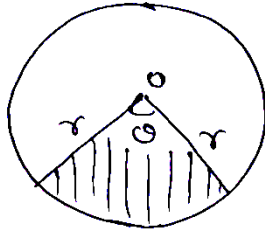
Semicircle

$$\text{Circumference} = \pi r + 2r$$

$$\text{Area} = \frac{\pi r^2}{2}$$

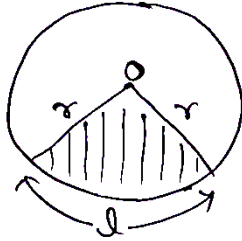
Area of Sector :-

Case 1



$$\text{Area} = \frac{\pi r^2}{360^\circ} \times \theta$$

Case 2

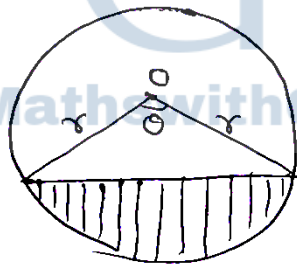


$$\text{Area} = \frac{\pi r^2}{2\pi r} \times l$$

OR

$$\text{Area} = \frac{l \times r}{2}$$

Area of Sector



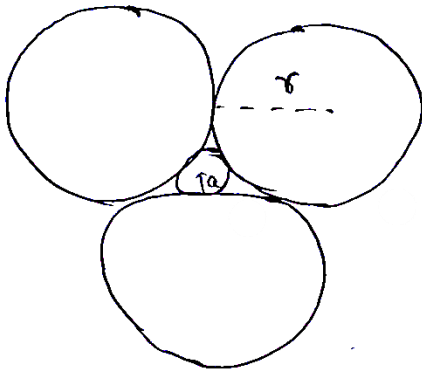
Minor Segment :-

$$\text{Area} = \frac{\pi r^2}{360^\circ} \times \theta - \frac{1}{2} r^2 \sin \theta$$

Major Segment :-

$$\text{Area} = \pi r^2 - (\text{minor segment Area})$$

1



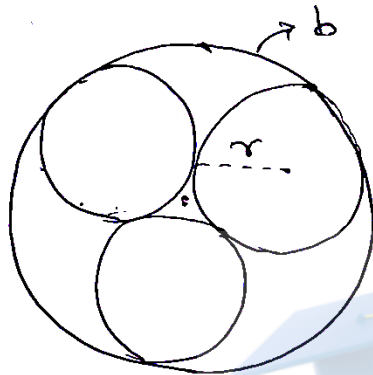
Big Equal Circle radius = r

Small Circle radius = a

then

$$a = \left(\frac{2}{\sqrt{3}} - 1 \right) r$$

2



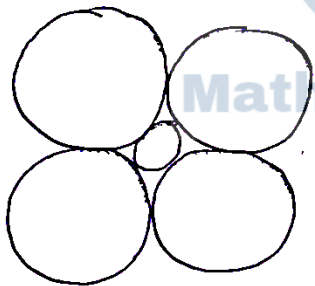
Big Circle radius = b

Small Equal radius = r

then

$$b = \left(\frac{2}{\sqrt{3}} + 1 \right) r$$

3

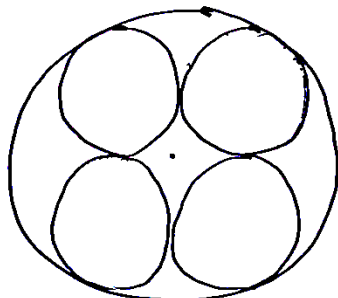


Small circle radius = a

Big circle radius = r

$$a = (\sqrt{2} - 1) r$$

4



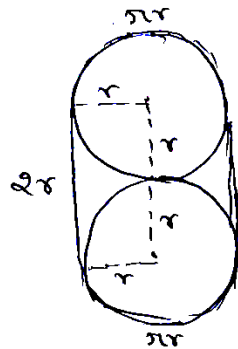
Small circle radius = r

Big circle radius = b

$$b = (\sqrt{2} + 1) r$$

length of Rubber/Rope

Case 1 :-



Rubber

$$l = 4r + 2\pi r$$

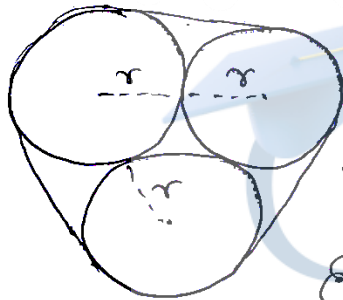
Trick

2 Circle

$$\rightarrow \text{double} = 2 \times 2r = 4r$$

$$\text{add} = 2\pi r$$

Case 2 :-



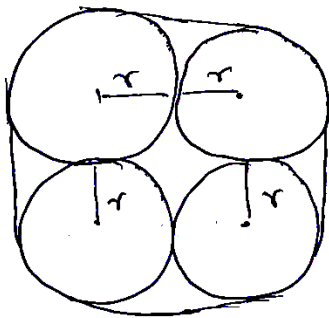
$$l = 6r + 2\pi r$$

Trick

3 Circle $\rightarrow 6r$

$$\text{add} \rightarrow + 2\pi r$$

Case 3 :-



$$l = 8r + 2\pi r$$

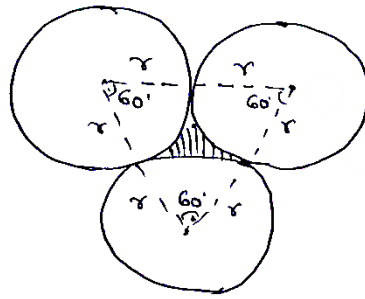
Trick

4 Circle $\rightarrow 8r$

$$\text{Add} \rightarrow + 2\pi r$$

Area

Case 1 :-



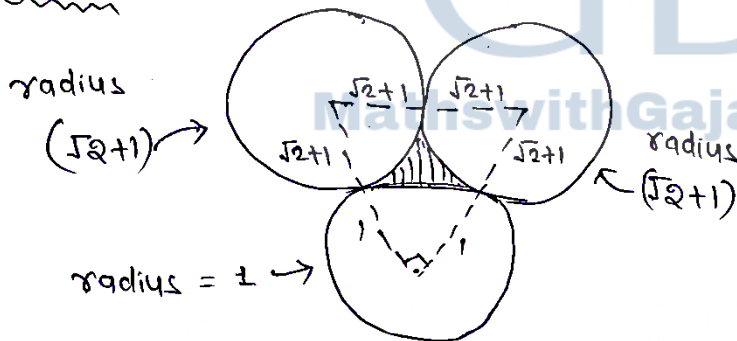
If
Same radius

$$\begin{aligned} \text{Area} &= (\Delta \text{ Area}) - (3 \text{ Sector } (60^\circ) \text{ Area}) \\ &= \frac{\sqrt{3}}{4} (2r)^2 - \left(3 \times \frac{\pi r^2}{360} \times 60^\circ \right) \\ &= \sqrt{3} r^2 - \frac{\pi r^2}{2} \end{aligned}$$

$$\text{Area} = r^2 \left[\sqrt{3} - \frac{\pi}{2} \right]$$

$$\text{Perimeter} = 3 \times \left(\frac{2\pi r}{360} \times 60^\circ \right) = \pi r$$

Case 2 :-



Check:-

$$a^2 + b^2 = c^2$$

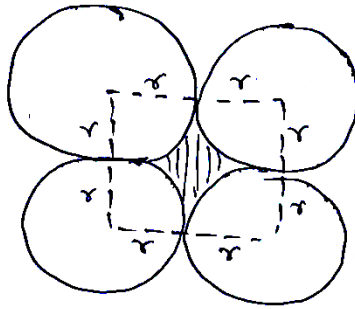
$$(\sqrt{2}+2)^2 + (\sqrt{2}+2)^2 = (2\sqrt{2}+2)^2$$

✓ ✓

Isosceles Right Triangle

$$\begin{aligned} \text{Area} &= \Delta \text{ Area} - \left[2 \times (45^\circ \text{ sector Area}) + (90^\circ \text{ sector Area}) \right] \\ &= \frac{1}{2} \times (2+\sqrt{2})^2 - \left[2 \times \frac{\pi (\sqrt{2}+1)^2}{360} \times 45^\circ + \frac{\pi (1)^2}{360} \times 90^\circ \right] \\ &= \frac{(2+\sqrt{2})^2}{2} - \left[\frac{\pi}{4} (\sqrt{2}+1)^2 + \frac{\pi}{4} \right] \end{aligned}$$

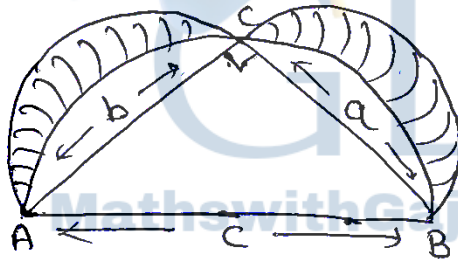
Case 3 :-



$$\begin{aligned} \text{Area} &= (2r)^2 - 4 \times \frac{\pi r^2}{360} \times 90^\circ \\ &= (\square \text{ Area}) - (4 \times \text{Sector } 90^\circ) \\ &= 4r^2 - \pi r^2 \end{aligned}$$

$$\text{Area} = r^2 [4 - \pi]$$

Case 4 :-

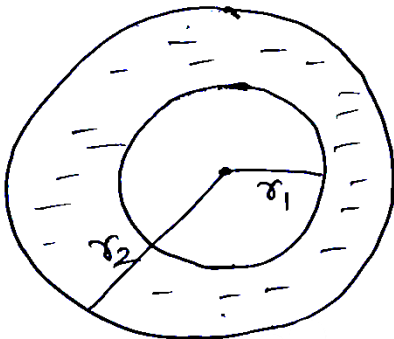


$$\text{Area} = \frac{1}{2} ab$$

OR

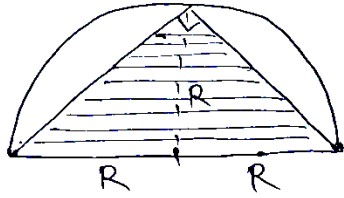
$$\text{Area} = (\triangle ABC) \text{ Area}$$

Case 5 :-



$$\text{Area} = \pi (r_2^2 - r_1^2)$$

⇒

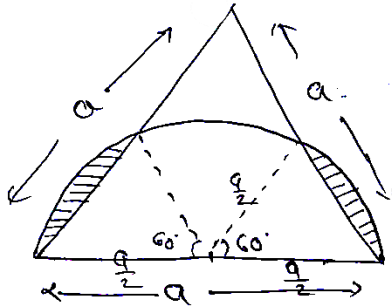


Maximum Area of Triangle.

$$\text{Area} = \frac{1}{2} \times 2R \times R$$

$$\boxed{\text{Area} = R^2}$$

⇒

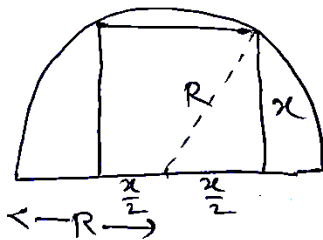


Area = ?

Equilateral Δ

$$\begin{aligned} \text{Area} &= 2 \left(\text{Sector } 60^\circ \text{ Area} - \Delta \text{ Area} \right) \\ &= 2 \left(\frac{\pi r^2}{360} \times 60^\circ - \frac{\sqrt{3} (\text{side})^2}{4} \right) \\ &= 2 \left[\frac{\pi \left(\frac{a}{2}\right)^2 \times \frac{1}{6}}{1} - \frac{\sqrt{3} \left(\frac{a}{2}\right)^2}{4} \right] \end{aligned}$$

⇒



side of square (x) = ?

$$\boxed{x = \frac{2}{\sqrt{5}} R}$$

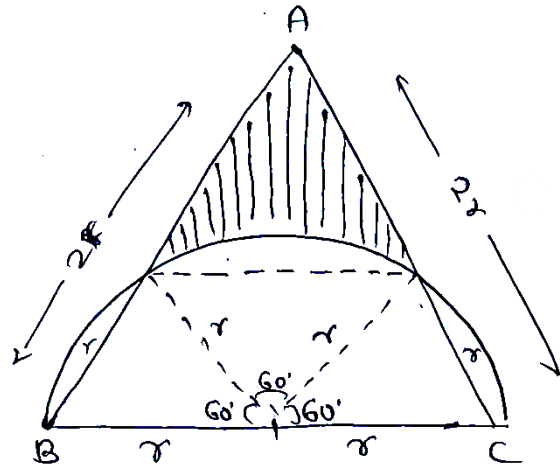
Solⁿ

$$x^2 + \left(\frac{x}{2}\right)^2 = R^2$$

$$\frac{5x^2}{4} = R^2$$

$$\boxed{x = \frac{2}{\sqrt{5}} R}$$

⇒



$$A_{\text{req}} = (\Delta ABC \text{ Area}) - [2 \text{ Equilateral } \Delta + \text{Sector } 60^\circ \text{ Area}]$$

$$= \frac{\sqrt{3}}{4} (2r)^2 - \left[2 \times \frac{\sqrt{3}}{4} (r)^2 + \frac{\pi r^2}{360} \times 60^\circ \right]$$

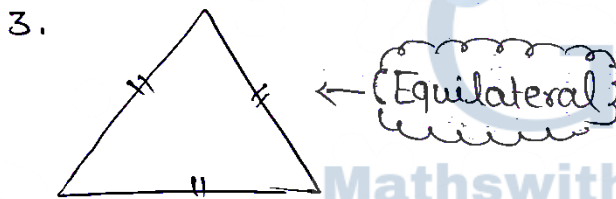
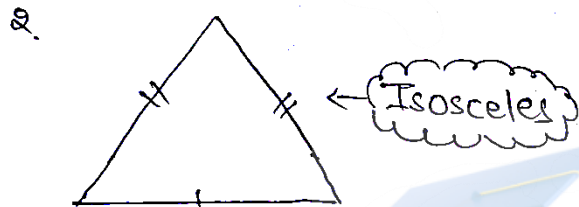
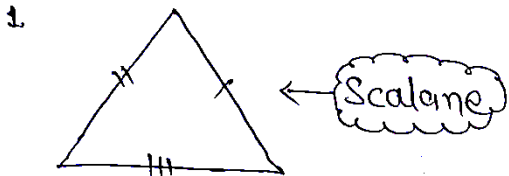
Remember

Radius	Circumference	Area
7	44	154
14	88	616
21	132	1386
28	176	2464
35	220	3850
42	264	5544

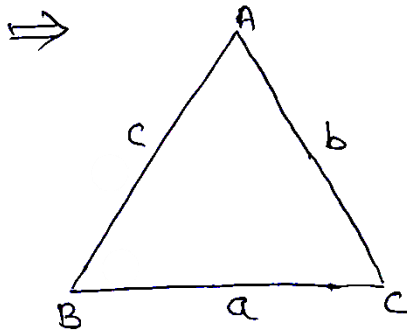
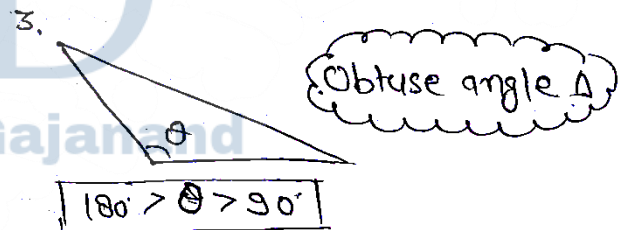
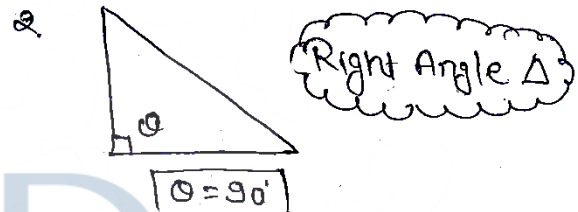
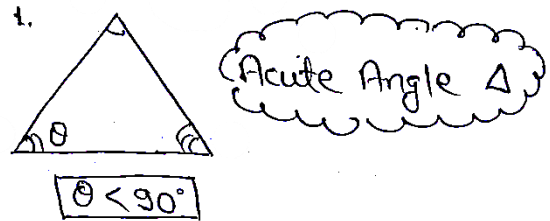
Triangle

Types of Triangle

Based on Side



Based on Angle



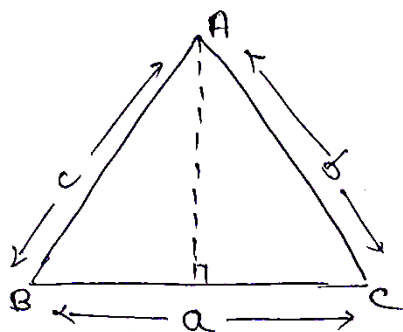
c is greatest side

$$a^2 + b^2 > c^2 \Rightarrow \text{Acute angle } \Delta$$

$$a^2 + b^2 = c^2 \Rightarrow \text{Right angle } \Delta$$

$$a^2 + b^2 < c^2 \Rightarrow \text{Obtuse angle } \Delta$$

⇒



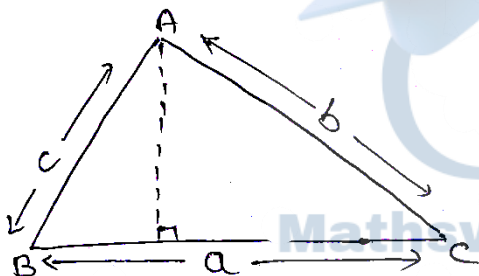
$$\text{Area} = \frac{1}{2} \times a \times h$$

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$s = \frac{a+b+c}{2}$$

⇒

Cosine Rule :-



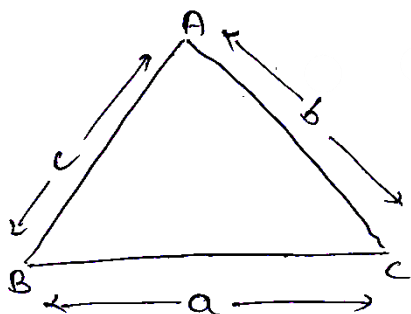
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

⇒

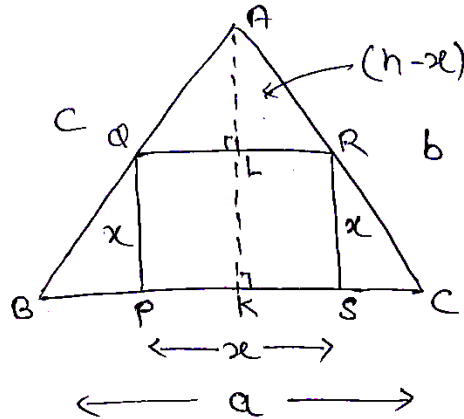
Sine Rule :-



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

R → Circum Radius

Maximum side of square in Triangle :-



$$\triangle ALQ \sim \triangle AKB$$

$$\frac{AL}{AK} = \frac{AQ}{AB} \quad \text{--- ①}$$

$$\triangle AQR \sim \triangle ABC$$

$$\frac{AQ}{AB} = \frac{QR}{BC} \quad \text{--- ②}$$

from ① & ②

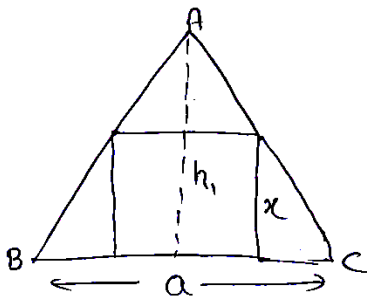
$$\frac{AL}{AK} = \frac{QR}{BC}$$

$$\frac{h-x}{h} = \frac{x}{a}$$

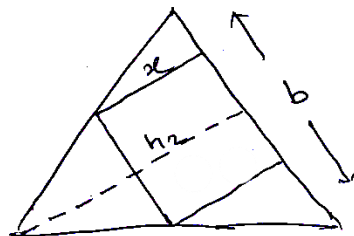
$$ah - ax = hx$$

$$x = \frac{ah}{a+h}$$

$$* \quad x = \frac{\text{base} \times \text{height}}{\text{base} + \text{height}} \quad *$$



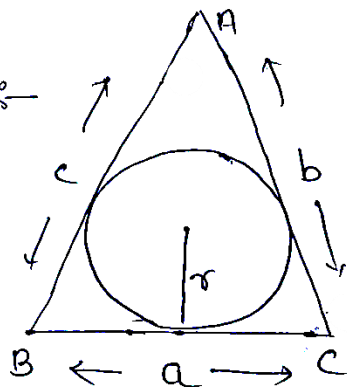
$$x = \frac{ah_1}{a+h_1}$$



$$x = \frac{bh_2}{b+h_2}$$

Maximum radius of Circle in Triangle :-

Case 1 :-



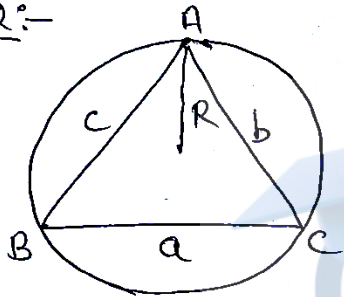
Inradius

$$r = \frac{\Delta}{S}$$

$\Delta \rightarrow$ Area of Triangle

$S \rightarrow$ Semiperimeter
 $\frac{(a+b+c)}{2}$

Case 2 :-



Circum-Radius

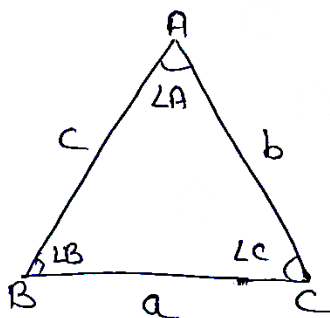
$$R = \frac{abc}{4\Delta}$$

$\Delta \rightarrow$ Area of $\triangle ABC$

MathswithGajanand

Area of Triangle

Case 1 :- If Two sides and Angle between them Given

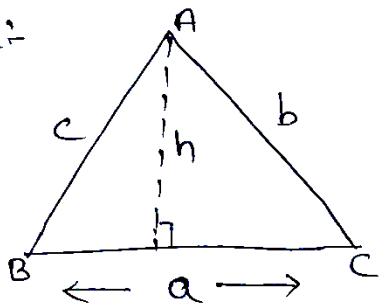


$$\text{Area} = \frac{1}{2} ab \sin C$$

$$\text{Area} = \frac{1}{2} bc \sin A$$

$$\text{Area} = \frac{1}{2} ac \sin B$$

Prove:-



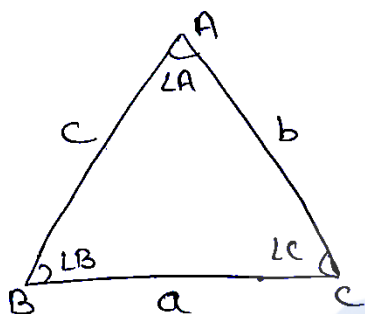
$$\sin B = \frac{h}{c}$$

$$h = c \sin B$$

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times a \times c \sin B = \frac{1}{2} ac \sin B$$

Case 2:- If One side and Two angle Given:-



$$\text{Area} = \frac{1}{2} a^2 \frac{\sin B \sin C}{\sin A}$$

$$\text{Area} = \frac{1}{2} b^2 \frac{\sin A \sin C}{\sin B}$$

$$\text{Area} = \frac{1}{2} c^2 \frac{\sin A \sin B}{\sin C}$$

Prove:-

Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

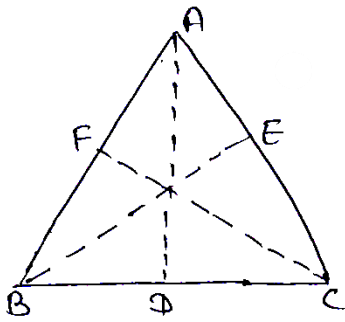
$$b = \frac{a \sin B}{\sin A}$$

$$\text{Area} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} a \times \frac{a \sin B}{\sin A} \times \sin C$$

$$= \frac{1}{2} a^2 \frac{\sin B \sin C}{\sin A}$$

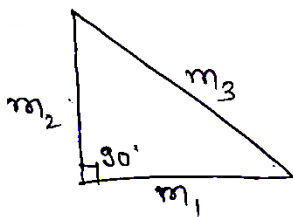
Case 3:- If length of medians are given:-



$$\text{Area} = \frac{4}{3} \sqrt{s(s-m_1)(s-m_2)(s-m_3)}$$

$$s = \frac{m_1 + m_2 + m_3}{2}$$

If m_1, m_2, m_3 makes Right angle Δ

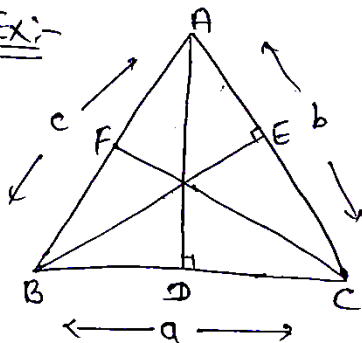


$$\text{Area} = \frac{4}{3} \left(\frac{1}{2} \times m_1 \times m_2 \right)$$

$\Rightarrow$ If the Altitudes of a Triangle is h_1, h_2 and h_3 then sides of Triangle

$$a : b : c = \frac{1}{h_1} : \frac{1}{h_2} : \frac{1}{h_3}$$

Ex:-



If $AD : BE : CF = 3 : 4 : 5$

then $a : b : c = ?$

$$\text{Sol}^n \quad \frac{1}{2} \times a \times 3 = \frac{1}{2} \times b \times 4 = \frac{1}{2} \times c \times 5$$

$$3a = 4b = 5c$$

$$\frac{a}{20} = \frac{b}{15} = \frac{c}{12}$$

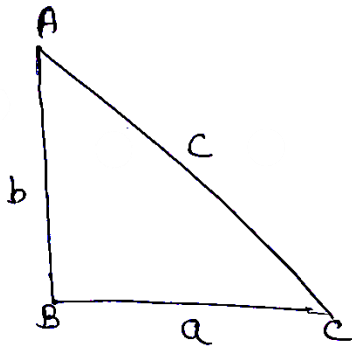
$$\boxed{20 : 15 : 12} \text{ Ans}$$

(Trick) $3 : 4 : 5$

$$a : b : c = \boxed{20 : 15 : 12}$$

Ans

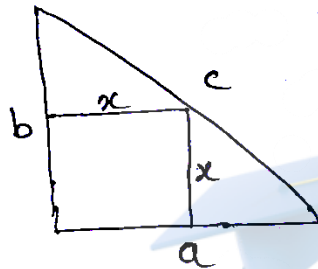
Right angle Triangle



$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{height}$$

Maximum side of Square in Right angle Δ

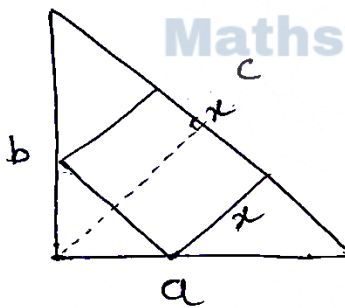
Case-1



$$x = \frac{\text{base} \times \text{height}}{(\text{base} + \text{height})}$$

$$\# \quad x = \frac{ab}{a+b} \quad \#$$

Case-2



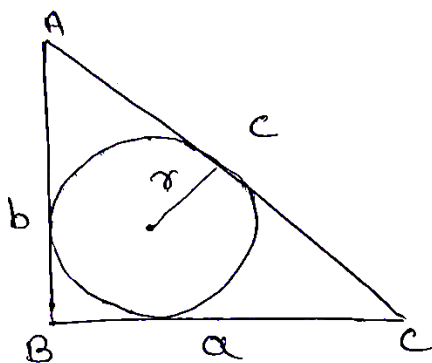
$$x = \frac{\text{base} \times \text{height}}{\text{base} + \text{height}}$$

$$x = \frac{cxh}{c+h} \quad \left[\because h = \frac{ab}{c} \right]$$

$$= \frac{c \times \frac{ab}{c}}{c + \frac{ab}{c}}$$

$$\# \quad x = \frac{abc}{a^2 + b^2 + ab} \quad \#$$

Maximum radius of Circle in Right angle Δ :-



Inradius

$$\# \quad r = \frac{a+b-c}{2} \quad \#$$

OR

$$r = \frac{\Delta}{s}$$

Prove:- $r = \frac{\Delta}{s} = \frac{\frac{1}{2}ab}{\frac{a+b+c}{2}}$

$$r = \frac{ab}{a+b+c}$$

$$= \frac{(a+b+c)(a+b-c)}{2(a+b+c)}$$

$$r = \frac{a+b-c}{2}$$

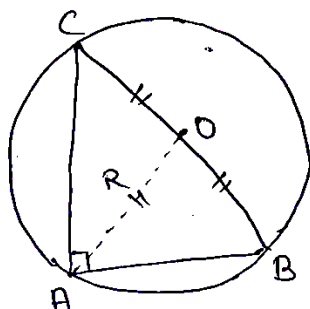
$$\because (a+b)^2 = a^2 + b^2 + 2ab$$

$$(a+b)^2 = c^2 + 2ab$$

$$\frac{(a+b)^2 - c^2}{2} = ab$$

$$ab = \frac{(a+b+c)(a+b-c)}{2}$$

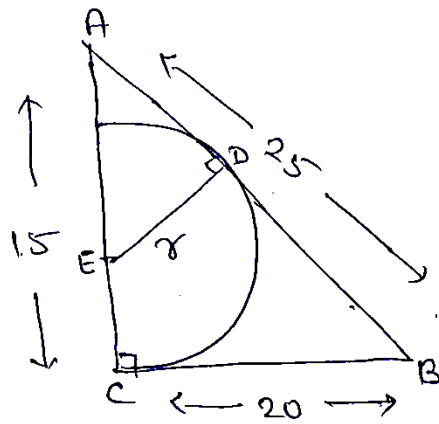
Circum-Radius



$$\# \quad R = \frac{\text{Hypotenous}}{2} \quad \#$$

$$R = \frac{BC}{2}$$

Example



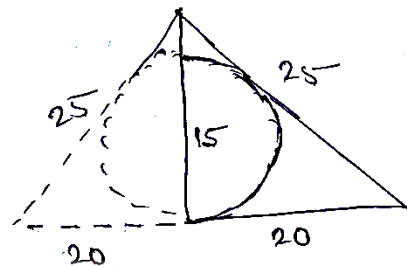
radius (r) = ?

Soln:- m-I $\triangle ADE \sim \triangle ACB$ m-II

$$\frac{AE}{AB} = \frac{DE}{CB}$$

$$\frac{15-r}{25} = \frac{r}{20}$$

$$\boxed{r = \frac{20}{3}}$$

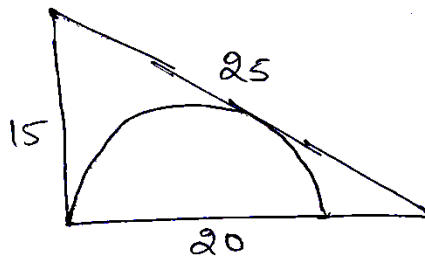


$$\text{Incentre} = r = \frac{\Delta}{s}$$

$$r = \frac{\frac{1}{2} \times 40 \times 15}{\frac{90}{2}}$$

$$\boxed{r = \frac{20}{3}}$$

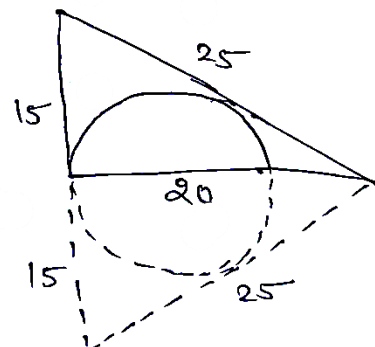
Ex:-



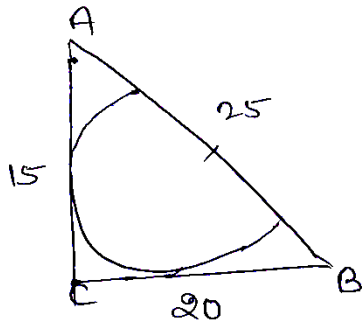
$r = ?$

Soln :- $r = \frac{\Delta}{s} = \frac{\frac{1}{2} \times 30 \times 20}{\frac{80}{2}}$

$$\boxed{r = \frac{15}{2}}$$



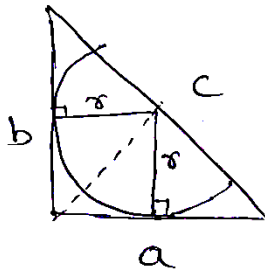
Ex:



$$\boxed{r = \frac{a \times b}{a+b}} = \frac{15 \times 20}{15+20} = \frac{300}{35} = \frac{60}{7}$$

Prove:

$$\boxed{r = \frac{a \times b}{a+b}}$$

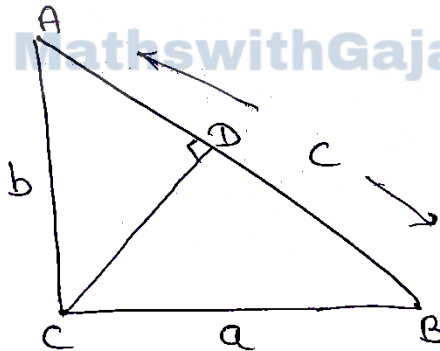


$$\frac{1}{2} \times b \times r + \frac{1}{2} \times a \times r = \frac{1}{2} \times a \times b$$

$$r(a+b) = ab$$

$$\boxed{r = \frac{ab}{a+b}}$$

Altitude in Right Angle



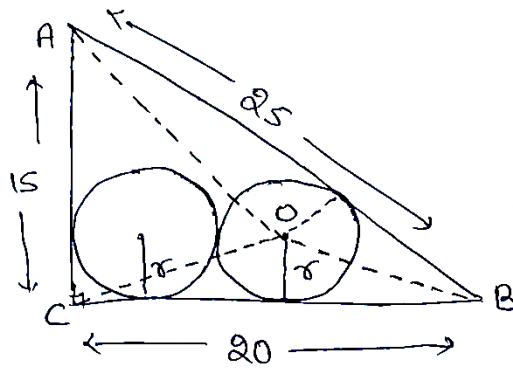
Altitude $\boxed{CD = \frac{a \times b}{c}}$

$$\boxed{CD^2 = AD \times BD}$$

$$\boxed{AD = \frac{b^2}{c}}$$

$$\boxed{BD = \frac{a^2}{c}}$$

Que.



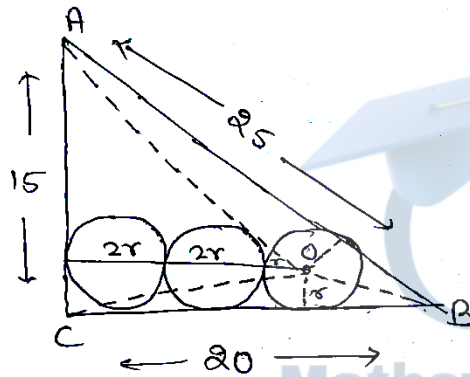
$$r = ?$$

Soln:-
$$\text{Ar}(\triangle AOC) + \text{Ar}(\triangle BOC) + \text{Ar}(\triangle AOB) = \text{Ar}(\triangle ABC)$$

$$\frac{1}{2} \times 15 \times 3r + \frac{1}{2} \times 20 \times r + \frac{1}{2} \times 25 \times r = \frac{1}{2} \times 20 \times 15$$

$$r = \frac{300}{90} = \frac{10}{3} \text{ Ans}$$

Que.



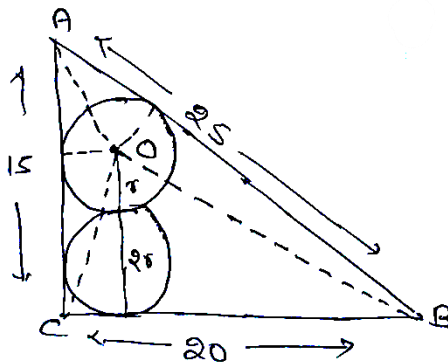
Soln
$$\text{Ar}(\triangle BOC) + \text{Ar}(\triangle AOC) + \text{Ar}(\triangle AOB) = \text{Ar}(\triangle ABC)$$

$$\frac{1}{2} \times 20 \times r + \frac{1}{2} \times 15 \times 5r + \frac{1}{2} \times 25 \times r = \frac{1}{2} \times 20 \times 15$$

$$120r = 20 \times 15$$

$$r = \frac{5}{2} \text{ Ans}$$

Que:



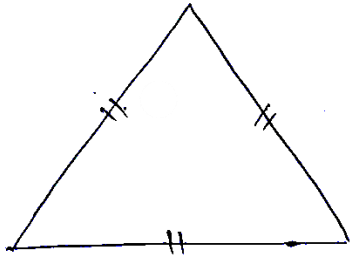
$$\text{Ar}(\triangle COB) + \text{Ar}(\triangle AOC) + \text{Ar}(\triangle AOB) = \text{Ar}(\triangle ABC)$$

$$\frac{1}{2} \times 20 \times 3r + \frac{1}{2} \times 15 \times r + \frac{1}{2} \times 25 \times r = \frac{1}{2} \times 20 \times 15$$

$$\boxed{r = 3} \text{ Ans,}$$

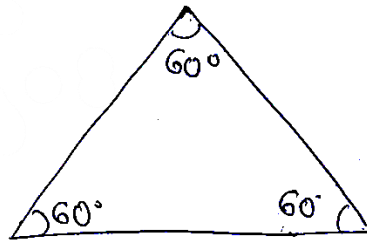
Equilateral Triangle

1.



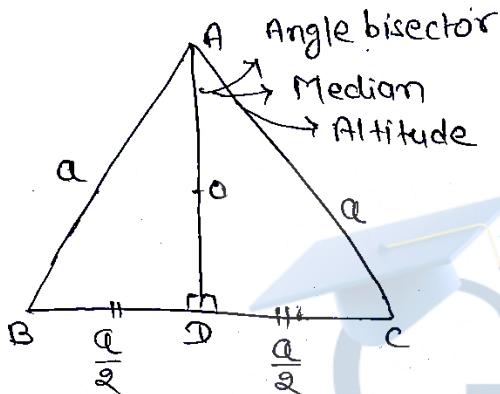
(Each side Equal)

2.



(each angle = 60°)

3.

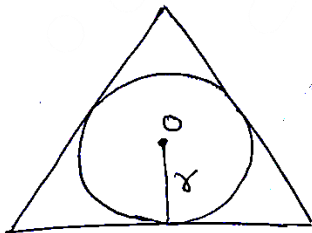


$$\# \text{ Area} = \frac{\sqrt{3}}{4} (\text{side})^2 \#$$

$$\# \text{ height} = \frac{\sqrt{3}}{2} (\text{side}) \#$$

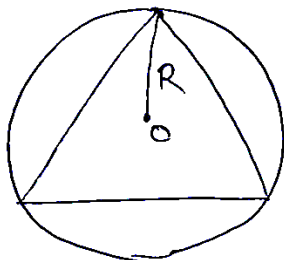
$$AO : OD = 2 : 1$$

4.



$$\text{Inradius } (r) = \frac{\text{side}}{2\sqrt{3}} \#$$

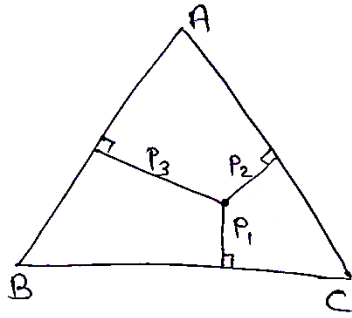
5.



$$\text{Circum-radius } R = \frac{\text{side}}{\sqrt{3}}$$

$\Rightarrow$ Note: 'O' is orthocentre, Circumcenter, Incentre, Centroid.

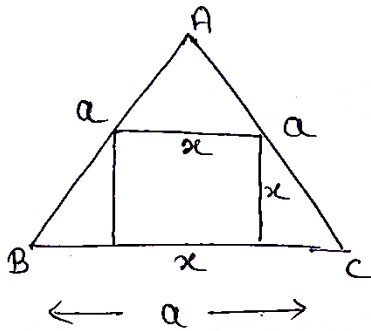
6.



$$\# \text{ side} = \frac{2}{\sqrt{3}} (P_1 + P_2 + P_3) \#$$

$$\# \text{ height} = P_1 + P_2 + P_3 \#$$

7.

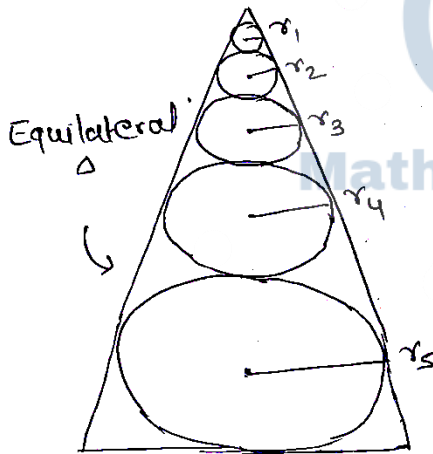


maximum side of square in Δ

$$x = \frac{\text{base} \times \text{height}}{\text{base} + \text{height}}$$

$$x = \frac{a \times \frac{\sqrt{3}a}{2}}{a + \frac{\sqrt{3}a}{2}} = \frac{\sqrt{3}a}{2 + \sqrt{3}}$$

8.



$$K = \frac{r_2}{r_1}, K = \frac{r_3}{r_2}, K = \frac{r_4}{r_3}, K = \frac{r_5}{r_4}$$

$$* K = \frac{r_3}{r_2} \Rightarrow r_2 = K r_1$$

$$K = \frac{r_3}{K r_1} \Rightarrow K^2 = \frac{r_3}{r_1}$$

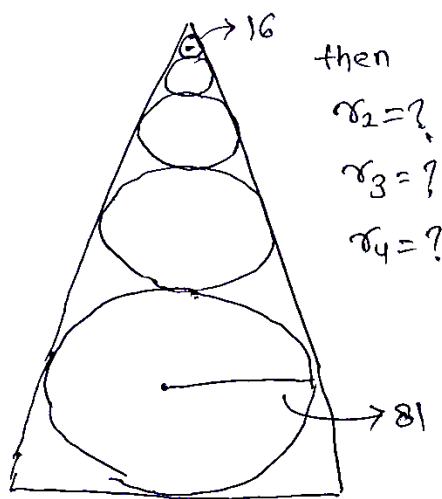
For 3 Circle

$$\frac{r_2}{r_1} = \frac{r_3}{r_2}$$

$$\# r_2 = \sqrt{r_1 r_3}$$

$$* K^2 = \frac{r_3}{r_1}, K^3 = \frac{r_4}{r_1}, K^4 = \frac{r_5}{r_1}$$

Que.



then
 $r_2 = ?$
 $r_3 = ?$
 $r_4 = ?$

Soln:-

$$K^4 = \frac{r_5}{r_1}$$

$$K^4 = \frac{81}{16} = \boxed{\frac{3}{2} = K}$$

$$K^3 = \frac{r_4}{r_1} \Rightarrow \frac{27}{8} = \frac{r_4}{16}$$

$$\boxed{r_4 = 54}$$

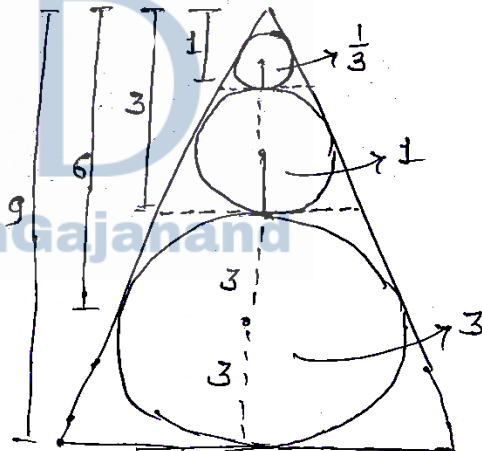
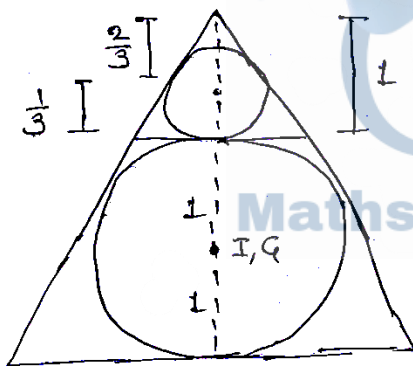
$$K^2 = \frac{r_3}{r_1} \Rightarrow \frac{9}{4} = \frac{r_3}{16}$$

$$\boxed{r_3 = 36}$$

$$K = \frac{r_2}{r_1} \Rightarrow \frac{3}{2} = \frac{r_2}{16}$$

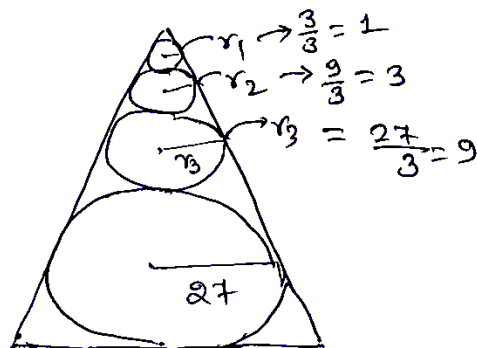
$$\boxed{r_2 = 24}$$

$\Rightarrow$ If Triangle is Equilateral :-

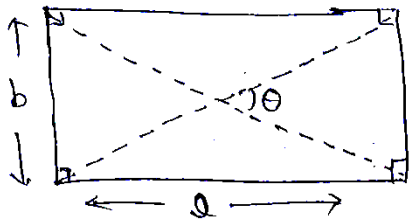


$\Rightarrow$ If Δ is an equilateral triangle then Ratio of radii of smaller and larger circle will be $\boxed{1:3}$

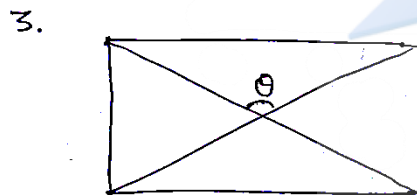
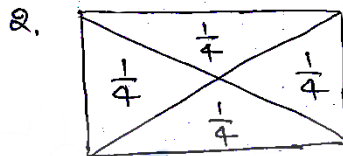
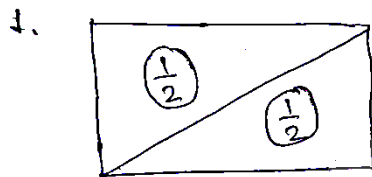
Ex:-



Rectangle



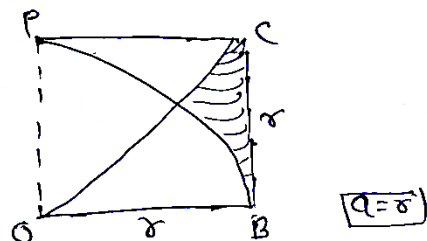
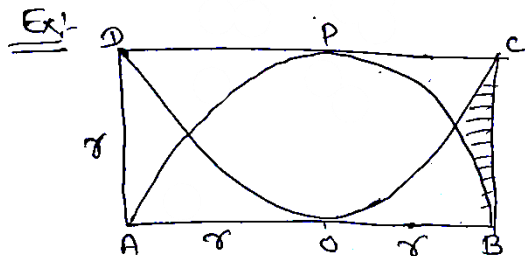
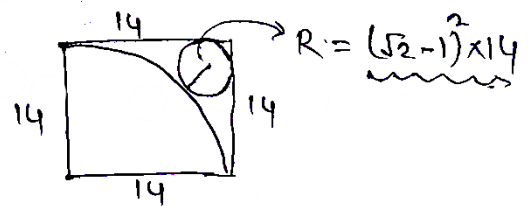
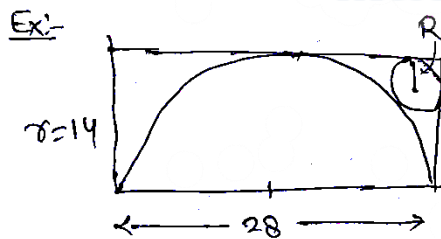
$$\begin{aligned} \text{Area} &= l \times b \\ \text{Perimeter} &= 2(l+b) \\ \text{Diagonal} &= \sqrt{l^2 + b^2} \\ \text{Area} &= \frac{1}{2} d^2 \sin \theta \end{aligned}$$

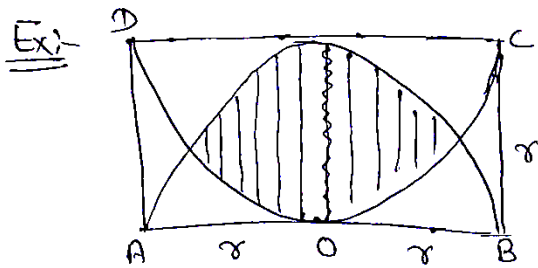


$$\theta \neq 90^\circ$$

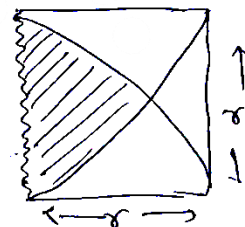
$$d_1 = d_2 \text{ (Both diagonals are equal)}$$

MathswithGajanand



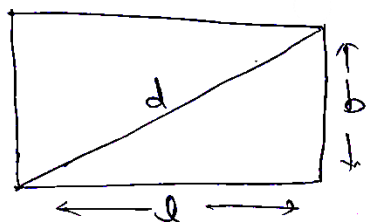


$\Rightarrow 2 \times$



$$2 \times \left[\frac{\pi r^2}{12} (4\pi - 3\sqrt{3}) \right]$$

Ex:-



Given $d = 13$

Area = 60

then $l, b = ?$

Soln :- $l^2 + b^2 = 169$ — (1)

$lb = 60$ — (2)

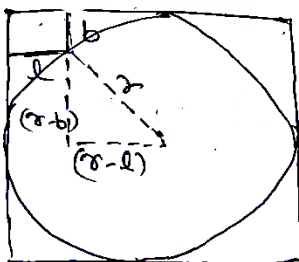
from (1) & (2) $l^2 + b^2 + 2lb = (l+b)^2 = 169 + 120 \Rightarrow 289$

$l^2 + b^2 - 2lb = (l-b)^2 = 169 - 120 \Rightarrow 49$

$l+b = 17$
 $l-b = 7$
 $\rightarrow l = \frac{17+7}{2} = 12 \text{ cm}$

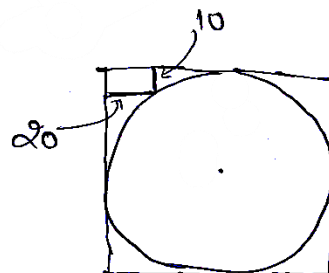
$b = \frac{17-7}{2} = \frac{10}{2} = 5 \text{ cm}$

$\Rightarrow$



$(x-l)^2 + (x-b)^2 = r^2$

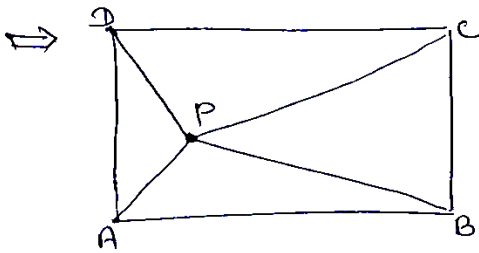
Ex:-



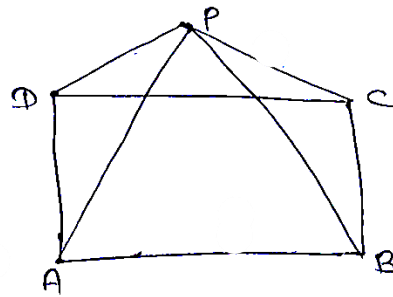
$(x-10)^2 + (x-20)^2 = r^2$

$x = 50$

$(40)^2 + (30)^2 = (50)^2 \checkmark$

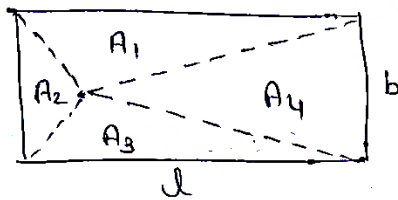


$$\boxed{PD^2 + BP^2 = PA^2 + PC^2}$$

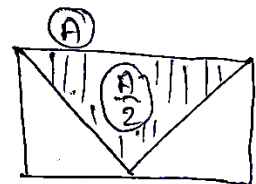
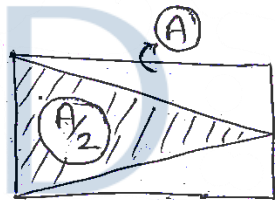
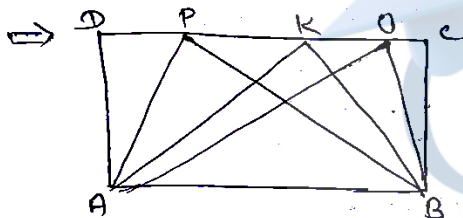


$$\# \boxed{PA^2 + PC^2 = PB^2 + PD^2}$$

⇒

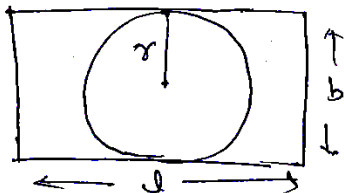


$$\boxed{A_1 + A_3 = A_2 + A_4}$$



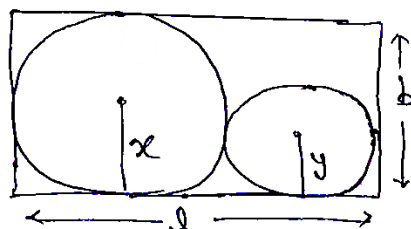
$$\boxed{A_s(\triangle APB) = A_s(\triangle AKB) = A_s(\triangle AOB) = \frac{1}{2}(\square ABCD)}$$

⇒



$$\boxed{r = \frac{b}{2}}$$

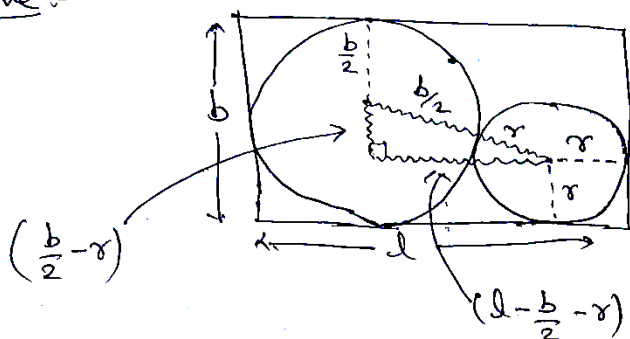
Type-1



$$\text{here } x = \frac{b}{2}$$

$$\# \boxed{(x+y)^2 = (l-x-y)^2 + (b-x-y)^2}$$

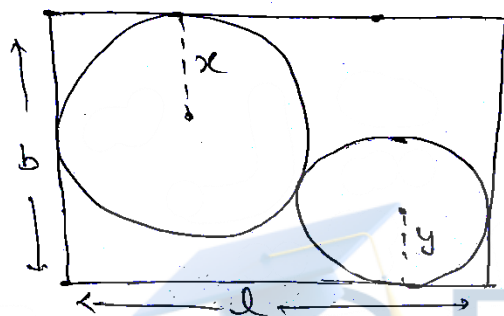
Prove:-



l and b Given and Ask $r=?$

$$\left[\left(\frac{b}{2} + r \right)^2 = \left(\frac{b}{2} - r \right)^2 + \left(l - \frac{b}{2} - r \right)^2 \right]$$

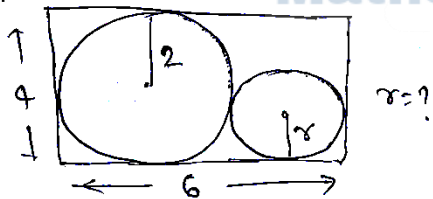
type-2



l and b Given
and one radius given

$$\# \left[(x+y)^2 = (b-x-y)^2 + (l-x-y)^2 \right] \#$$

Ex:-



Soln:-

$$(l-x-y)^2 + (b-x-y)^2 = (x+y)^2$$

$$(6-2-r)^2 + (4-2-r)^2 = (2+r)^2$$

$$(4-r)^2 + (2-r)^2 = (2+r)^2$$

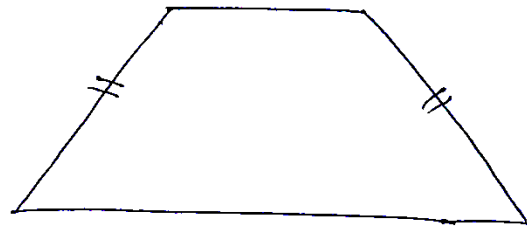
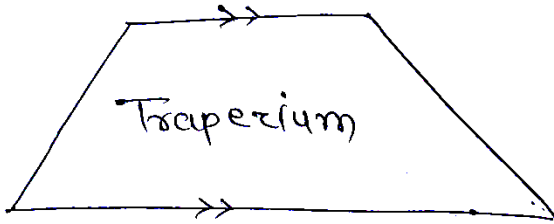
$$16+r^2-8r + 4+r^2-4r = 4+r^2+4r$$

$$r^2-16r+16=0$$

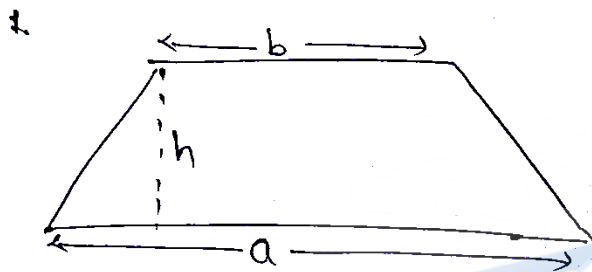
$$r = \frac{16 \pm \sqrt{256-64}}{2} \Rightarrow 8 \pm 4\sqrt{3}$$

$$\boxed{r = 8 - 4\sqrt{3}} \text{ Any}$$

Trapezium



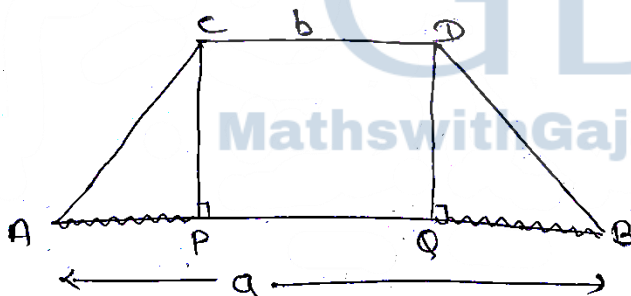
(Isosceles trapezium)



#

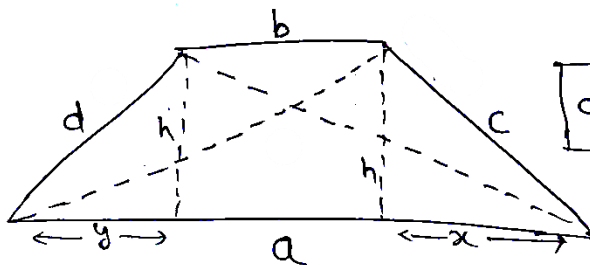
$$\text{Area} = \frac{1}{2}(a+b) \times h$$

2. In isosceles trapezium



$$AP = BQ = \frac{a-b}{2}$$

3.



$$d_1^2 + d_2^2 = c^2 + d^2 + 2ab$$

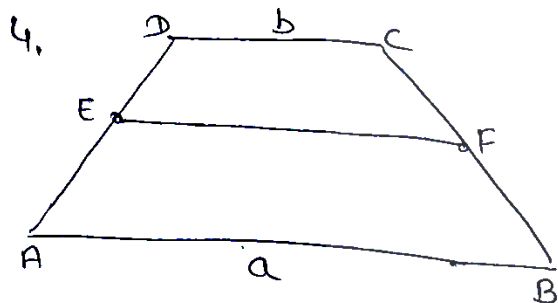
Prove:

$$d_1^2 = h^2 + (a-x)^2 \quad \text{--- ①}$$

$$d_2^2 = h^2 + (a-y)^2 \quad \text{--- ②}$$

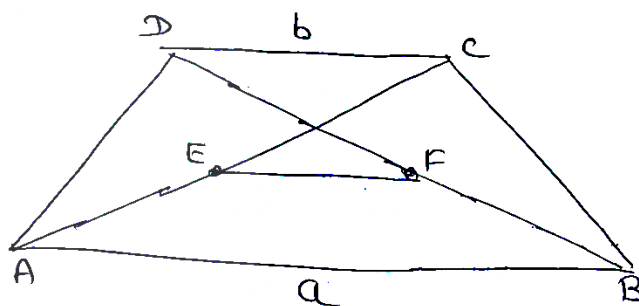
from ① & ②

$$d_1^2 + d_2^2 = c^2 + d^2 + 2ab$$



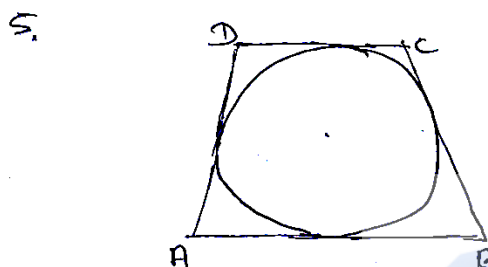
$$\# \quad EF = \frac{a+b}{2}$$

(E & F are mid point)



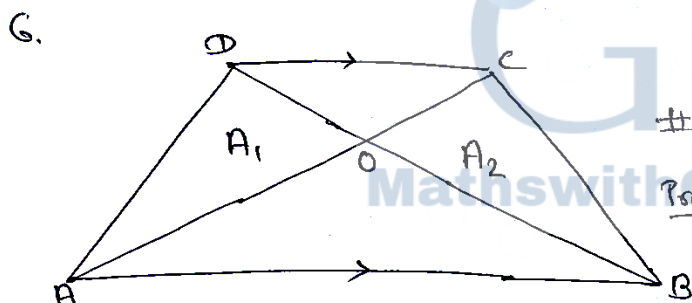
$$\# \quad EF = \frac{a-b}{2}$$

(E and F are mid point)



Imp.

$$AB + CD = BC + AD$$

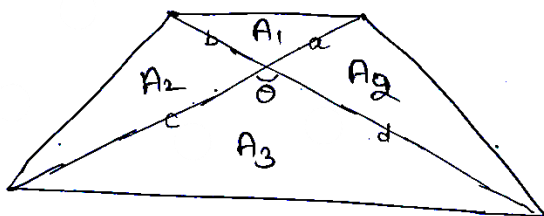


$$\# \quad A_1 = A_2$$

Prove

$$\begin{aligned} \text{Area}(\triangle ABD) &= \text{Area}(\triangle ABC) \\ - \text{Ar}(\triangle AOB) &\quad - \triangle AOB \\ \hline \text{Ar}(\triangle AOD) &= \text{Ar}(\triangle BOC) \end{aligned}$$

7.



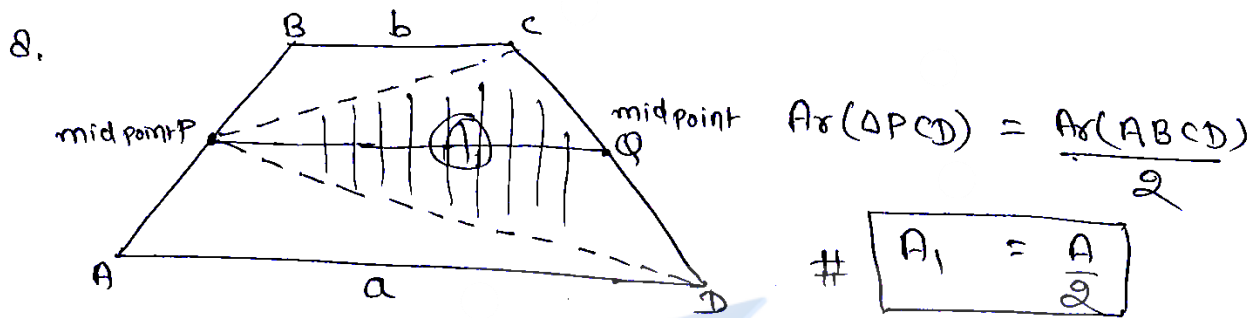
$$\# \quad A_2^2 = A_1 \times A_3$$

Prove

$$A_1 = \frac{1}{2} ab \sin \theta, \quad A_2 = \frac{1}{2} bc \sin(180-\theta)$$

$$A_3 = \frac{1}{2} cd \sin \theta, \quad A_4 = \frac{1}{2} ad \sin(180-\theta)$$

$$A_1 \times A_3 = A_2^2$$



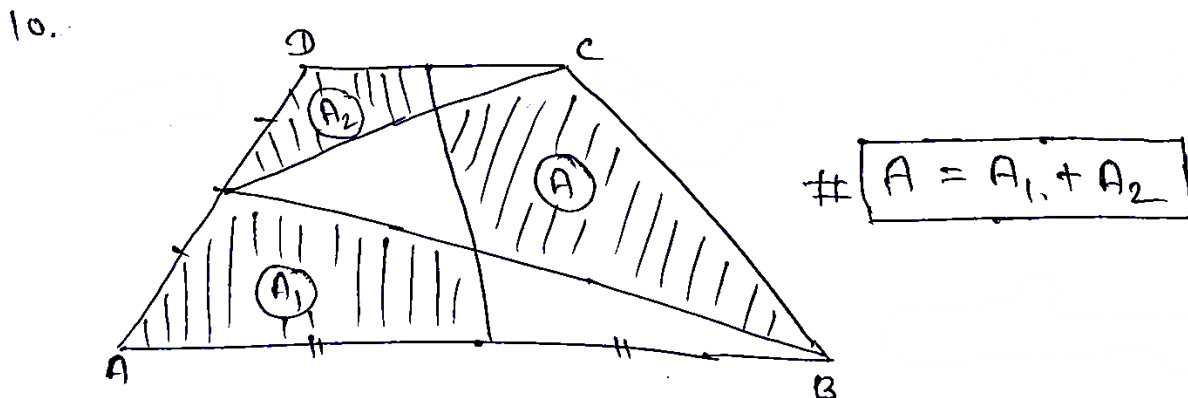
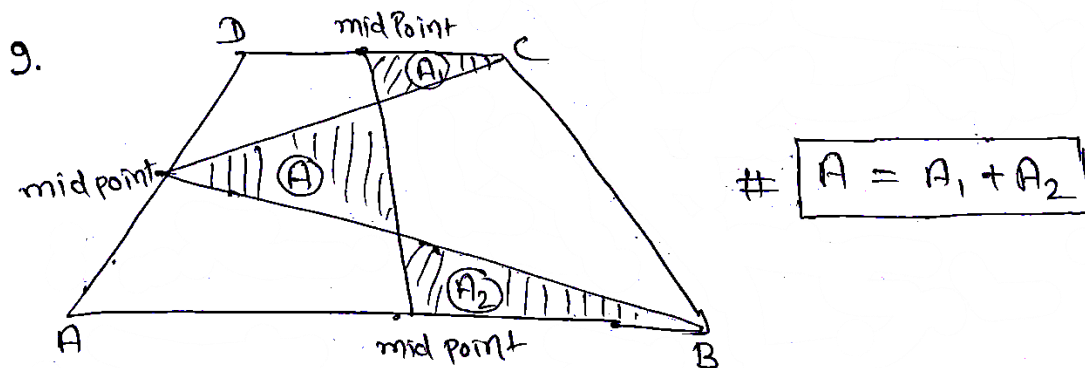
Prove

$$\triangle APD + \triangle PBC = \frac{1}{2} \times a \times \frac{h}{2} + \frac{1}{2} \times b \times \frac{h}{2}$$

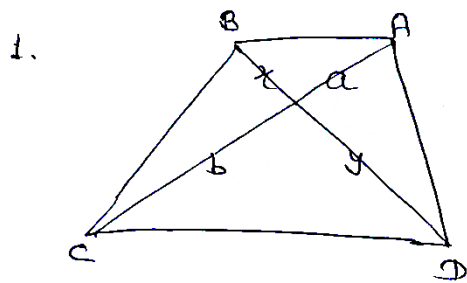
$$A_1 = \frac{1}{4}(a+b) \times h$$

$$\text{Total Area} = \frac{1}{2}(a+b) \times h$$

$$[A_1 = \frac{A}{2}]$$



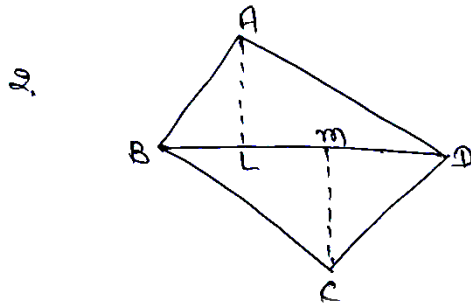
Quadrilateral



(i) $\angle A + \angle B + \angle C + \angle D = 360^\circ$

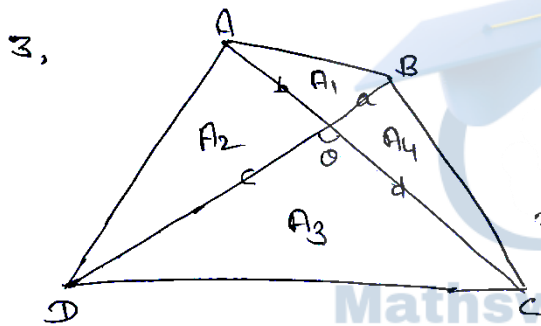
(ii) $Ar(\triangle ABD) : Ar(\triangle CBD) = a : b$

(iii) $Ar(\triangle ACB) : Ar(\triangle ACD) = x : y$



$Area(ABCD) = \frac{1}{2} BD \times m + \frac{1}{2} BD \times n$

$Area = \frac{1}{2} (AL + m) \times BD$



$A_1 A_3 = A_2 A_4$

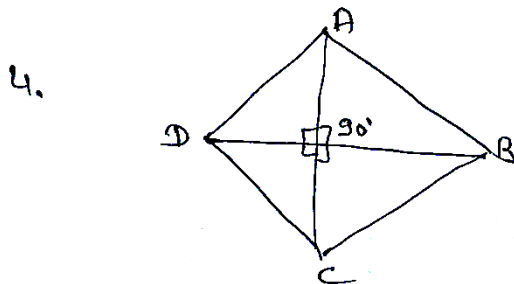
Prove $A_1 = \frac{1}{2} ab \sin \theta$

$A_3 = \frac{1}{2} cd \sin \theta$

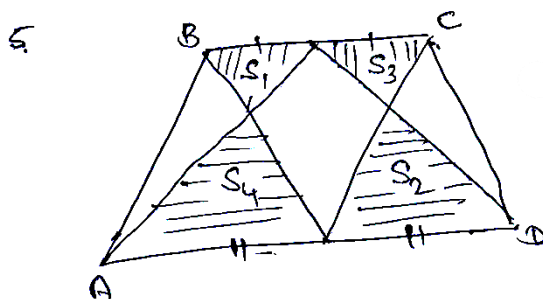
$A_2 = \frac{1}{2} bc \sin \theta$

$A_4 = \frac{1}{2} ad \sin \theta$

$[A_1 A_3 = A_2 A_4]$

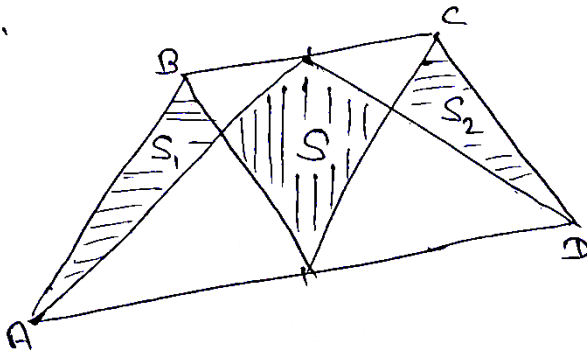


$AB^2 + CD^2 = AD^2 + BC^2$



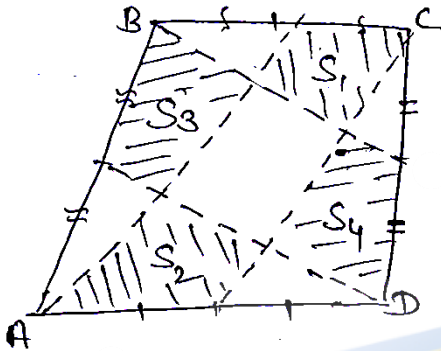
$S_1 + S_2 = S_3 + S_4$

6.



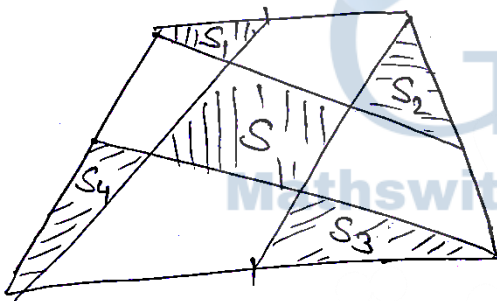
$$\# \boxed{S = S_1 + S_2} \#$$

7.



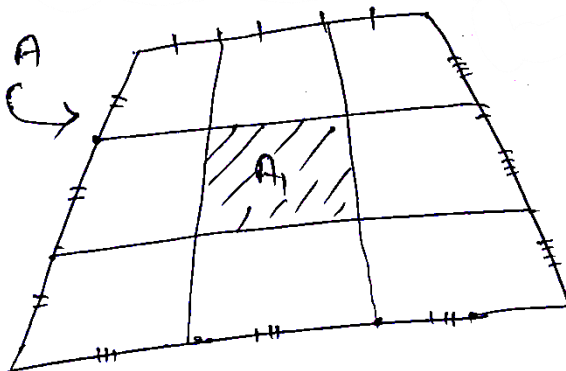
$$\# \boxed{S_1 + S_2 = S_3 + S_4} \#$$

8.



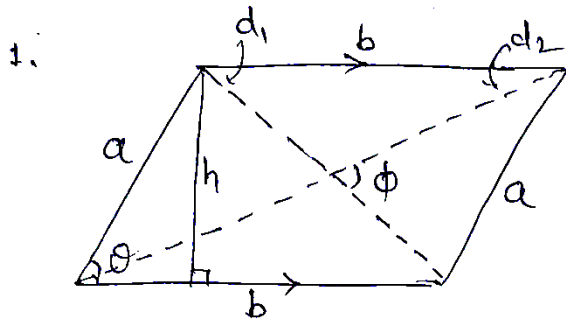
$$\boxed{S = S_1 + S_2 + S_3 + S_4}$$

9.



$$\# \boxed{A_1 = \frac{A}{9}} \#$$

Parallelogram



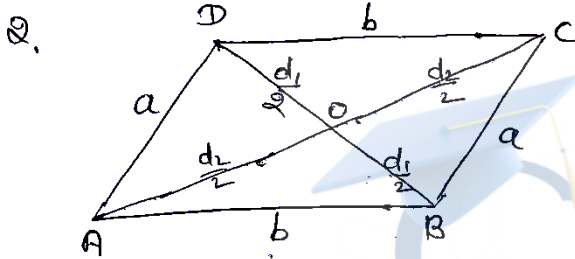
$$d_1 \neq d_2$$

$$\text{Perimeter} = 2(a+b)$$

$$\text{Area} = b \times h$$

$$\text{Area} = a \times b \times \sin \theta$$

$$\text{Area} = \frac{1}{2} \times d_1 \times d_2 \times \sin \phi$$



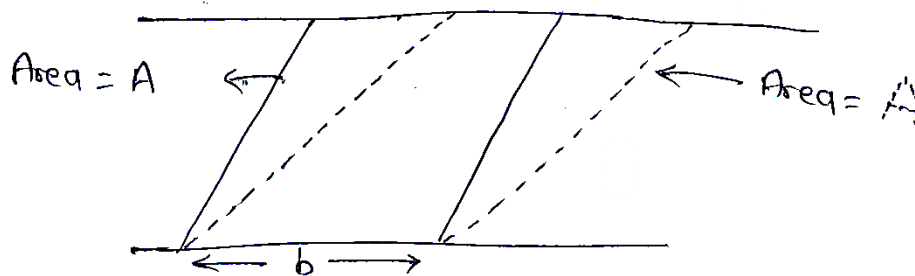
$$\# d_1^2 + d_2^2 = 2(a^2 + b^2)$$

Prove by Apollonius theorem in $\triangle OCD$

$$a^2 + b^2 = 2 \left[\frac{d_1^2}{4} + \frac{d_2^2}{4} \right]$$

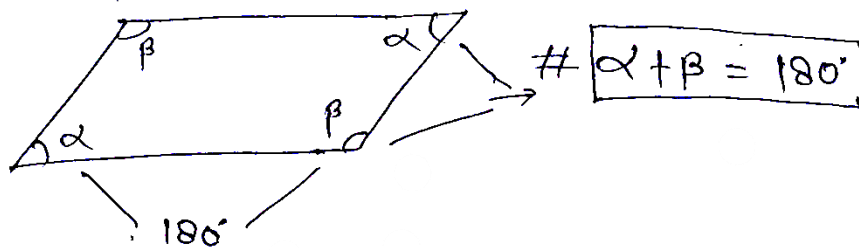
$$2(a^2 + b^2) = d_1^2 + d_2^2$$

3. Any parallelogram built on the same base between two parallel lines has the same Area.

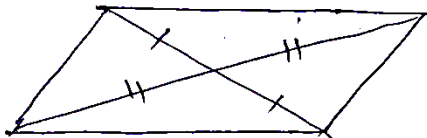


$$A = A'$$

4. Opposite Angle are equal.

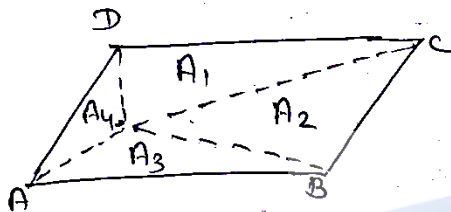


5.



$d_1 \neq d_2$, $d_1 \not\perp d_2$

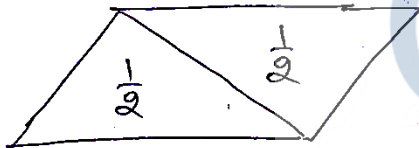
6.



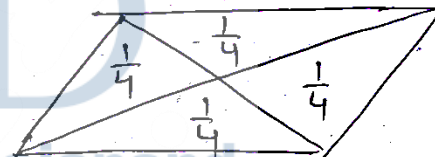
$A_1 + A_3 = A_2 + A_4$

$AO^2 + OC^2 \neq BO^2 + OD^2$

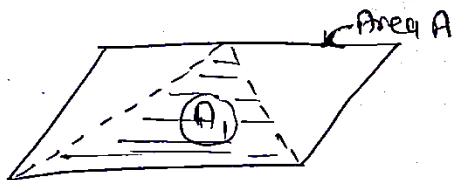
7.



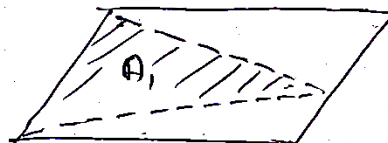
8.



9.



$A_1 = \frac{A}{2}$

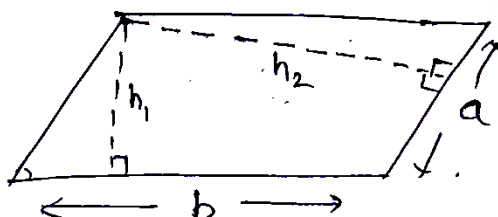


$A_1 = \frac{A}{2}$



$A_1 = \frac{A}{2}$

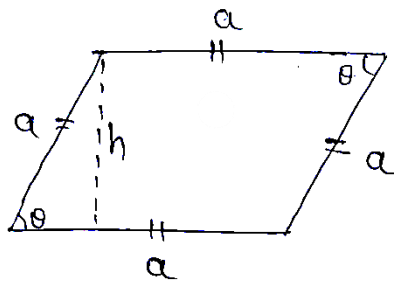
10.



$b \times h_1 = a \times h_2$

Rhombus

1.



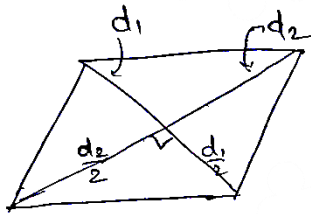
$$\text{Perimeter} = 4a$$

$$\text{Area} = \text{Side} \times \text{height}$$

$$\text{Area} = (\text{Side})^2 \times \sin \theta$$

$$\text{Area} = \frac{1}{2} \times d_1 \times d_2$$

2.



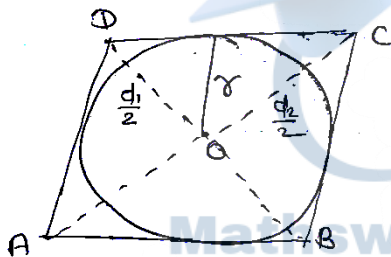
$$\# d_1^2 + d_2^2 = 4(\text{side})^2$$

Prove:- $\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = (\text{side})^2$

#

$$d_1 \perp d_2$$

3.



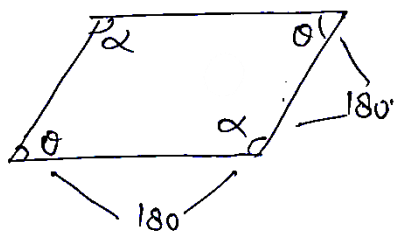
#

$$r = \frac{d_1 \times d_2}{4 \times \text{side}}$$

Prove

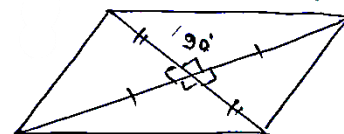
$$r = \frac{ab}{c} = \frac{\frac{d_1}{2} \times \frac{d_2}{2}}{\text{side}}$$

4.



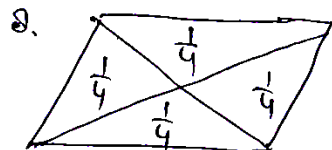
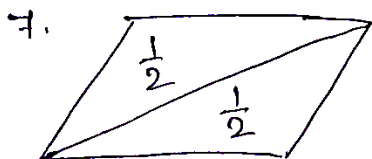
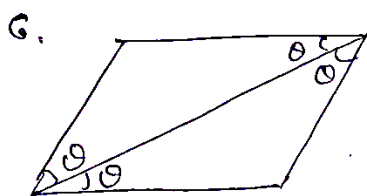
Opposite angle are equal
and $\theta + \alpha = 180^\circ$

5.

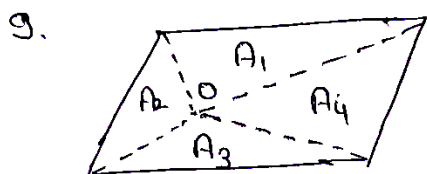


$$d_1 \perp d_2$$

d_1 and d_2 intersect each other equally



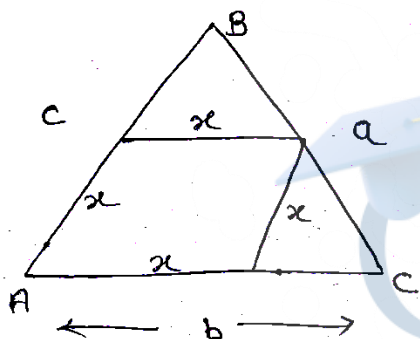
diagonal is Angle bisector



$$\# A_1 + A_3 = A_2 + A_4$$

$$\# AO^2 + OC^2 = BO^2 + OD^2$$

Maximum length Rhombus in a Triangle:



$$\# x = \frac{b \times c}{b + c}$$

MathswithGajanand

Prove:- $\triangle BED \sim \triangle BAC$ | $\triangle CFD \sim \triangle CAB$

$$\frac{c-x}{c} = \frac{x}{b}$$

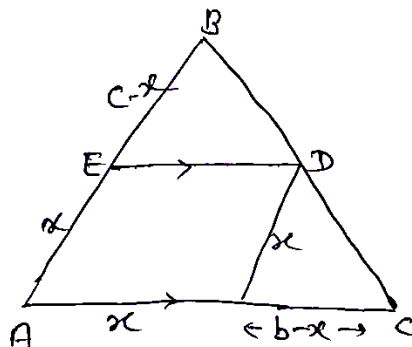
$$\frac{b-x}{b} = \frac{x}{c}$$

$$\frac{b(c-x)}{c} = \frac{c(b-x)}{b}$$

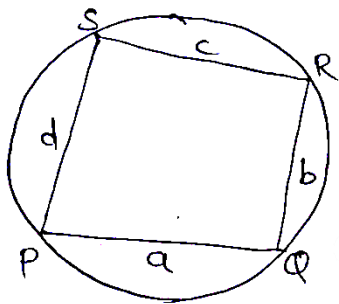
$$b^2c - b^2x = c^2b - c^2x$$

$$bc(b-c) = (b^2 - c^2)x$$

$$x = \frac{bc}{b+c}$$



Cyclic Quadrilateral



$$\langle i \rangle \angle P + \angle R = 180^\circ$$

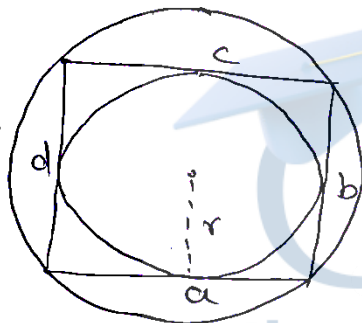
$$\angle S + \angle Q = 180^\circ$$

(Sum of opposite Angle = 180°)

$$\langle ii \rangle \text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

$$\text{where } s = \frac{a+b+c+d}{2}$$

Bicentric Quadrilateral:-



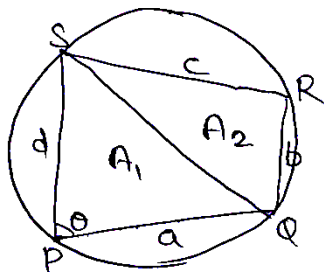
$$\# a+c = b+d$$

$$s = b+d / s = a+c$$

$$\text{Area} = \sqrt{abcd}$$

$$\gamma = \frac{\Delta}{s} = \frac{\sqrt{abcd}}{s}$$

#



$$\# \frac{A_1}{A_2} = \frac{ad}{bc}$$

Proof:-

$$A_1 = \frac{1}{2} ad \sin \theta$$

$$A_2 = \frac{1}{2} bc \sin(180-\theta)$$

$$\frac{A_1}{A_2} = \frac{ad}{bc}$$

Ex:- In a cyclic Quadrilateral whose two side are 2 and 5 and third side is 3, then fourth side is: and $\angle = 60^\circ$

Soln:- $\cos 60^\circ = \frac{5^2 + 2^2 - BD^2}{2 \times 2 \times 5}$

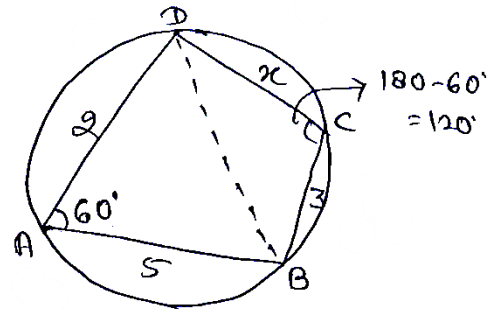
$$\frac{1}{2} = \frac{29 - BD^2}{20}$$

$$BD^2 = 19$$

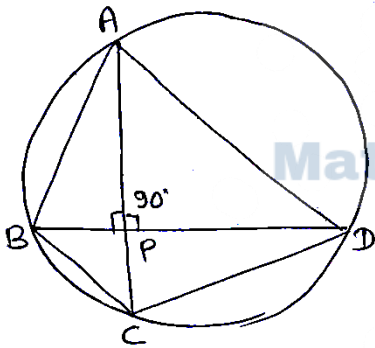
$$\cos 120^\circ = \frac{x^2 + 9 - BD^2}{2 \times x \times 3}$$

$$-\frac{1}{2} = \frac{x^2 + 9 - 19}{6x}$$

$$x^2 + 3x - 10 = 0 \rightarrow x = -5, \boxed{2}$$



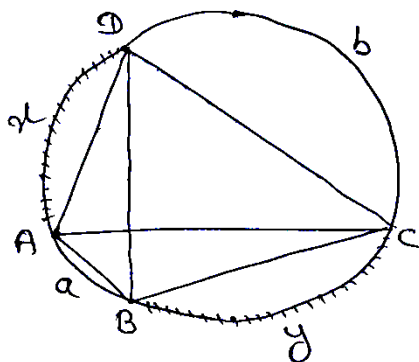
$\Rightarrow$ IF Diagonals are perpendicular in cyclic Quadrilateral



$$AP^2 + BP^2 + CP^2 + DP^2 = 4R^2$$

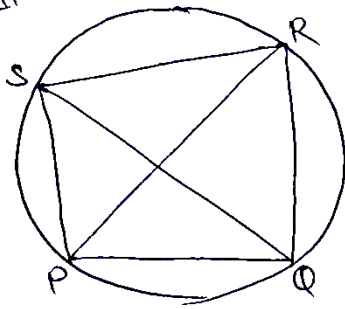
$$AB^2 + BC^2 + CD^2 + DA^2 = 8R^2$$

$\Rightarrow$



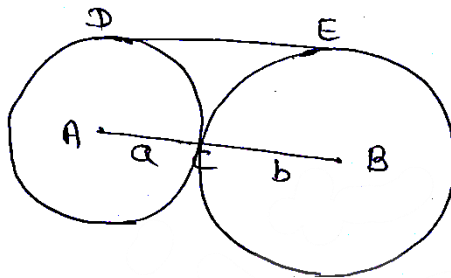
$$\# \boxed{x + y = a + b}$$

⇒ Imp.



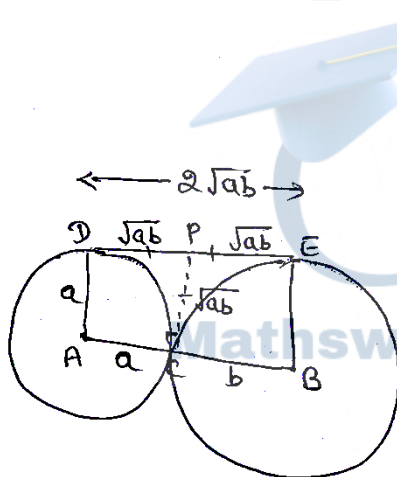
$$PR \times SQ = SR \times PQ + RQ \times SP$$

Ex:-



find length of $CD = ?$

Soln:-



$$AP \text{ (diagonal)} = \sqrt{a^2 + ab}$$

$$AP = \sqrt{a(a+b)}$$

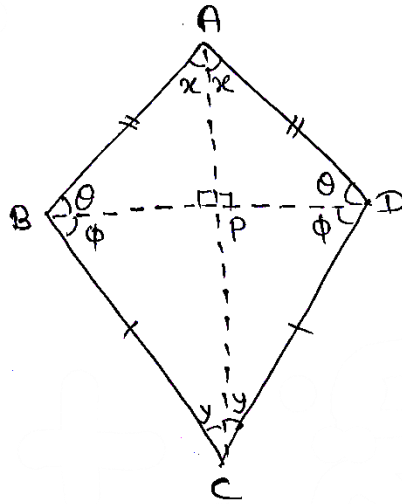
$$AP \times CD = AC \times DP + AD \times CP$$

$$\sqrt{a(a+b)} \times CD = a \times \sqrt{ab} + a \times \sqrt{ab}$$

$$\sqrt{a} \times \sqrt{a+b} \times CD = 2a\sqrt{ab}$$

$$CD = \frac{2a\sqrt{b}}{\sqrt{a+b}} \quad \underline{\underline{\text{Ans}}}$$

Kite



(i) $AB = AD$

$BC = CD$

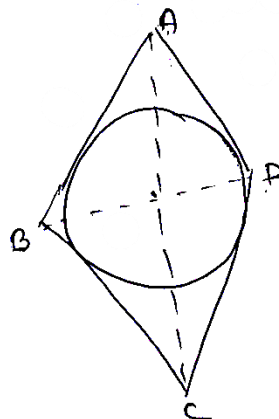
(ii) $BP = PD$

(iii) $\angle B = \angle D$

(iv) $\text{Area} = \frac{1}{2} \times d_1 \times d_2$

(v) $AC \perp BD$

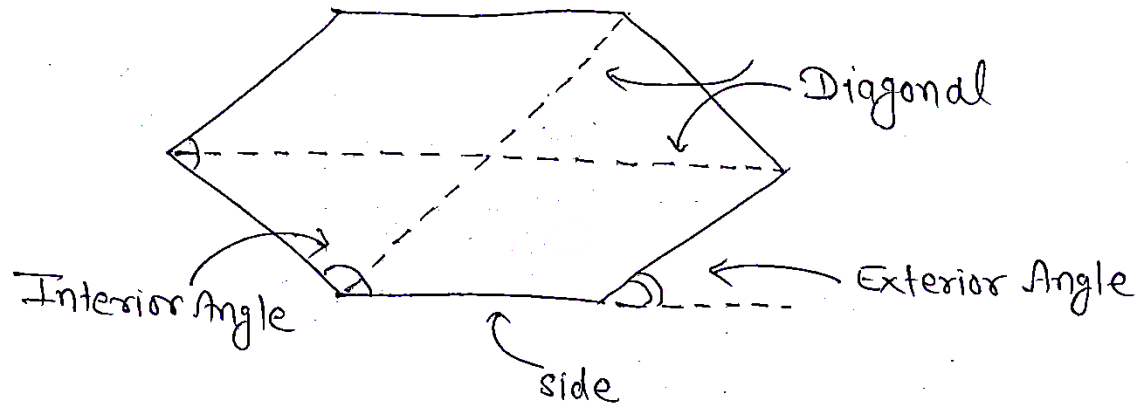
$\Rightarrow$ It is a Quadrilateral inside which a circle can be drawn.



Polygon

A polygon is a closed shape figure that has a minimum of three sides and three vertices.

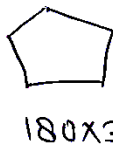
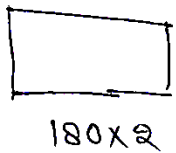
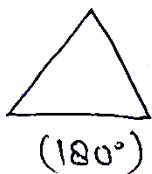
Ex:-



If $n \rightarrow$ Numbers of sides/no. of vertices

$$1. \text{ Diagonals} = \frac{n(n-3)}{2}$$

$$2. \text{ Sum of Interior Angle} = (n-2) \times 180^\circ$$



$$180 \times (n-2)$$

$$3. \text{ Sum of Exterior Angle} = 360^\circ$$

$$4. \text{ Sum of Interior Angle and Exterior Angle} = 180 \times n$$

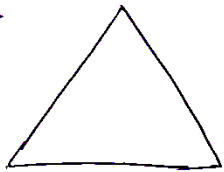
$$5. \text{ Perimeter} = a + b + c + d + \dots$$

$$6. \text{ Each Interior Angle (In Regular Polygon)} = \frac{(n-2) \times 180^\circ}{n}$$

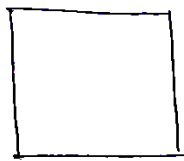
Regular Polygon

⇒ A polygon with equal sides and Congruent interior angles.

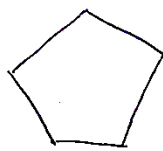
⇒



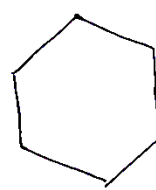
Triangle
(Equilateral)



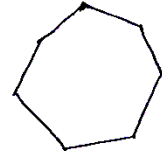
Square



Pentagon

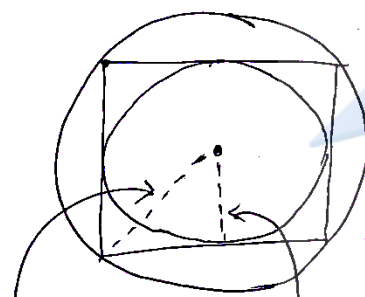


Hexagon



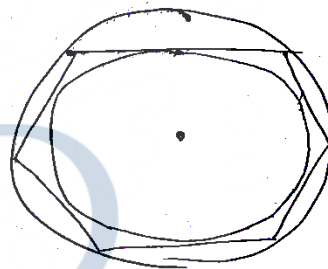
Heptagon

⇒ In a Regular polygon we can draw two circle (inside and outside) with center.



Circumradius
(R)

Inradius
(r)

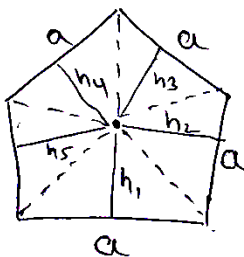


⇒ In a n sides Regular polygon:

$$h_1 + h_2 + h_3 + h_4 + \dots + h_n = n \times r$$

if $n=5$ $[h_1 + h_2 + h_3 + h_4 + h_5 = 5 \times r]$

Prove:



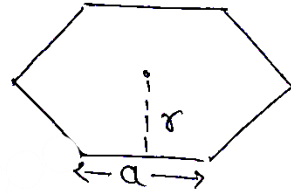
$$\frac{1}{2} \times a \times h_1 + \frac{1}{2} \times a \times h_2 + \frac{1}{2} \times a \times h_3 + \dots + \frac{1}{2} \times a \times h_n = \frac{1}{2} \times a \times r \times n$$

$$[h_1 + h_2 + h_3 + \dots + h_n = n \times r]$$

Area of Polygon (Regular)

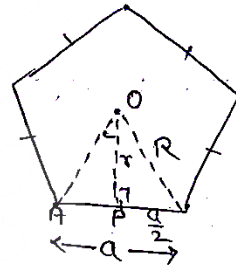
Case-1 :- If Side and radius (r) given:-

$$\text{Area} = n \times \left(\frac{1}{2} \times a \times r \right)$$



Case-2 :- If Only side is given:-

$$\text{Area} = \frac{n}{4} a^2 \cot \frac{\pi}{n}$$



Prove:

$$A = n \times \frac{1}{2} \times a \times \frac{a}{2} \cot \frac{\pi}{n}$$
$$= n \times \frac{1}{4} \times a^2 \cot \frac{\pi}{n}$$

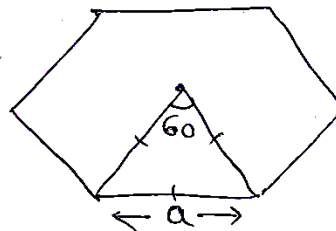
In $\triangle OAP$

$$\tan\left(\frac{\pi}{n}\right) = \frac{a}{2r}$$

$$r = \frac{a}{2} \cot\left(\frac{\pi}{n}\right)$$

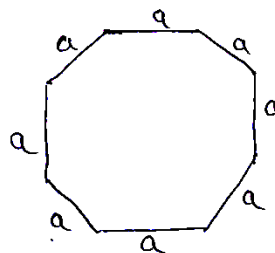
If $n=6$ (Hexagon)

$$\text{Area} = 6 \times \left(\frac{\sqrt{3}}{4} \times a^2 \right)$$

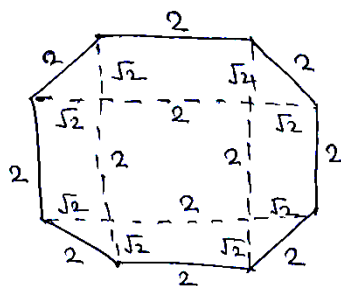


If $n=8$ (Octagon)

$$\text{Area} = 2(\sqrt{2}+1)a^2$$



Ex:-



If $a=2$ then Area = ?

Soln:- m-I Area = $2(\sqrt{2}+1)a^2$
 $= \boxed{8(\sqrt{2}+1)}$ Ans

m-II Basic Area = $4\left(\frac{1}{2} \times \sqrt{2} \times \sqrt{2}\right) + 4 \times (2 \times \sqrt{2}) + 2^2$
 $= (8 + 8\sqrt{2})$ Ans

Case 3:- When only Inradius given:-

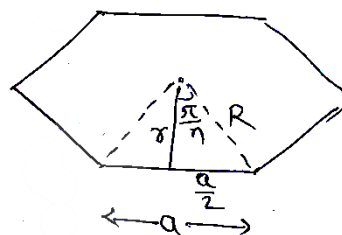
$$\text{Area} = nr^2 \tan \frac{\pi}{n}$$

Prove $\tan \frac{\pi}{n} = \frac{a}{2r}$

$$a = 2r \tan \left(\frac{\pi}{n}\right)$$

$$\text{Area} = \frac{1}{4} na^2 \cot \frac{\pi}{n}$$

$$= \frac{1}{4} n (4r^2 \tan^2 \left(\frac{\pi}{n}\right)) \cot \frac{\pi}{n} = nr^2 \tan \frac{\pi}{n}$$



Case 4:- When Circum-radius (R) are given:-

$$\text{Area} = \frac{nR^2}{2} \sin \left(\frac{2\pi}{n}\right)$$

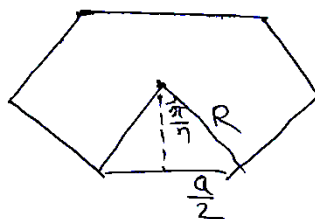
Prove: $\sin \left(\frac{\pi}{n}\right) = \frac{a}{2R}$

$$a = 2R \sin \frac{\pi}{n}$$

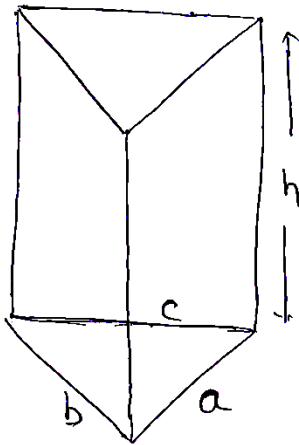
$$\text{Area} = \frac{1}{4} na^2 \cot \frac{\pi}{n}$$

$$= \frac{1}{4} \times n \times 4R^2 \sin^2 \left(\frac{\pi}{n}\right) \times \frac{\cos \left(\frac{\pi}{n}\right)}{\sin \left(\frac{\pi}{n}\right)} \Rightarrow nR^2 \sin \frac{\pi}{n} \cos \frac{\pi}{n}$$

$$\Rightarrow \left[\frac{nR^2}{2} \sin \left(\frac{2\pi}{n}\right) \right]$$



Prism



Lateral Surface Area

$$L.S.A = \text{Base Perimeter} \times \text{height}$$

Total Surface Area

$$T.S.A = L.S.A + 2(\text{Area of Base})$$

$$\text{Volume} = (\text{Base Area}) \times (\text{height})$$

[1] A right prism has triangular base. If v be the number of vertices, e be the number of edges and f be the number of faces of the prism. The value of $\frac{v+e-f}{2}$ is:

Soln:- v = number of vertices of prism = 6

e = edges of prism = 9

f = faces of the prism = 5

$$= \frac{6+9-5}{2}$$

$$= \boxed{5} \text{ Ans}$$

[2] A right prism has height 18cm and its base is a triangle with sides 5cm, 8cm and 12cm. What is its lateral surface area?

Soln:- $L.S.A = (5+8+12) \times 18$

$$= 25 \times 18$$

$$\Rightarrow \boxed{450 \text{ cm}^2} \text{ Ans}$$

- ③ The base of a right prism is a triangle whose sides are 8cm, 15cm and 17cm and its lateral surface area is 480 cm^2 . What is the volume of the prism?

Soln:- 8, 15, 17 are triplet. Its Right Angle Δ

$$\text{L.S.A} = 480 \text{ cm}^2$$

$$(8+15+17) \times \text{height} = 480$$

$$\text{height} = 12 \text{ cm}$$

$$\text{Volume} = (\text{Base Area}) \times \text{height}$$

$$= \left(\frac{1}{2} \times 8 \times 15\right) \times 12 = \boxed{720 \text{ cm}^3} \text{ Ans.}$$

- ④ A prism has base a right angled triangle whose side adjacent to the right angles are 10cm and 12cm long. The height of the prism is 20cm. the density of the material of the prism is 6gm/cubic cm. the weight of the prism is?

Soln Volume = $\frac{1}{2} \times 10 \times 12 \times 20 = 1200 \text{ cm}^3$

$$\text{weight} = \text{volume} \times \text{density}$$

$$= 1200 \times 6 = 7200 \text{ gm} = \boxed{7.2 \text{ kg}} \text{ Ans}$$

- ⑤ Let ABCDEF be a prism whose base is a right angled triangle where sides adjacent to 90° are 9cm and 12cm, if the cost of painting the prism is Rs. 151.20 at the rate of 20 paise per sqcm then the height of the prism is?

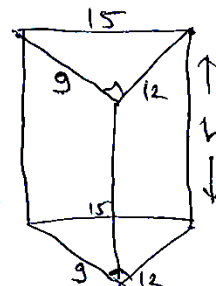
Soln:- Total Painted Area = $151.20 \times 5 = 756 \text{ cm}^2$

$$\text{T.S.A} = \text{L.S.A} + 2(\text{Base Area})$$

$$756 = (15+9+12) \times h + 2 \times \frac{1}{2} \times 9 \times 12$$

$$756 = 36h + 108$$

$$\boxed{h = 18 \text{ cm}} \text{ Ans}$$



[6] A right prism stands on a base of 6cm side equilateral triangle and its volume is $81\sqrt{3} \text{ cm}^3$, the height of the prism is:

Soln

$$\text{Volume} = (\text{Base Area}) \times \text{height}$$

$$81\sqrt{3} = \left[\frac{\sqrt{3}}{4} \times (6)^2 \right] \times h$$

$$81\sqrt{3} = 9\sqrt{3} \times h$$

$$\boxed{h = 9 \text{ cm}} \text{ Ans}$$

[7] The base of a right prism is an equilateral triangle. If the lateral surface area and volume are 120 cm^2 , $40\sqrt{3} \text{ cm}^3$ respectively then the side of base of the prism is:

Soln

$$\text{L.S.A} = 120$$

$$3a \times h = 120$$

$$a \times h = 40 \text{ --- ①}$$

$$\text{Volume} = 40\sqrt{3}$$

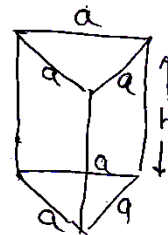
$$\left(\frac{\sqrt{3}}{4} a^2 \right) \times h = 40\sqrt{3}$$

$$a^2 \times h = 160 \text{ --- ②}$$

$$\text{②} \div \text{①} \quad \frac{a^2 \times h}{a \times h} = \frac{160}{40}$$

$$\boxed{a = 4 \text{ cm}}$$

Ans



[8] Height of a prism-shaped part of machine is 8cm and its base is an isosceles triangle, whose each of the equal sides is 5cm and remaining side is 6cm. The volume of the part is:

Soln:-

$$\text{Volume} = (\text{Base Area}) \times \text{height}$$

$$\text{Area} = \frac{b}{4} \sqrt{4a^2 - b^2} = \frac{6}{4} \sqrt{4(5)^2 - (6)^2} = 12 \text{ cm}^2$$

$$\text{Volume} = 12 \times 8 = \boxed{96 \text{ cm}^3} \text{ Ans}$$

- [9] Base of a right prism is a rectangle, the ratio of whose length and breadth is 3:2. If the height of the prism is 12 cm and total surface area is 288 cm^2 the volume of the prism is:

Soln:- $l : b = 3 : 2$

Let length = $3x$, breadth = $2x$

$$\text{T.S.A} = (\text{Base Perimeter}) \times h + 2(\text{base Area})$$

$$288 = 10x \times 12 + 2(6x^2)$$

$$12x^2 + 120x - 288 = 0$$

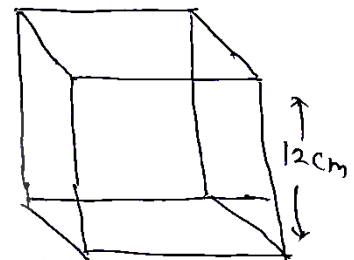
$$x^2 + 10x - 24 = 0$$

$$\boxed{x=2}$$

$$l = 6, b = 4$$

$$\text{Area of Base} = 6 \times 4 = 24$$

$$\text{Volume} = 24 \times 12 = \boxed{288 \text{ cm}^3} \text{ Ans}$$



- [10] The base of a right prism is a square having side of 15 cm. If its height is 8 cm, then find the total surface area.

Soln:- $\text{T.S.A} = (\text{Base Perimeter}) \times h + 2(\text{Base Area})$

$$= (4 \times 15) \times 8 + 2(15^2)$$

$$= 480 + 450$$

$$= \boxed{930 \text{ cm}^2} \text{ Ans}$$

[11] The base of a solid right prism of height 10cm is a square and its volume is 160cm^3 . What is the total surface area of the prism.

Soln:- $T.S.A = I.S.A + 2(\text{Base Area})$

$$\text{Volume} = \text{Base Area} \times \text{height}$$

$$160 = a^2 \times 10$$

$$\boxed{a = 4}$$

$$T.S.A = 16 \times 10 + 2(16) = \boxed{192} \text{ And}$$

[12] The height of a right prism with a square base is 15cm. If the area of the total surface of the prism is 608cm^2 its volume is:

Soln:- Let side = a

$$T.S.A = (4a \times h) + 2(a^2)$$

$$608 = 4a \times 15 + 2a^2$$

$$a^2 + 30a - 304 = 0$$

$$a = 8\text{cm}$$

$$\text{Volume} = (8)^2 \times 15 = \boxed{960\text{cm}^3} \text{ Any}$$

[13] The base of a right prism is a trapezium whose length of parallel sides are 25cm and 11cm and the perpendicular distance between the parallel sides is 16cm. If the height of the prism is 10cm, then the volume of the prism is:

Soln:- $\text{Volume} = (\text{base Area}) \times \text{height}$

$$= \frac{1}{2}(25+11) \times 16 \times 10$$

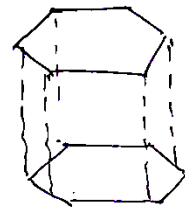
$$= 18 \times 16 \times 10$$

$$= \boxed{2880\text{cm}^3} \text{ Any}$$

[14] The base of a right prism is a regular hexagon of side 5cm. If its height is $12\sqrt{3}$ cm, then its volume is:

Soln

$$\begin{aligned}\text{Volume} &= (\text{Base Area}) \times \text{height} \\ &= \left(6 \times \frac{\sqrt{3}}{4} \times 5^2\right) \times 12\sqrt{3} \\ &= 18 \times 3 \times 25 \\ &= \boxed{1350 \text{ cm}^3} \text{ Ans}\end{aligned}$$



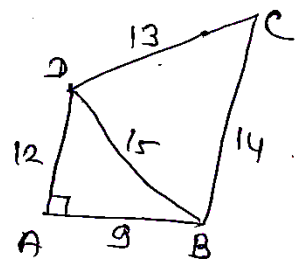
[15] The base of a right prism is a quadrilateral ABCD, given that AB = 9cm, BC = 14cm, CD = 13cm, DA = 12cm and $\angle DAB = 90^\circ$ if the volume of the prism be 2070 cm^3 , then the area of the lateral surface is:

Soln:-

$$\text{Volume} = (\text{Base Area}) \times \text{height}$$

$$BD = 15 \text{ cm (triplet - 9, 12, 15)}$$

$$\begin{aligned}\text{Area of } \triangle ABD &= \frac{1}{2} \times 9 \times 12 = 54 \text{ cm}^2 \\ (\triangle ABD)\end{aligned}$$



$$\begin{aligned}\text{Area of } \triangle BCD &= \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{13+14+15}{2} = 21 \\ &= \sqrt{21 \times 8 \times 7 \times 6} \\ &= 84 \text{ cm}^2\end{aligned}$$

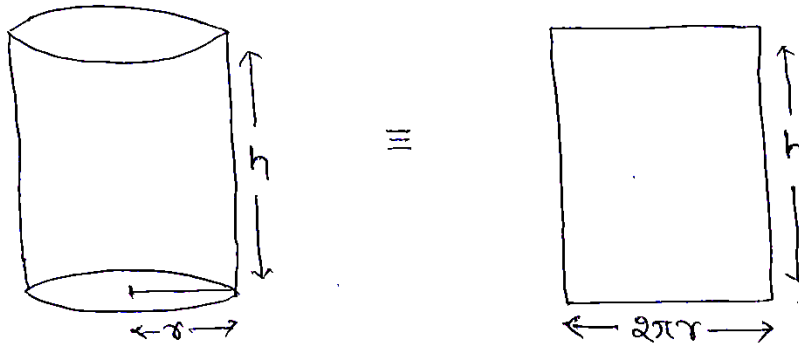
$$\text{Area of Base (ABCD)} = 84 + 54 = 138 \text{ cm}^2$$

$$\text{Volume} = 138 \times h$$

$$h = \frac{2070}{138} = 15 \text{ cm}$$

$$\begin{aligned}\text{L.S.A} &= (9 + 14 + 13 + 12) \times 15 \\ &= 48 \times 15 \\ &= \boxed{720 \text{ cm}^2} \text{ Ans}\end{aligned}$$

Cylinder



Curved Surface Area (C.S.A) = $2\pi rh$

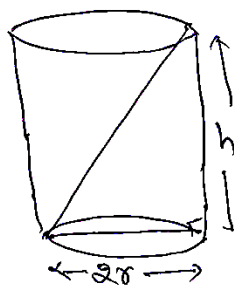
Total Surface Area (T.S.A) = $2\pi rh + 2\pi r^2$

$T.S.A = 2\pi r(r+h)$

Volume = Base Area $\times$ height

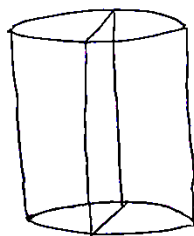
$V = \pi r^2 h$

Diagonal:-

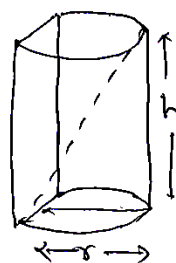


$d = \sqrt{h^2 + 4r^2}$

Half Cylinder:-



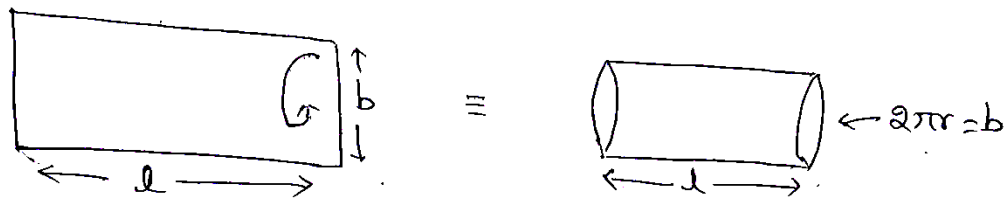
=



$d = \sqrt{r^2 + h^2}$

* A Cylinder is made by rolling a rectangle :-

Case 1:- About its breadth :-



$$\begin{aligned} 2\pi r &= b \\ h &= l \end{aligned}$$

$$\text{Volume} = \pi \left(\frac{b}{2\pi} \right)^2 \times l$$

Case 2:- About its length :-



$$\begin{aligned} 2\pi r &= l \\ h &= b \end{aligned}$$

$$\text{Volume} = \pi \left(\frac{l}{2\pi} \right)^2 \times b$$

Ex:- A cylinder is made by rolling a rectangle about its breadth. Breadth and length of the rectangle are 7m and 11m respectively. Find the volume of the created cylinder.

Soln:-

$$2\pi r = b$$

$$2\pi r = 7$$

$$r = \frac{7}{2 \times 22}$$

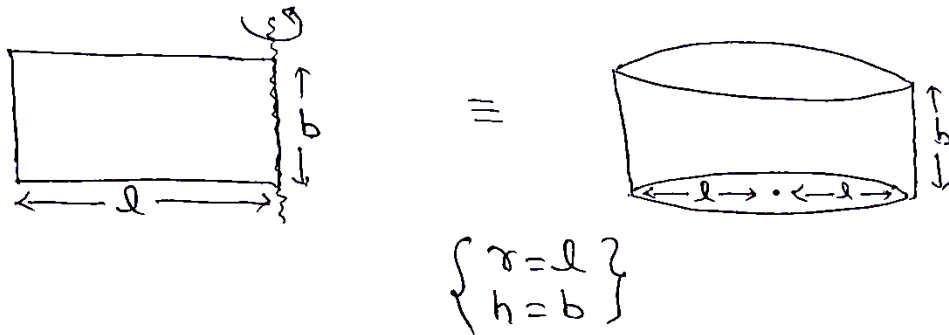
$$r = \frac{49}{44} \text{ m}$$

$$\text{Volume} = \pi \left(\frac{49}{44} \right)^2 \times 11$$

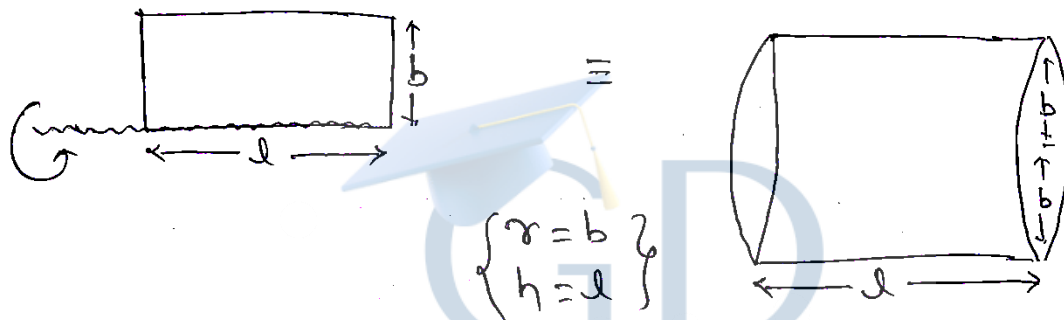
$$= \boxed{\frac{343}{8} \text{ m}^3} \text{ Ans}$$

* A Cylinder is made by rotating a rectangle

Case 1 :- About its breadth :-



Case 2 :- About its length :-



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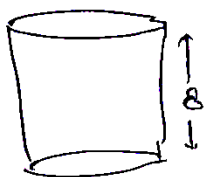
* The minimum length of rope needed to wrap around a cylinder

$$l = \sqrt{(C \times n)^2 + h^2}$$

C = Circumference

n = no. of round

Ex:- Circumference = 3 cm, height = 8, 5 round, minimum length ?



$$\text{length} = \sqrt{(3 \times 5)^2 + (8)^2}$$

$$= \boxed{17 \text{ cm}} \text{ Ans}$$

* The Area covered by roller

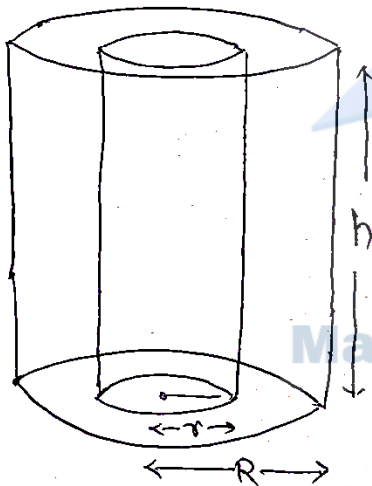
$$\text{Area} = (2\pi r h \times n)$$

$n \rightarrow$ no. of revolutions

* The distance covered by roller:-

$$D = (2\pi r \times n) \quad n \rightarrow \text{no. of revolutions}$$

* Hollow Cylinder :-

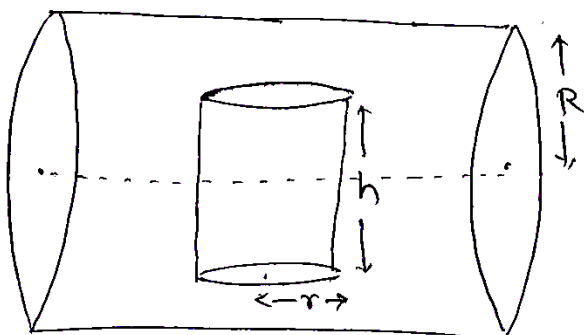


$$\text{Volume} = \pi R^2 h - \pi r^2 h$$

$$V = \pi (R^2 - r^2) h$$

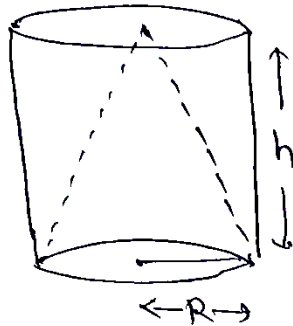
$$\text{T.S.A} = 2\pi R h + 2\pi r h + 2\pi (R^2 - r^2)$$

*

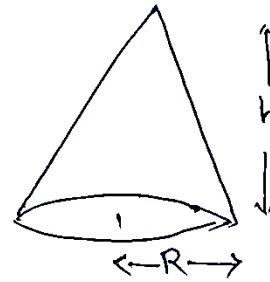


$$r^2 + \left(\frac{h}{2}\right)^2 = R^2$$

* Cutting Cylinder



$\equiv$

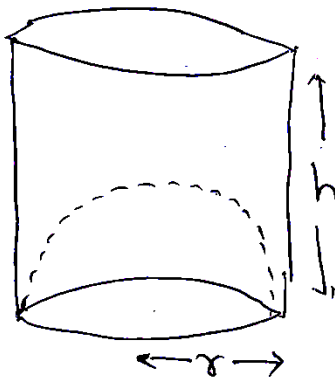


Volume = $\pi R^2 h - \frac{1}{3} \pi R^2 h$

$V = \frac{2}{3} \pi R^2 h$

T.S.A = $2\pi R h + \pi R^2 + \pi R h$

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Volume = $\pi r^2 h - \frac{2}{3} \pi r^3$

T.S.A = $2\pi r h + \pi r^2 + 2\pi r^2$

[16] The radius of a circular cylinder is 5cm and its height is 14cm. Then the volume of cylinder in cm^3 .

Soln:- $V = \pi r^2 h$
 $= \frac{22}{7} \times (14)^2 \times (5)$
 $= \boxed{1100 \text{ cm}^3}$ Ans

[17] The diameter of a cylinder is 7cm and its height is 16cm. Using the value of $\pi = \frac{22}{7}$, the lateral surface area of the cylinder is:

Soln:- L.S.A = $2\pi rh$
 $= 2 \times \frac{22}{7} \times \frac{7}{2} \times 16 = \boxed{352 \text{ cm}^2}$ Ans

[18] The curved surface area of volume of a cylindrical object are 88 cm^2 and 132 cm^2 respectively. The height of the cylindrical object is:

Soln:- $\frac{\pi r^2 h}{2\pi rh} = \frac{132}{88} \Rightarrow \frac{r}{2} = \frac{3}{2} \Rightarrow r = 3$

$2\pi rh = 88$
 $h = \frac{88}{2 \times \frac{22}{7} \times 3} = \frac{14}{3} = \boxed{4\frac{2}{3} \text{ cm}}$ Ans

[19] The height and the total surface area of a right circular cylinder are 4cm and $8\pi \text{ cm}^2$ respectively. The radius of the base of cylinder is:

- (a) $(2\sqrt{2}-2) \text{ cm}$ (b) $(\sqrt{2}-2) \text{ cm}$ (c) 2 cm (d) $\sqrt{2} \text{ cm}$

Soln: $2\pi r(r+h) = 8\pi$

$$r(4+r) = 4$$

$$r^2 + 4r - 4 = 0 \Rightarrow r = \frac{-4 \pm \sqrt{16+16}}{2} = -2 \pm 2\sqrt{2}$$

$$r = \boxed{(2\sqrt{2} - 2) \text{ cm}} \text{ Ans}$$

20 The radius of two circular cylinder are in the ratio 3:2 and the ratio of their volumes is 27:16. What is the ratio of their heights?

Soln:-

$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^2 \left(\frac{h_1}{h_2}\right)$$

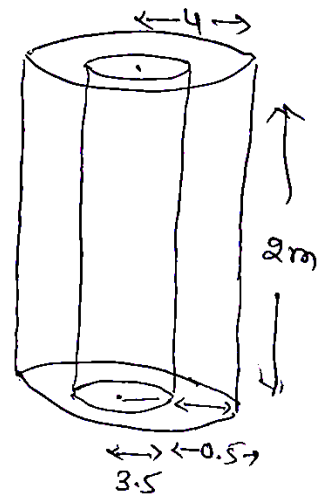
$$\frac{27}{16} = \left(\frac{3}{2}\right)^2 \frac{h_1}{h_2} \Rightarrow \frac{h_1}{h_2} = \frac{3}{4}$$

$$\frac{h_1}{h_2} = \frac{3}{4} \text{ Ans}$$

21 A cylindrical tube, open at both ends, is made of a metal sheet which is 0.5cm thick. Its outer radius is 4cm and length is 2m. How much metal (in cm^3) has been used in making the tube?

Soln:-

$$\begin{aligned} V &= \pi(R^2 - r^2) \times h \\ &= \pi(4^2 - 3.5^2) \times 200 \\ &= \boxed{750\pi} \text{ Ans} \end{aligned}$$



Q2 The external and the internal radii of a hollow right circular cylinder of height 15cm are 6.75cm and 5.25cm respectively. If it is melted to form a solid cylinder of height half of the original cylinder, then the radius of the solid cylinder is:

Soln:- Volume (hollow) = Volume (solid)

$$\pi(r_1^2 - r_2^2) \times h = \pi R^2 \times \frac{h}{2}$$

$$(6.75^2 - 5.25^2) = \frac{R^2}{2}$$

$$(6.75 + 5.25)(6.75 - 5.25) = \frac{R^2}{2}$$

$$12 \times 1.5 \times 2 = R^2$$

$$\boxed{R = 6\text{cm}} \text{ Ans}$$

Q3 If the radius of the base of a right circular cylinder is increased by 20% and the height is decreased by 30%, then what is the percentage increase/decrease in the volume?

Soln:-

$$V \propto r^2 h$$

$$20\% = \frac{1}{5}$$

$$r \rightarrow 5 : 6$$

$$30\% = \frac{3}{10}$$

$$r^2 \rightarrow 25 : 36$$

$$h \rightarrow 10 : 7$$

$$V = r^2 h = 250 : 252$$



$$\uparrow \% = \frac{2}{250} \times 100$$

$$\Rightarrow 0.8\% \text{ Increased}$$

24 The radius of the base of a right circular cylinder is increased by 20%. By what percent should its height be reduced so that its volume remains the same as before?

Soln:-

$$r \rightarrow 5 : 6$$

$$r^2 = 25 : 36$$

$$h = 36 : 25$$

$$\frac{11}{36} \times 100 = \frac{275}{9} = 30\frac{5}{9} \% \quad \underline{\text{Ans}}$$

25 A well 20m in diameter is dug 14m deep and the earth taken out is spread all around it to a width of 5m to a width of form an embankment. The height of the embankment is:

Soln:- Volume of well = volume of embankment

$$\pi(10^2) \times 14 = \pi[15^2 - 10^2] \times H$$

$$H = \frac{100 \times 14}{125} = 11.2\text{m}$$

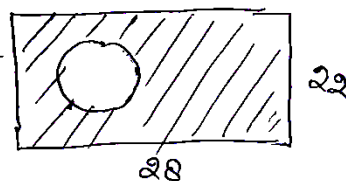
26 A cylindrical well of height 80m and radius 7m is dug in a field 28m long and 22m wide. The earth taken out is spread evenly on the field. What is the increased in the level of the field?

Soln:- Volume of well = shaded Area $\times$ height

$$\pi r^2 h = [a \times b - \pi r^2] \times H$$

$$\frac{22}{7} \times 7 \times 7 \times 80 = [28 \times 22 - \frac{22}{7} \times 7 \times 7] \times H$$

$$\text{Height} = \frac{\frac{22}{7} \times 7 \times 7 \times 80}{22 \times 7 (41)} = 26.66\text{m} \quad \underline{\text{Ans}}$$



Q7 From a solid cylindrical wooden block of height 18cm and radius 7.5cm, a conical cavity of the same height and same radius is taken out. What is total surface area of the remaining solid?

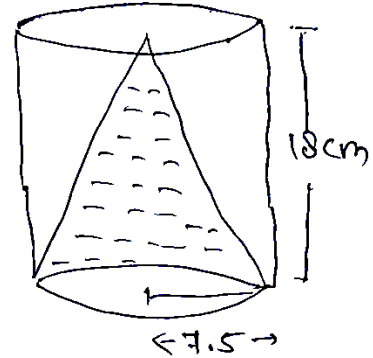
Soln:-

$$l^2 = r^2 + h^2$$

$$l^2 = (7.5)^2 + (18)^2$$

$$= \frac{225}{4} + 324$$

$$l = \sqrt{\frac{1521}{4}} = \frac{39}{2}$$



$$T.S.A = \pi r l + 2\pi r h + \pi r^2$$

$$= \pi r (l + 2h + r)$$

$$= \pi \times \frac{15}{2} \left(\frac{39}{2} + 36 + \frac{15}{2} \right) = \left(\frac{945}{2} \right) \pi = \boxed{472.5\pi}$$

Q8 The height of a solid cylinder is 30cm and the diameter of its base is 10cm. Two identical conical holes each of radius 5cm and height 12cm are drilled out. What is the surface area of the remaining solid?

Soln:-

$$L.S.A = 2\pi r h$$

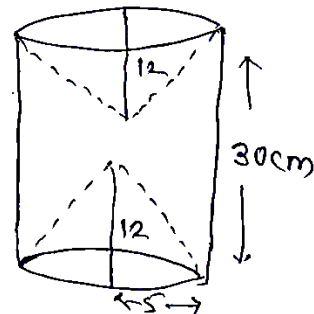
$$= 2\pi \times 5 \times 30$$

$$= 300\pi$$

$$L.S.A \text{ of cone} = 2(\pi r l)$$

$$= 2\pi \times 5 \times 13$$

$$= 130\pi$$



$$T.S.A \text{ of remaining part} = 300\pi + 130\pi$$

$$= \boxed{430\pi} \text{ Ans}$$

29 A sphere of diameter 6cm is dropped in a right circular cylindrical vessel partly filled with water. The diameter of the cylindrical vessel is 10cm. If the sphere is just completely submerged in water, then the rise of water level in the cylindrical vessel is;

Soln - If the water level rises by 'H' cm

$$\pi R^2 H = \text{Volume of sphere}$$

$$\pi (6)^2 H = \frac{4}{3} \pi (3)^3$$

$$\boxed{H = 1 \text{ cm}} \text{ Ans}$$

30 A right circular cylinder is partially filled with water. Two iron spherical balls are completely immersed in the water so that the height of the water in the cylinder rises by 4cm. If the radius of one ball is half of the other and the diameter of the cylinder is 18cm, then the radii of the spherical balls are.

Soln - Volume of 2 balls = volume of cylinder of height (4cm)

$$\frac{4}{3} \pi r_1^3 + \frac{4}{3} \pi r_2^3 = \pi (9)^2 \times 4$$

$$r_1 = 2r_2 \text{ (Given)}$$

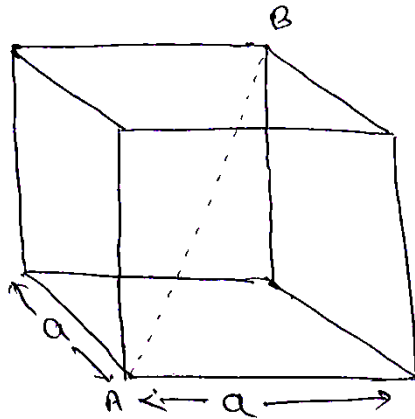
$$\frac{4}{3} \pi (8r_2^3 + r_2^3) = \pi (9)^2 \times 4$$

$$9r_2^3 = 9 \times 9 \times 3$$

$$r_2 = \boxed{3 \text{ cm}} \text{ Ans.}$$

$$r_1 = 2r_2 = 2 \times 3 = \boxed{6 \text{ cm}} \text{ Ans}$$

Cube And Cuboid



Total Surface Area

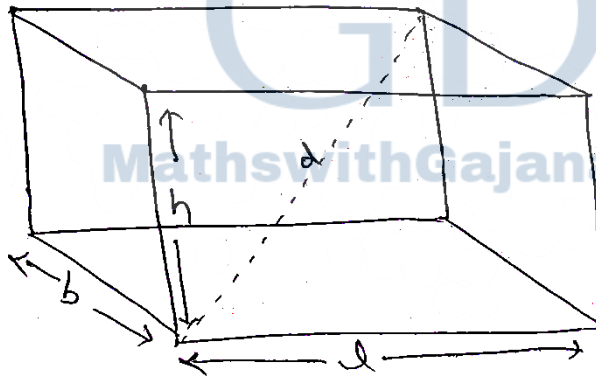
$$T.S.A = 6 \times (\text{side})^2$$

$$\text{Volume} = (\text{side})^3$$

$$\text{Diagonal} = \sqrt{3}(\text{side})$$

(AB)

Cuboid



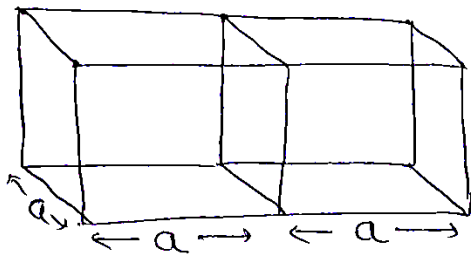
$$\text{Total Surface Area (T.S.A)} = 2(lb + bh + hl)$$

$$\text{Area of four wall} = 2(l+b) \times h$$

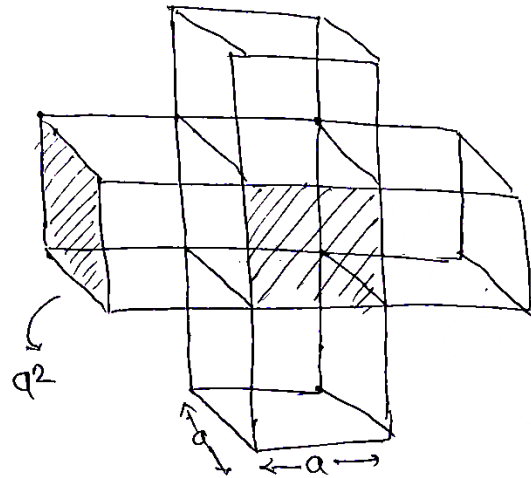
$$\text{Volume} = l \times b \times h$$

$$\text{diagonal (d)} = \sqrt{l^2 + b^2 + h^2}$$

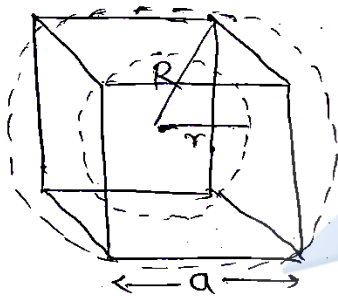
⇒



$$\text{Total Surface Area} = 10a^2$$



⇒



$$r = \frac{a}{2}$$

$$R = \frac{\text{diagonal}}{2} = \frac{\sqrt{3}a}{2}$$

Total Surface Area

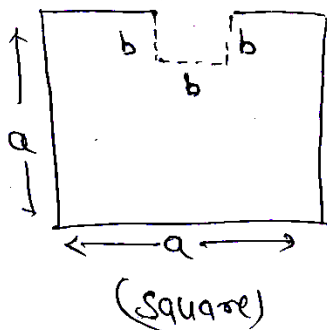
$$= a^2 \times (5 \times 4 + 2)$$

↑ Surface cube ↓ middle cube

$$= 22a^2$$

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⇒



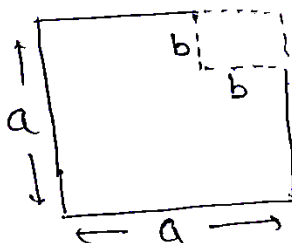
Remaining

$$\text{Area} = a^2 - b^2$$

$$\text{Perimeter} = 4a - b + 3b = 4a + 2b$$

↑
increased

⇒

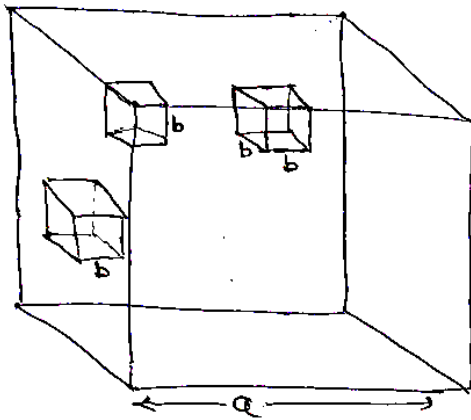


$$\text{area} = a^2 - b^2$$

$$\text{Perimeter} = 4a - 2b + 2b = 4a$$

No changes

Remove cube from a Big Cube



Remove from Corner

$$T.S.A = 6a^2 - 3b^2 + 3b^2 = 6a^2$$

No effect

Remove from edge

$$T.S.A = 6a^2 - 2b^2 + 4b^2 = 6a^2 + 2b^2$$

Remove from surface

$$T.S.A = 6a^2 - b^2 + 5b^2 = 6a^2 + 4b^2$$

- [31] The areas of three adjacent faces of cuboidal tank are 3m^2 , 12m^2 and 16m^2 . The capacity of the tank, in litres is:

Soln:- $lb = 3$, $bh = 12$, $hl = 16$

$$V = lbh$$

$$\hookrightarrow l^2 b^2 h^2 = 3 \times 12 \times 16$$

$$lbh = \sqrt{3 \times 12 \times 16} = 24\text{m}^3 = 24000\text{L}$$

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- [32] The internal measure of a cuboidal room are with length as 12m , breadth as 8m and height as 10m . The total cost of white washing all four walls of the room and also the ceiling of the room, if the cost of white washing is Rs. 25 per m^2 is:

Soln:- Area of 4 walls = $2(l+b) \times h$

$$= 2(12+8) \times 10 = 400\text{m}^2$$

$$\text{Area of roof} = 12 \times 8 = 96\text{m}^2$$

$$\text{Total Area} = 400 + 96 = 496\text{m}^2$$

$$\text{Cost} = 496 \times 25 = \boxed{\text{Rs. } 12400}$$

- [33] A solid cube of volume 13824 cm^3 is cut into 8 cubes of equal volumes. The ratio of the surface area of the original cube of the sum of the surface areas of three of the smaller cubes is:

Soln:-

	1 small cube	:	Original cube
$V \rightarrow$	1	:	8
side $\rightarrow$	1	:	2
Area $\rightarrow$	1	:	4

$\downarrow \times 3 \text{ cube}$

$[3 : 4] \text{ Ans}$

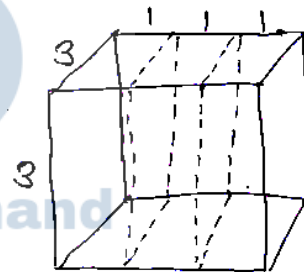
- [34] A solid cube is cut into three cuboids of same volumes. What is the ratio of the surface area of the cube to the surface sum of the surface areas of any two of the cuboids so formed?

Soln:- Let the side of cube is 3m

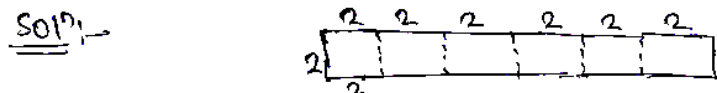
Surface area of cube = $6 \times 3^2 = 54$

Surface area of one cuboid
 $= 2(lb + bh + hl) = 2(1 \times 3 + 3 \times 3 + 3 \times 1)$
 $= 30$

Cube : 2 cuboid = $54 : 2 \times 30 = [9 : 10] \text{ Ans}$



- [35] Six cubes each edge 2cm, are joined end to end. What is the total surface area of the resulting cuboid in cm^2 ?



$l = 12, b = 2, h = 2$

Surface Area = $2(lb + bh + hl)$
 $= 2(12 \times 2 + 2 \times 2 + 2 \times 12)$
 $= 104 \text{ cm}.$

[36] The areas of three adjacent faces of a cuboidal solid block of wax are 216 cm^2 , 96 cm^2 and 144 cm^2 . It is melted and 8 cubes of the same size are formed from it. What is the lateral surface area of 3 such cubes?

Soln:- $l \times b = 216$

$$b \times h = 96$$

$$h \times l = 144$$

multiply $(l b h)^2 = 216 \times 96 \times 144$

$$l b h = 36 \times 4 \times 12 = 1728$$

$$\text{Volume of Cuboid} = 8 \times \text{volume of cube}$$

$$l b h = 8 \times a^3$$

$$1728 = 8 \times a^3$$

$$a = 6$$

$$\begin{aligned} \text{Lateral surface area of 3 such cubes} &= 3 \times 4a^2 = 3 \times 4 \times 6^2 \\ &= 432 \text{ cm}^2 \end{aligned}$$

[37] A hemispherical depression of diameter 4cm is cut out from each face of a cubical block of sides 10cm. Find the surface area of the remaining solid

Soln:- Remaining surface area of each face of cubical block

$$= a^2 - \pi r^2 + 2\pi r^2$$

$$= a^2 + \pi r^2$$

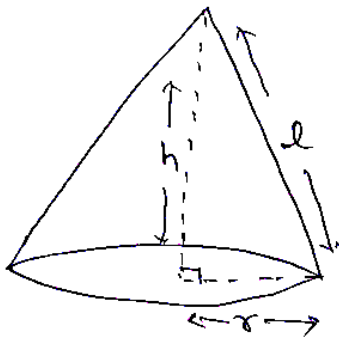
$$= (10)^2 + \frac{22}{7} \times (2)^2 = \frac{788}{7}$$

$$\text{Total surface area} = 6 \times \frac{788}{7}$$

$$= \frac{4728}{7}$$

$$= \left[675 \frac{3}{7} \text{ cm}^2 \right] \underline{\underline{\text{Ans}}}$$

Cone



$$\text{Slant height (l)} = \sqrt{h^2 + r^2}$$

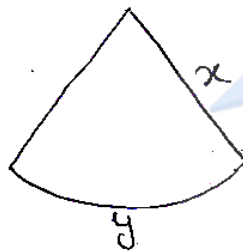
$$\# \text{ C.S.A} = \pi r l$$

$$\text{T.S.A} = \pi r l + \pi r^2$$

$$\# \text{ T.S.A} = \pi r (l + r)$$

$$\# \text{ Volume} = \frac{1}{3} \pi r^2 h$$

$\Rightarrow$ Cone is made by sector



$\equiv$

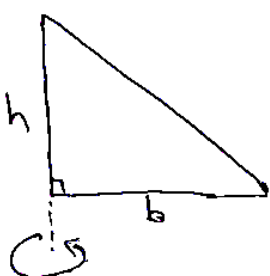


$$l = x$$

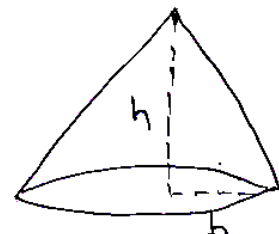
$$2\pi r = y$$

$\Rightarrow$ Cone is made by Rotating a Triangle

Case 1



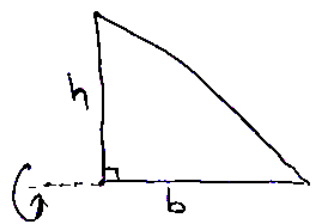
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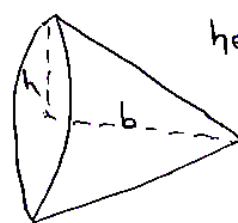
height of cone = h

radius of cone = b

Case 2



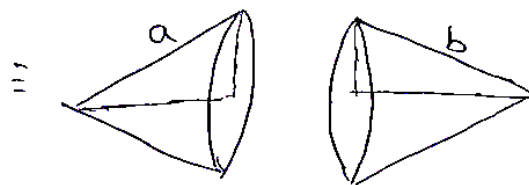
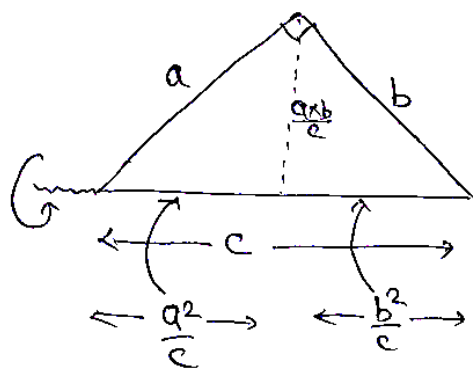
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height of cone = b

radius of cone = h

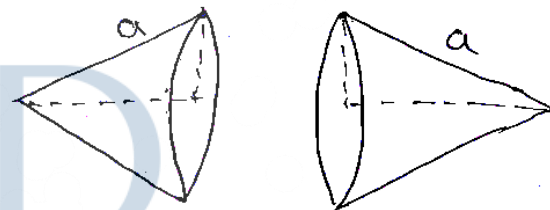
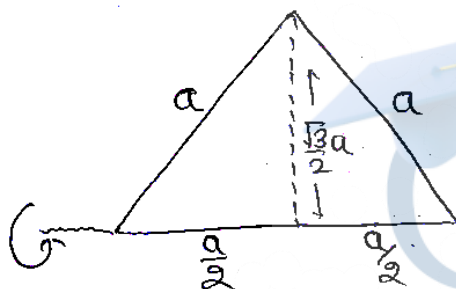
Case 3:- Rotating by Hypotenuse



$$\begin{array}{l|l} r = \frac{ab}{c} & r = \frac{ab}{c} \\ h = \frac{a^2}{c} & h = \frac{b^2}{c} \end{array}$$

$$\text{Volume} = \frac{1}{3} \pi \left(\frac{a \times b}{c} \right)^2 \times c$$

Case 4:- Equilateral Triangle

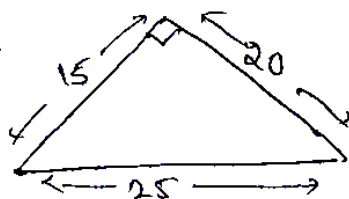


Cone of height = $\frac{a}{2}$

Cone of Radius = $\frac{\sqrt{3}a}{2}$

$$\text{Volume} = 2 \times \left[\pi \left(\frac{\sqrt{3}a}{2} \right)^2 \times \frac{a}{2} \right]$$

Ex:-

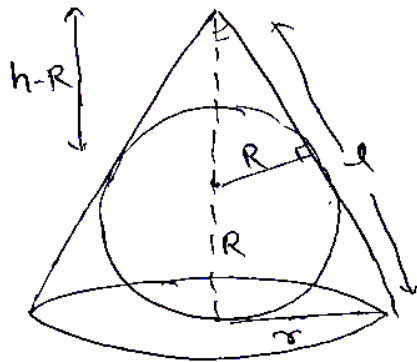


= Volume of Cone = ?

Soln:- $r = \frac{ab}{c} = \frac{15 \times 20}{25} = 12$

$$\text{Volume} = \frac{1}{3} \pi (12)^2 \times 25$$

⇒ Maximum radius of sphere in Cone :-

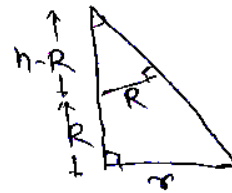


by Similarity

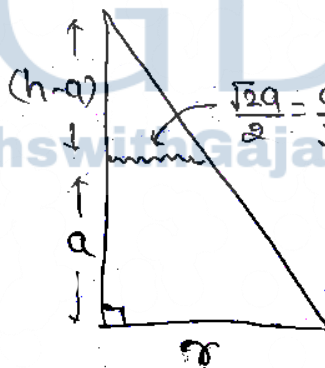
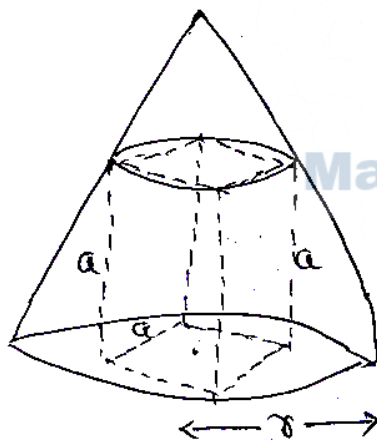
$$\frac{R}{r} = \frac{h-R}{l}$$

$$Rl = hr - Rr$$

$$\# \boxed{R = \frac{hr}{l+r}}$$



⇒ Maximum side of Cube in Cone :-



by Similarity

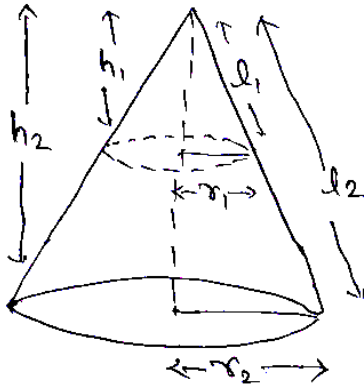
$$\frac{h-a}{h} = \frac{a}{\sqrt{2}r}$$

$$h\sqrt{2}r - a\sqrt{2}r = ah$$

$$h\sqrt{2}r = a(h + \sqrt{2}r)$$

$$\# \boxed{a = \frac{\sqrt{2}rh}{\sqrt{2}r + h}}$$

Cutting of Cone



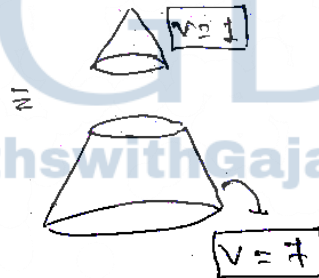
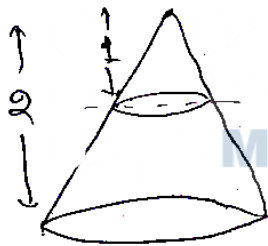
$$\frac{h_1}{h_2} = \frac{r_1}{r_2} = \frac{l_1}{l_2}$$

$$\frac{V_1}{V_2} = \frac{\frac{1}{3}\pi r_1^2 h_1}{\frac{1}{3}\pi r_2^2 h_2} = \frac{r_1^3}{r_2^3} = \frac{h_1^3}{h_2^3} = \frac{r_1^2 h_1}{r_2^2 h_2}$$

$$\frac{V_1}{V_2} = \frac{r_1^3}{r_2^3} = \frac{h_1^3}{h_2^3}$$

$$\left[\frac{C.S.A_1}{C.S.A_2} = \frac{\pi r_1 l_1}{\pi r_2 l_2} = \frac{l_1^2}{l_2^2} = \frac{r_1^2}{r_2^2} = \frac{h_1^2}{h_2^2} \right]$$

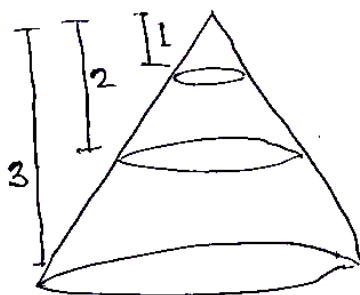
Special Case :- Cut from mid points



$$h_1 : h_2 = 1 : 2$$

$$V_1 : V_2 = (1)^3 : (2)^3 = 1 : 8$$

⇒ Cut in three equal parts



$$h_1 : h_2 : h_3 = 1 : 2 : 3$$

$$V_1 : V_2 : V_3 = 1 : 8 : 27$$

$$C.S.A_1 : C.S.A_2 : C.S.A_3 = 1 : 4 : 9$$



Volume → 1 : 7 : 19

[38] The radius of a circular cone is 6cm and its height is 7cm. Then the volume of cone in cm^3 is:

Soln:- $V = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \times \frac{22}{7} \times (6)^2 \times 7 = \boxed{264 \text{ cm}^3} \text{ Ans}$

[39] The area of the base of a right circular cone is $81\pi \text{ cm}^2$ its height is 12cm. What is curved surface area of the cone?

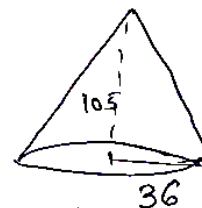
Soln:- $\pi r^2 = 81\pi$
 $r = 9$
 $l = \sqrt{12^2 + 9^2}$
 $l = 15$

$\left. \begin{array}{l} \pi r^2 = 81\pi \\ r = 9 \end{array} \right\} \begin{array}{l} \text{C.S.A} = \pi r l \\ = \pi \times 9 \times 15 \\ = \boxed{135\pi} \text{ Ans} \end{array}$

[40] The radius of a solid right circular cone is 36 cm and its height is 105 cm. The total surface area of the cone is:

Soln:- T.S.A = $\pi r(r+l)$
 $l = 111$ (Triplet $\rightarrow 36, 105, 111$)
 $12 \quad 35 \quad 37$

T.S.A = $\pi \times 36(36+111) = \boxed{5292\pi} \text{ Ans}$



[41] The radius and the height of a right circular cone are in the ratio 5:12. Its curved surface area is 816.4 cm^2 . What is the volume of the cone?

Soln:- $r:h = 5:12$ (Triplet: 5, 12, 13)

$r = 5a, h = 12a, l = 13a$

$\pi r l = 816.4$
 $3.14 \times 5a \times 13a = 816.4$
 $\boxed{a = 2}$

$\rightarrow r = 10, h = 24$
 $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times 3.14 \times 10^2 \times 24$
 $= \boxed{2512} \text{ Ans}$

42 If the radius of the base of a cone is doubled, and the volume of the new cone is three times the volume of the original cone, then what will be the ratio of the height of the original cone to that of the new cone?

Soln:- $R \rightarrow 1:2$
 $V \rightarrow 1:3$

height $\rightarrow h:H$

$$\frac{V_1}{V_2} = \frac{r_1^2 \times h_1}{r_2^2 \times h_2}$$

$$\frac{1}{3} = \frac{1}{4} \times \frac{h}{H}$$

$$\frac{h}{H} = \frac{4}{3} \Rightarrow h:H = \boxed{4:3} \text{ Ans}$$

43 The curved surface area of a right circular cone is 2310 cm^2 and its radius is 21 cm . If its radius is increased by 100% and height is reduced by 50% , then its capacity will be (Correct to one decimal)

Soln:-

C.S.A. = 2310

height = 28 cm (Triplet $\rightarrow 21, 28, 35$)

$$\frac{22}{7} \times 21 \times l = 2310$$

$$l = 35 \text{ cm}$$

New radius = 42 cm

New height = $28 - 14 = 14 \text{ cm}$

$$\text{Volume} = \frac{1}{3} \times \frac{22}{7} \times 42 \times 42 \times 14$$

$$= 25872 \text{ cm}^3$$

$$\text{Capacity} = \frac{25872}{1000}$$

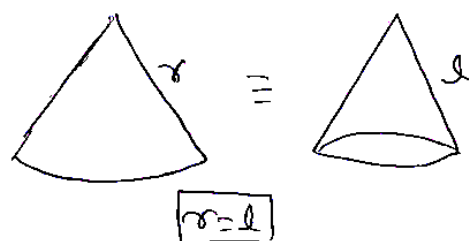
$$= \boxed{25.9 \text{ litre}}$$

[44] A sector is formed by opening out a cone of base radius 8cm and height 6cm. Then the radius of the sector is:

Soln.:-

$$l = \sqrt{8^2 + 6^2} = 10$$

$$r = l = \boxed{10\text{cm}} \text{ Ans}$$



[45] A sector of radius 10.5 cm with the central angle 120° is folded to form a cone by joining the two bounding radii of the sector. What is the volume of the cone so formed?

Soln.:-

$$l = \frac{2\pi r}{360} \times 120$$

$$l = 2 \times \frac{22}{7} \times 10.5 \times \frac{1}{3}$$

$$l = 22$$

$$l = 2\pi R = 22$$

$$R = 3.5\text{cm}$$

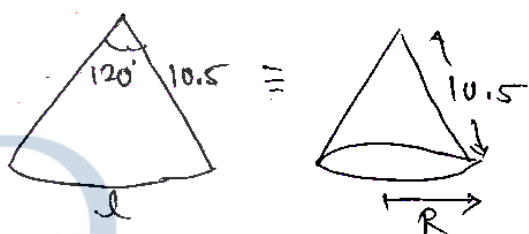
$$h = \sqrt{(10.5)^2 - (3.5)^2} = 7\sqrt{2}$$

$$V = \frac{1}{3} \pi R^2 h$$

$$= \frac{1}{3} \pi \times (3.5)^2 \times 7\sqrt{2}$$

$$\boxed{V = \frac{343\sqrt{2}}{12} \pi}$$

Ans



- [46] A Conical tent has to accommodate 25 persons. Each Person must have 4m^2 of space on the ground and 80m^3 of air to breathe. Find the height of the tent.

Soln.:- Base Area = $25 \times 4\text{m}^2$
 $= 100\text{m}^2$

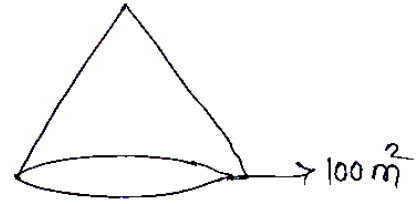
$$\pi r^2 = 100$$

$$r^2 = \frac{100}{\pi}$$

$$V = \frac{1}{3}\pi r^2 h$$

$$80\text{m}^3 \times 25 = \frac{1}{3} \times \pi \times \frac{100}{\pi} \times h$$

$$\boxed{h = 60\text{m}} \text{ Ans}$$



- [47] The radius of the base of a conical tent is 16 metres. If $427\frac{3}{7}\text{m}^2$ canvas is required to construct the tent, then the slant height of the tent is:

Soln.:- $\pi r l = 427\frac{3}{7}$

$$\frac{22}{7} \times 16 \times l = \frac{2992}{7} \Rightarrow l = \boxed{8.5 \text{ metre}}$$

- [48] The volume of a conical tent is 1232m^3 and the area of its base is 154m^2 . Find the length of the canvas required to build the tent, if the canvas is 2m in width?

Soln.:- $\pi r^2 = 154$

$$r = 7\text{m}$$

$$V = 1232$$

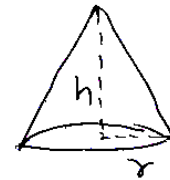
$$\frac{1}{3}\pi r^2 h = 1232$$

$$h = \frac{1232 \times 3}{154} = 24\text{m}$$

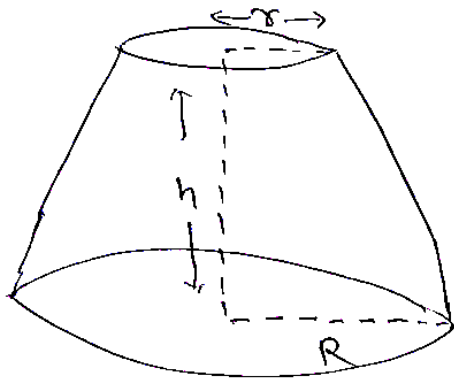
$$\text{Area of canvas} = \pi r l = \pi r (\sqrt{r^2 + h^2})$$

$$= \frac{22}{7} \times 7 \times \sqrt{24^2 + 7^2} = \frac{22}{7} \times 7 \times 25 = 550\text{m}^2$$

$$l \times 2 = 550 \Rightarrow l = \boxed{275\text{m}} \text{ Ans}$$



Frustum of Cone



$$l = \sqrt{h^2 + (R-r)^2}$$

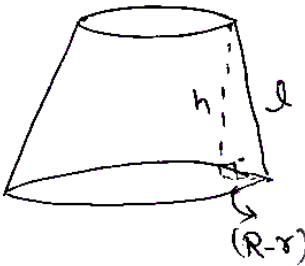
$$C.S.A = \pi(R+r)l$$

$$T.S.A = \pi(R+r)l + \pi R^2 + \pi r^2$$

$$T.S.A = \pi[R^2 + r^2 + (R+r)l]$$

$$Volume = \frac{1}{3} \pi (R^2 + r^2 + Rr)h$$

Proof



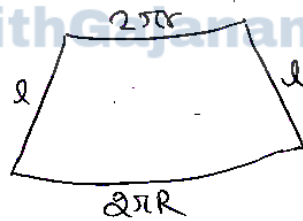
$$l = \sqrt{h^2 + (R-r)^2}$$

C.S.A

Proof:

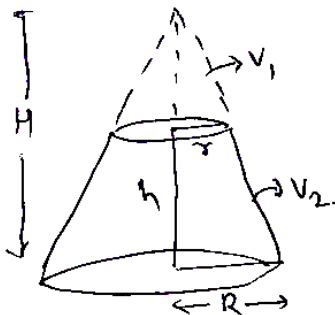


$\equiv$



$$\begin{aligned} \text{Area} &= \frac{1}{2} (2\pi R + 2\pi r) \times l \\ &= \frac{2\pi}{2} (R+r)l \\ &= \pi(R+r)l \end{aligned}$$

Volume



$$\begin{aligned} V_2 &= V - V_1 \\ &= \frac{1}{3} \pi R^2 H - \frac{1}{3} \pi r^2 (H-h) \end{aligned}$$

$$\begin{aligned} \frac{r}{R} &= \frac{H-h}{H} \\ Hr &= HR - hR \\ H' &= \frac{hR}{R-r} \end{aligned}$$

$$V_2 = \frac{1}{3} \pi \left[R^2 \left(\frac{hR}{R-r} \right) - r^2 \left(\frac{hR}{R-r} - h \right) \right]$$

$$V_2 = \frac{1}{3} \pi \left[\frac{R^3}{R-r} - \frac{r^3}{R-r} \right]$$

$$V_2 = \frac{1}{3} \pi (R^2 + r^2 + Rr)h \quad \checkmark$$

49 The radii of two circular faces of the frustum of a cone of height 21 cm are 3 cm and 2 cm respectively. What is the volume of the frustum of cone in cm^3 .

Soln.:-

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

$$= \frac{1}{3} \times \frac{22}{7} (3^2 + 2^2 + 3 \times 2) \times 21$$

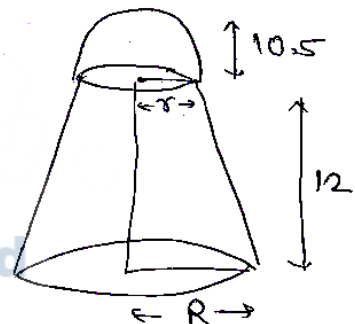
$$= \frac{1}{3} \times \frac{22}{7} \times 19 \times 21 = \boxed{418 \text{ cm}^3} \text{ Ans}$$

50 A 22.5 m high tent is in the shape of a frustum of a cone surmounted by a hemisphere. If the diameters of the upper and the lower circular ends of the frustum are 21 m and 39 m, respectively, then find the area of cloth used to make the tent (ignoring the wastage).

Soln.:- $r = \frac{21}{2} = 10.5$, $R = \frac{39}{2}$

$$\begin{aligned} \text{C.S.A of Hemisphere} &= 2\pi r^2 \\ &= 2 \times \frac{22}{7} \times \left(\frac{21}{2}\right)^2 \end{aligned}$$

$$= \boxed{693 \text{ cm}^2}$$



$$\text{C.S.A of Frustum of Cone} = \pi(R+r)l$$

$$l = \sqrt{12^2 + \left(\frac{39}{2} - \frac{21}{2}\right)^2}$$

$$= \sqrt{144 + 81} = \sqrt{225} = 15 \text{ cm}$$

$$\text{C.S.A} = \frac{22}{7} \times \left(\frac{39}{2} + \frac{21}{2}\right) \times 15$$

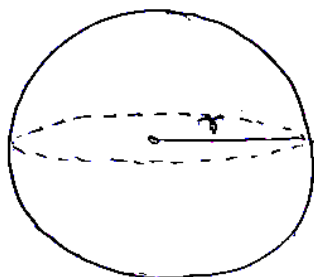
$$= \frac{22}{7} \times 30 \times 15 = \frac{9900}{7} = \boxed{1414\frac{2}{7} \text{ cm}^2}$$

$$\text{Total Area} = \left(693 + 1414\frac{2}{7}\right)$$

$$= \boxed{2107\frac{2}{7} \text{ cm}^2} \text{ Ans}$$

Sphere and Hemisphere

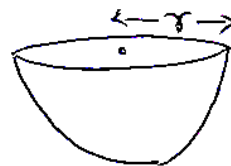
Sphere



$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$\text{T.S.A} = 4\pi r^2$$

Hemisphere

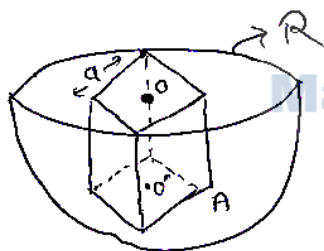


$$\text{Volume} = \frac{2}{3} \pi r^3$$

$$\text{C.S.A} = 2\pi r^2$$

$$\text{T.S.A} = 3\pi r^2$$

⇒ Maximum side of cube in Hemisphere :-

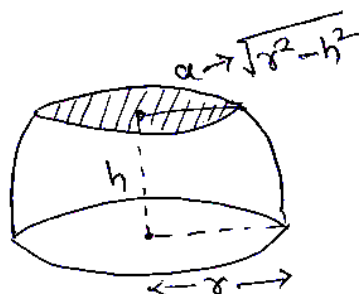


$$R^2 = a^2 + \frac{a^2}{2}$$

$$R^2 = \frac{3a^2}{2}$$

$$a = \sqrt{\frac{2}{3}} R$$

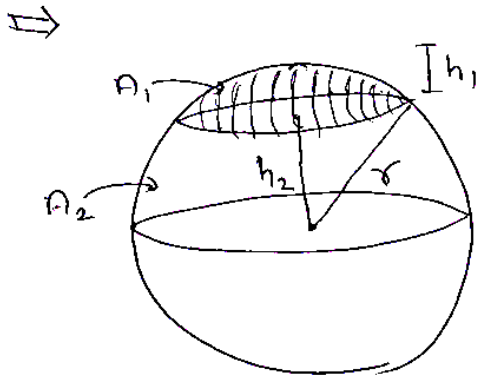
⇒



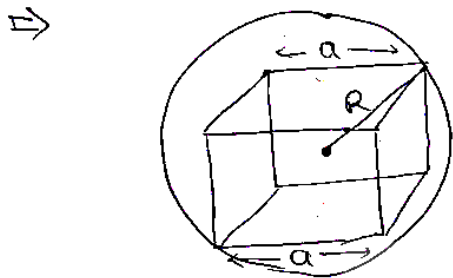
$$\text{Volume} = \frac{1}{3} \pi (3r^2 - h^2) h$$

$$\text{C.S.A} = 2\pi r h$$

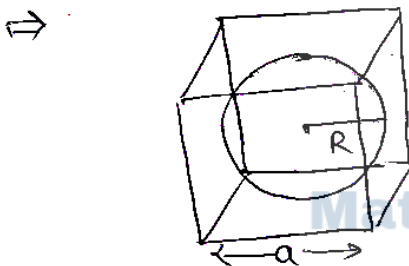
$$\text{T.S.A} = 2\pi r h + \pi r^2 + \pi(r^2 - h^2)$$



$$\left[\begin{array}{l} A_1 = 2\pi r h_1 \\ A_2 = 2\pi r h_2 \\ A = 2\pi r (h_1 + h_2) \\ A = 2\pi r^2 \end{array} \right]$$



$$R = \frac{\sqrt{3}a}{2}$$



$$R = \frac{a}{2}$$

51. Find the volume of a sphere whose radius is 7.5 cm,

Soln.

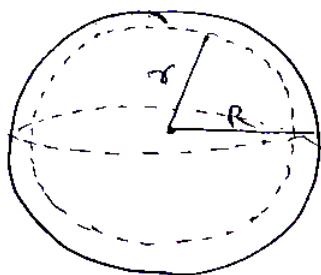
$$V = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times \left(\frac{15}{2}\right)^3$$

$$= \frac{4}{3} \times \frac{22 \times 15 \times 15 \times 15}{7 \times 8}$$

$$= \frac{11 \times 5 \times 15 \times 15}{7} = \boxed{1767.85 \text{ cm}^3} \quad \underline{\text{Ans}}$$

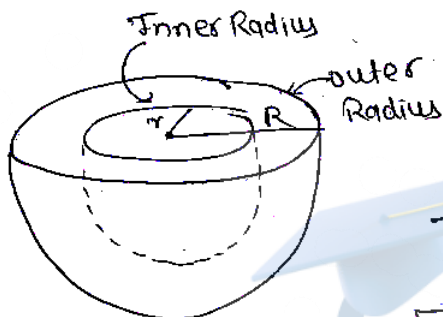
Hollow Sphere



$$\text{Volume} = \frac{4}{3}\pi(R^3 - r^3)$$

$$\begin{aligned}\text{T.S.A} &= 4\pi R^2 + 4\pi r^2 \\ &= 4\pi(R^2 + r^2)\end{aligned}$$

Hollow hemisphere



$$\left[\text{Volume} = \frac{2}{3}\pi(R^3 - r^3) \right]$$

$$\text{T.S.A} = 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$$

$$* \left[\text{T.S.A} = \pi(3R^2 + r^2) \right]$$

52 The volume of a hemisphere is $2425\frac{1}{2} \text{ cm}^3$. Find the radius.

Soln.:-

$$\frac{2}{3}\pi r^3 = 2425\frac{1}{2}$$

$$\frac{2}{3} \times \frac{22}{7} \times r^3 = \frac{4851}{2}$$

$$r^3 = \frac{3 \times 7 \times 441}{8}$$

$$r = \frac{21}{2}$$

$$r = \boxed{10.5 \text{ cm}} \text{ Ans}$$

[53] The volume of a solid hemisphere is 19404 cm^3 . Its total surface area is:

Soln:-

$$\frac{2}{3} \times \frac{22}{7} \times r^3 = 19404$$
$$r^3 = \frac{19404 \times 7 \times 3}{2 \times 22}$$
$$r^3 = 441 \times 21$$
$$r = 21$$

$$\text{T.S.A} = 3\pi r^2 = 3 \times \frac{22}{7} \times 21 \times 21 = \boxed{4158} \text{ Any}$$

[54] The internal diameter of a hollow hemispherical vessel is 24 cm . It is made of a steel sheet which is 0.5 cm thick. What is the total surface area of the vessel?

Soln:-

$$\begin{aligned} \text{T.S.A} &= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2) \\ &= \pi[3R^2 + r^2] \\ &= \pi[3(12.5)^2 + (12)^2] = \boxed{612.75 \text{ cm}^2} \text{ Any} \end{aligned}$$

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[55] The surface area of a sphere is 221.76 cm^2 . Its volume is: (Correct to one decimal)

Soln:-

$$4\pi r^2 = 221.76$$
$$4 \times \frac{22}{7} \times r^2 = 221.76$$
$$r^2 = 17.64$$
$$r = 4.2 \text{ cm}$$
$$\begin{aligned} \text{Volume} &= \frac{4}{3} \times \frac{22}{7} \times 4.2 \times 4.2 \times 4.2 \\ &= 310.464 \\ &\approx \boxed{310.5} \text{ Any} \end{aligned}$$

[56] The radius of a sphere is reduced by 40%. By what Percent will its volume decrease?

Soln:-

$$V \propto r^3$$

Initial Now

$$(5)^3 : (3)^3$$

$$125 : 27$$



$$\text{decrease \%} = \frac{98}{125} \times 100 = 78.4\%$$

$$40\% = \frac{2}{5}$$

[57] If the radius of a sphere is increased by 4cm, its surface area is increased by $464\pi \text{ cm}^2$. What is the volume of the original sphere?

Soln

$$4\pi [(R+4)^2 - R^2] = 464\pi$$

$$4(R+4+R)(R+4-R) = 464$$

$$16(2R+4) = 464$$

$$R = \frac{25}{2}$$

$$\text{Volume} = \frac{4}{3} \pi \left(\frac{25}{2}\right)^3 = \frac{15625}{6} \pi$$

[58] The radius of a spherical balloon is inflated from 3.5cm to 4.9cm by pushing air into it. What is the percentage increase in the volume of the original balloon?

Soln:-

$$r \rightarrow 3.5 : 4.9$$

$$5 : 7$$

$$V \rightarrow (5)^3 : (7)^3$$

$$125 : 343$$



$$\frac{218}{125} \times 100 = \boxed{174.4\%} \text{ Ans}$$

[59] The ratio of the total surface area and volume of a sphere is $2:7$. Its radius is:

Soln:-
$$\frac{T.S.A}{V} = \frac{4\pi r^2}{\frac{4}{3}\pi r^3} = \frac{2}{7}$$

$$r = \frac{21}{2} = \boxed{10.5 \text{ cm}} \text{ Ans.}$$

[60] A sphere of maximum volume is cut out from a solid hemisphere. What is the ratio of the volume of the sphere to that of the remaining solid?

Soln:- Sphere : Hemisphere

$$\frac{4}{3}\pi\left(\frac{R}{2}\right)^3 : \frac{2}{3}\pi R^3$$

$$\boxed{1:4}$$

Sphere : Remaining Solid = $\boxed{1:3}$ Ans.



[61] A solid cone of height 9cm with diameter of its base 18cm is cut out from a wooden solid sphere of Radius 9cm. The percentage of wood wastage is:

Soln:- Volume of sphere = $\frac{4}{3}\pi(9)^3 = 972\pi$

Volume of cone = $\frac{1}{3}\pi(9)^2 \times 9 = 243\pi$

$$\% \text{ of wasted wood} = \frac{972\pi - 243\pi}{972\pi} \times 100$$

$$= \boxed{75\%} \text{ Ans}$$

[62] How many balls of radius 10cm can be made by melting a bigger ball of diameter 80cm?

Soln:- Basic

Radius of small ball = 10cm
(r)

radius of big ball (R) = $\frac{80}{2}$
= 40cm

Volume of (bigger) = Volume of (small)

$$\frac{4}{3}\pi(40)^3 = n \times \frac{4}{3}\pi(10)^3$$

$$n = \frac{(40)^3}{(10)^3} = \boxed{64}$$

Exam

$$\begin{aligned} \text{No. of ball} &= \left(\frac{R}{r}\right)^3 \\ &= \frac{(40)^3}{(10)^3} = \boxed{64} \text{ Ans} \end{aligned}$$

[63] Three solid metallic spheres whose radii are 1cm, xcm and 8cm, are melted and recast into a single solid sphere of diameter 18cm. The surface area of the sphere with radius xcm is.

Soln:-

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi r_1^3 + \frac{4}{3}\pi r_2^3 + \frac{4}{3}\pi r_3^3 + \dots$$

$$R^3 = r_1^3 + r_2^3 + r_3^3$$

$$9^3 = (1)^3 + (x)^3 + (8)^3$$

$$729 = 1 + x^3 + 512$$

$$x^3 = 216$$

$$x = 6$$

$$\text{Surface Area} = 4\pi x^2$$

$$= 4\pi \times (6)^2$$

$$= \boxed{144\pi} \text{ Ans}$$

Mix. Questions

[64] If V_1 , V_2 and V_3 be the volumes of a right circular cone, a sphere and a right circular cylinder having the same radius and same height then

(a) $V_1 = \frac{V_2}{4} = \frac{V_3}{3}$ (b) $\frac{V_1}{2} = \frac{V_2}{3} = V_3$ (c) $\frac{V_1}{3} = \frac{V_2}{2} = V_3$ (d) $6V_1 = 3V_2 = 2V_3$

Soln.:- $V_1 = \frac{1}{3}\pi r^2 h$, $V_2 = \frac{4}{3}\pi r^3$, $V_3 = \pi r^2 h$
($2r=h$)

$$V_2 \Rightarrow \frac{2}{3}\pi r^2 h$$

$$\pi r^2 h = 3V_1 = \frac{3V_2}{2} = V_3 \Rightarrow \boxed{6V_1 = 3V_2 = 2V_3} \text{ Ans}$$

[65] A solid metallic sphere of radius 8 cm is melted and drawn into a wire of uniform cross-section. If the length of the wire is 2400 cm, then its radius (in mm) is:

Soln.:- Sphere wire (Cylinder)

$$\frac{4}{3}\pi r^3 = \pi R^2 H$$

$$\frac{4}{3}(8)^3 = r^2(2400 \text{ cm})$$

$$r^2 = \frac{4 \times 8 \times 8 \times 8}{3 \times 2400}$$

$$r = \frac{16}{30} \text{ cm}$$

$$r = \frac{16}{30} \times 10 \text{ mm} = \frac{16}{3} \text{ mm}$$

$$\boxed{r = 5\frac{1}{3} \text{ mm}} \text{ Ans}$$

Q66 A cylindrical vessel with radius 6cm and height 5cm is to be made by melting a number of spherical metal balls of diameter 2cm. The minimum number of balls needed is:

Soln:-

$$\pi r^2 h = n \times \frac{4}{3} \pi R^3$$

$$6^2 \times 5 = \frac{4}{3} \times (1)^3 \times n$$

$$\boxed{n = 135} \text{ Ans}$$

Q67 A hemispherical bowl of internal diameter 36cm is full of a liquid. This liquid is to be filled into cylindrical bottles each of radius 3cm and height 12cm. How many such bottles are required to empty the bowl?

Soln:-

$$\frac{2}{3} \pi r^3 = \pi R^2 H \times n$$

$$\frac{2}{3} \times (18)^3 = (3)^2 \times 12 \times n$$

$$\boxed{n = 36} \text{ Ans}$$

Q68 A child reshape a cone made up of clay of height 24cm and radius 6cm into a sphere. The radius of the sphere is:

Soln:-

Cone → sphere

$$\frac{1}{3} \pi r^2 h = \frac{4}{3} \pi R^3$$

$$(6)^2 \times 24 = 4 R^3$$

$$\boxed{R = 6 \text{ cm}}$$

69] A hollow sphere of internal and external diameter 6cm and 10cm respectively is melted into a right circular cone of diameter 8cm. The height of the cone is:

Soln :-

$$\frac{4}{3}\pi(5^3 - 3^3) = \frac{1}{3}\pi(4)^2 \times h$$

$$98 = 4h$$

$$h = \boxed{24.5\text{cm}} \text{ Ans}$$

70] A Solid wooden toy is in the shape of a right circular cone mounted on a hemisphere. If the radius of the hemisphere is 4.2cm and the total height of the toy is 10.2cm. Find the volume of wooden toy (nearly)?

Soln :-

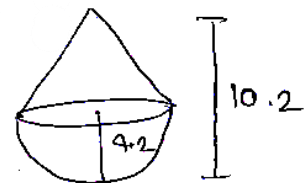
$$\text{Height of Cone} = 10.2 - 4.2 = 6\text{cm}$$

$$\text{Volume} = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

$$= \frac{1}{3}\pi r^2 (h + 2r)$$

$$= \frac{1}{3} \times \frac{22}{7} \times (4.2)^2 [6 + 2 \times 4.2]$$

$$= \frac{1}{3} \times \frac{22}{7} \times (4.2)^2 \times 14.4 = 266\text{cm}^3$$



71] A Solid cylinder of base radius 12cm and height 15cm is melted and recast into n toys each in the shape of a right circular cone of height 9cm mounted on a hemisphere of radius 3cm. The value of n is:

Soln :-

$$\pi R^2 H = \left[\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 \right] \times n$$

$$(12)^2 \times 15 = n \left[\frac{1}{3} \times 3^2 \times 9 + \frac{2}{3} \times 3^3 \right]$$

$$n = \boxed{48} \text{ Ans}$$



72 A rectangular block of metal has dimensions 21cm, 77cm and 24cm. The block has been melted into a sphere. The radius of the sphere is:

Soln:-

$$l b h = \frac{4}{3} \pi r^3$$

$$21 \times 77 \times 24 = \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$r^3 = 7 \times 7 \times 7 \times 3 \times 3 \times 3$$

$$r = \boxed{21 \text{ cm}} \text{ Ans}$$

73 A solid cone of radius 7cm and height 7cm was melted along with two solid spheres of radius 7cm each to form a solid cylinder of radius 7cm. What is the curved surface area of the cylinder?

Soln:-

$$\text{Cylinder} = \text{Cone} + 2 \times \text{sphere}$$

$$\pi (7)^2 \times h = \frac{1}{3} \pi (7)^2 \times 7 + 2 \left(\frac{4}{3} \pi (7)^3 \right)$$

$$h = \frac{7}{3} + \frac{56}{3}$$

$$h = 21 \text{ cm}$$

$$\text{C.S.A} = 2 \times \frac{22}{7} \times 21 \times 7 = \boxed{924 \text{ cm}^2} \text{ Ans}$$

74 Balls of marbles of diameter 1.4cm are dropped into a cylindrical beaker containing some water and fully submerged. The diameter of the beaker is 7cm. Find how many marbles have been dropped in it if the water rises by 5.6cm?

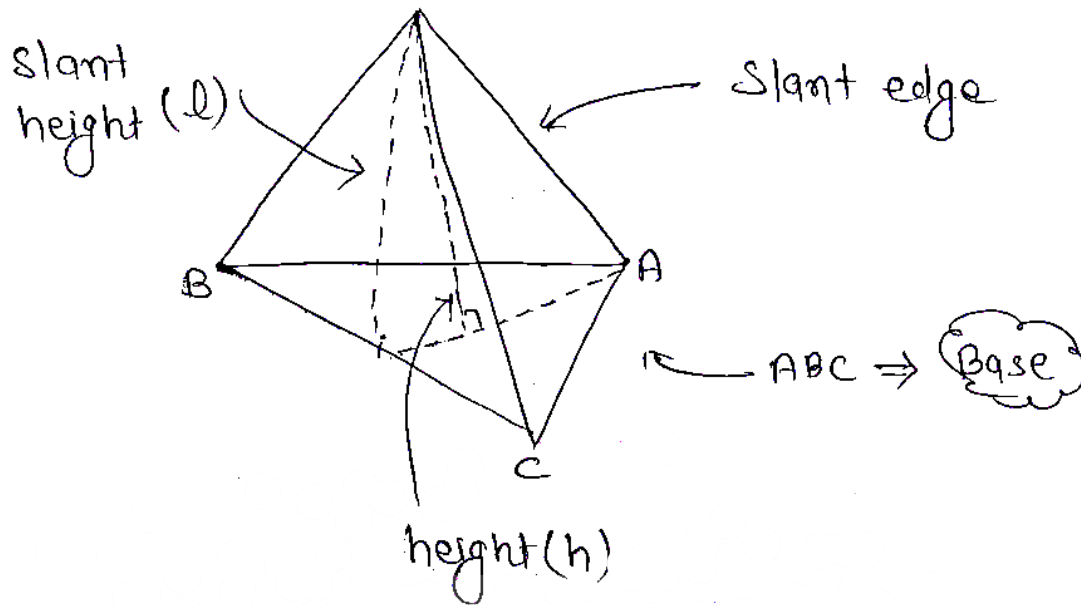
Soln:-

$$n \times \left(\frac{4}{3} \pi (0.7)^3 \right) = \pi \left(\frac{7}{2} \right)^2 \times 5.6$$

$$n \times \frac{4}{3} \times \frac{7}{10} \times \frac{7}{10} \times \frac{7}{10} = \frac{7}{2} \times \frac{7}{2} \times \frac{56}{10}$$

$$n = \boxed{150} \text{ Ans}$$

Pyramid



$$\# \text{ L.S.A} = \frac{1}{2} \times (\text{Perimeter of Base}) \times (\text{slant height})$$

$$\# \text{ T.S.A} = \text{L.S.A} + (\text{Area of Base})$$

$$\# \text{ Volume} = \frac{1}{3} \times (\text{Area of Base}) \times \text{height}$$

Ex 5 Area of the base of a pyramid is 57 cm^2 and height is 10 cm , then its volume is:

Soln. $\text{Volume} = \frac{1}{3} \times 57 \times 10$

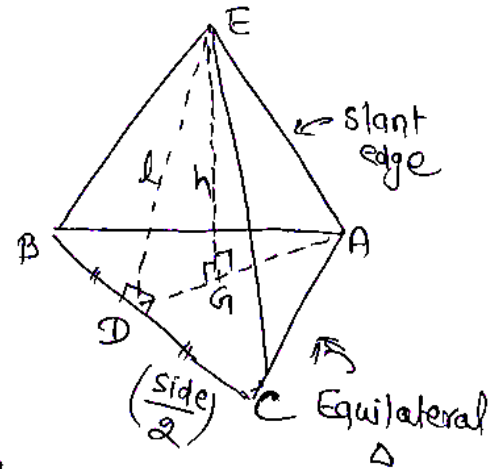
$$= \boxed{190 \text{ cm}^3} \text{ Ans}$$

Pyramid :- If Base is Equilateral Triangle

$$* h^2 + (DG)^2 = l^2$$

$$* h^2 + (GA)^2 = (\text{slant edge})^2$$

$$* l^2 + \left(\frac{\text{side}}{2}\right)^2 = (\text{slant edge})^2$$



$$\left\{ DG = \frac{\text{side}}{2\sqrt{3}} \right\}$$

$$\left\{ AG = \frac{\text{side}}{\sqrt{3}} \right\}$$

[76] The base of a right pyramid is an equilateral triangle with side 8cm, and height is $30\sqrt{3}$ cm. The volume of the pyramid is:

Soln:-

$$\text{Volume} = \frac{1}{3} \times (\text{Base Area}) \times \text{height}$$

$$= \frac{1}{3} \times \frac{\sqrt{3}}{4} \times (8)^2 \times 30\sqrt{3} = \boxed{480 \text{ cm}^3} \text{ Ans}$$

[77] The volume of a right pyramid is $45\sqrt{3} \text{ cm}^3$ and its base is an equilateral triangle with side 6cm. What is the height of the pyramid?

Soln:-

$$45\sqrt{3} = \frac{1}{3} \times \frac{\sqrt{3}}{4} \times (6)^2 \times h$$

$$45\sqrt{3} = 3\sqrt{3} h$$

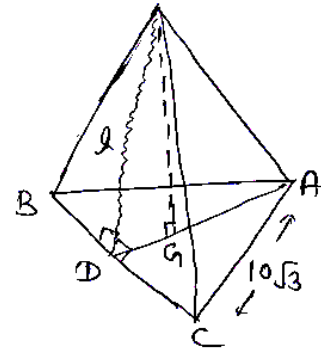
$$h = \boxed{15 \text{ cm}} \text{ Ans}$$

78 The base of a right pyramid is equilateral triangle of side $10\sqrt{3}$ cm. If the total surface area of the pyramid is $270\sqrt{3}$ cm², its height is:

Soln:- $OG = \frac{\text{side}}{2\sqrt{3}} = \frac{10\sqrt{3}}{2\sqrt{3}} = 5 \text{ cm}$

$$l^2 = h^2 + OG^2$$

$$l = \sqrt{h^2 + 25}$$



$$\text{T.S.A} = \frac{1}{2} (\text{Base Perimeter}) \times \text{slant height} + (\text{Base Area})$$

$$270\sqrt{3} = \frac{1}{2} (3 \times 10\sqrt{3}) \times \sqrt{h^2 + 25} + \frac{\sqrt{3}}{4} (10\sqrt{3})^2$$

$$270\sqrt{3} = 15\sqrt{3} \sqrt{h^2 + 25} + 75\sqrt{3}$$

$$\sqrt{h^2 + 25} = 13$$

$$h^2 + 25 = 169$$

$$h^2 = 144 = \boxed{12 \text{ cm}} \text{ Ans}$$

79 The base of right pyramid is an equilateral triangle, each side of which is 20 cm. Each slant edge is 30 cm. The verticle height of the pyramid is:

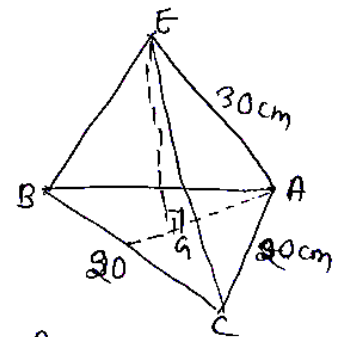
Soln:- $h^2 + (GA)^2 = (\text{slant edge})^2$

$$h^2 + \left(\frac{20}{\sqrt{3}}\right)^2 = (30)^2$$

$$h^2 = 900 - \frac{400}{3}$$

$$h^2 = \frac{2300}{3}$$

$$h = 10\sqrt{\frac{23}{3}} \text{ Ans}$$



$$GA = \frac{\text{side}}{\sqrt{3}} = \frac{20}{\sqrt{3}}$$

80 The base of a right pyramid is an equilateral triangle with area $16\sqrt{3} \text{ cm}^2$. If the area of one of its lateral faces is 30 cm^2 , then its height is:

Soln:-

$$\frac{\sqrt{3}}{4} a^2 = 16\sqrt{3}$$

$$a^2 = 64$$

$$\text{side} = \boxed{a = 8 \text{ cm}}$$

$$\frac{1}{2} \times 8 \times (\text{slant height}) = 30$$

$$l = \frac{15}{2}$$

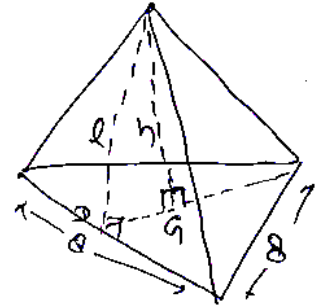
$$DQ = \frac{\text{side}}{2\sqrt{3}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$$

$$l^2 = h^2 + (DQ)^2$$

$$\frac{225}{4} = h^2 + \left(\frac{4}{\sqrt{3}}\right)^2$$

$$h^2 = \frac{225}{4} - \frac{16}{3}$$

$$\boxed{h = \sqrt{\frac{611}{12}}} \text{ Ans}$$



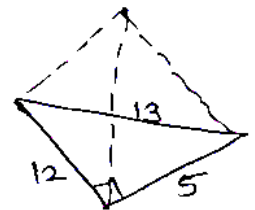
81 If the base of a right pyramid is triangle of sides 5 cm, 12 cm and 13 cm and its volume is 330 cm, then its height will be ?

Soln

$$V = \frac{1}{3} \times (\text{base Area}) \times \text{height}$$

$$330 = \frac{1}{3} \times \left(\frac{1}{2} \times 5 \times 12\right) \times h$$

$$h = \boxed{33 \text{ cm}} \text{ Ans}$$

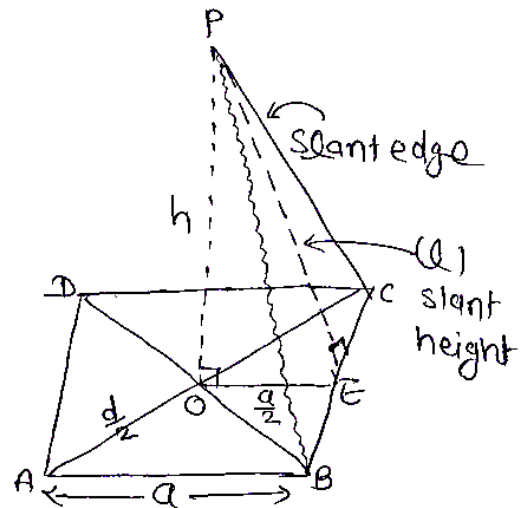


Pyramid :- If Base is Square

$$h^2 + \left(\frac{a}{2}\right)^2 = l^2$$

$$h^2 + \left(\frac{d}{2}\right)^2 = (\text{slant edge})^2$$

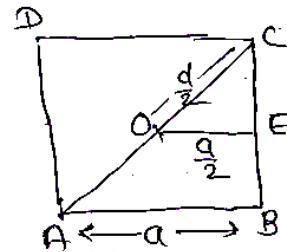
$$l^2 + \left(\frac{a}{2}\right)^2 = (\text{slant edge})^2$$



$$\left[\text{Volume} = \frac{1}{3} \times (a^2) \times h \right]$$

$$\left[\text{L.S.A} = \frac{1}{2} \times (4a) \times l \right]$$

$$\left[\text{T.S.A} = \text{L.S.A} + a^2 \right]$$



[82] The base of a right pyramid is a square of side 10cm. If its height is 10cm, then the area of its lateral surface is:

Soln:-

$$\begin{aligned} \text{L.S.A} &= \frac{1}{2} \times (4a) \times l & \left| \quad h^2 + \left(\frac{a}{2}\right)^2 = l^2 \right. \\ &= \frac{1}{2} \times 40 \times 5\sqrt{5} & 100 + 25 = l^2 \\ &= \boxed{100\sqrt{5}} \text{ Ans} & l = \sqrt{125} = 5\sqrt{5} \end{aligned}$$

[83] Base of a right pyramid is a square of side 10cm. If the height of the pyramid is 12cm, then its total surface area is:

Soln:-

$$\begin{aligned} h^2 + \left(\frac{a}{2}\right)^2 &= l^2 \\ 144 + 25 &= l^2 \\ l &= \sqrt{169} = 13\text{cm} \end{aligned}$$

$$\begin{aligned}
 T.S.A &= \frac{1}{2}(40) \times 13 + (10)^2 \\
 &= 260 + 100 \\
 &= \boxed{360 \text{ cm}^2} \text{ Ans.}
 \end{aligned}$$

[84] A right pyramid 6m high has a square base of which the diagonal is $\sqrt{1152}$ m. Volume of the pyramid is

Soln:-

$$\begin{aligned}
 \text{Volume} &= \frac{1}{3} \times (\text{base Area}) \times \text{height} \\
 &= \frac{1}{3} \times \left(\frac{1}{2} d^2\right) \times h \\
 &= \frac{1}{3} \times \frac{1}{2} \times 1152 \times 6 = \boxed{1152 \text{ m}^3} \text{ Ans}
 \end{aligned}$$

[85] The base of a right pyramid is a square of side $8\sqrt{2}$ cm and each of its slant edge is of length 10 cm. What is the volume of the pyramid?

Soln:-

$$\begin{aligned}
 \text{diagonal (d)} &= \sqrt{2} a \\
 &= \sqrt{2} \times 8\sqrt{2} = 16 \text{ cm}
 \end{aligned}$$

$$h^2 + OB^2 = AB^2$$

$$h^2 + \left(\frac{16}{2}\right)^2 = (10)^2$$

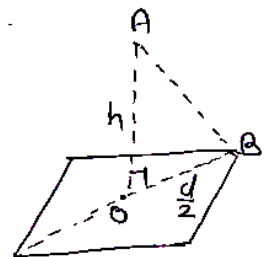
$$h^2 = 36$$

$$h = 6 \text{ cm}$$

$$\text{Volume} = \frac{1}{3} \times (8\sqrt{2})^2 \times 6$$

$$= \frac{1}{3} \times 128 \times 6$$

$$= \boxed{256 \text{ cm}^3} \text{ Ans}$$

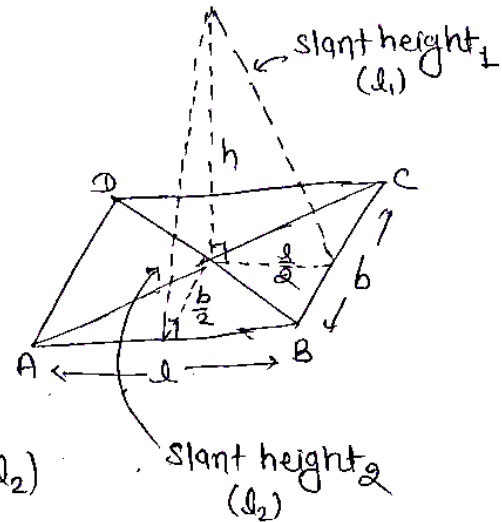


Pyramid :- If Base is Rectangle

$$h^2 + \left(\frac{l}{2}\right)^2 = (\text{slant height}_1)^2$$

$$h^2 + \left(\frac{b}{2}\right)^2 = (\text{slant height}_2)^2$$

$$h^2 + \left(\frac{d}{2}\right)^2 = (\text{slant edge})^2$$

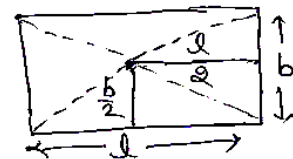


$$l.s.A = 2\left(\frac{1}{2} \times b \times l_1\right) + 2\left(\frac{1}{2} \times l \times l_2\right)$$

$$l.s.A = b l_1 + l l_2$$

$$T.S.A = l.s.A + l \times b$$

$$\text{Volume} = \frac{1}{3} \times (l \times b) \times h$$



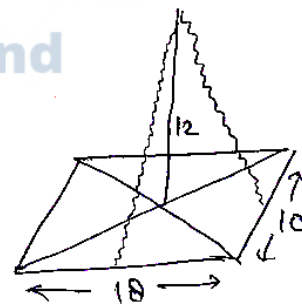
Ex:- length = 18cm, breadth = 10cm, height = 12cm, then

(i) Slant height on breadth?

$$\text{Soln:- } l_1 = \sqrt{(12)^2 + (9)^2} = \boxed{15\text{cm}} \text{ Ans}$$

(ii) Slant height on length?

$$\text{Soln:- } l_2 = \sqrt{(12)^2 + (5)^2} = \boxed{13\text{cm}} \text{ Ans}$$



(iii) $l.s.A = ? \Rightarrow (18 \times 13 + 10 \times 15) = \boxed{284\text{cm}^2} \text{ Ans}$

(iv) $T.S.A = 284 + (18 \times 10) = \boxed{464\text{cm}^2} \text{ Ans}$

(v) $\text{Volume} = \frac{1}{3} \times (18 \times 10) \times 12 = \boxed{720\text{cm}^3} \text{ Ans}$

(vi) $\text{slant edge} = \sqrt{h^2 + \left(\frac{d}{2}\right)^2} = \sqrt{(12)^2 + 106} = \sqrt{250} = 5\sqrt{10}$
 $(d = \sqrt{10^2 + 18^2} = 2\sqrt{106})$

Pyramid :- If Base Regular Hexagon

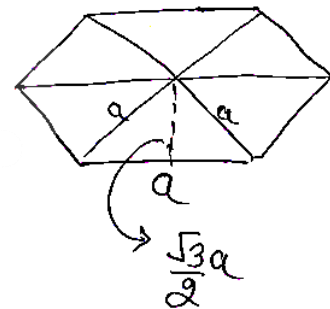
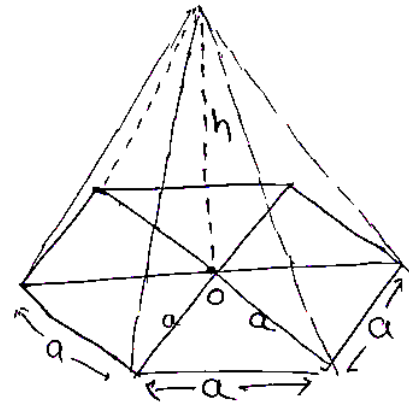
$$h^2 + (\text{side})^2 = (\text{slant edge})^2$$

$$h^2 + \left(\frac{\sqrt{3}}{2} \text{side}\right)^2 = (\text{slant height})^2$$

$$[L.S.A = \frac{1}{2} \times (6a) \times \text{slant height}]$$

$$[T.S.A = L.S.A + 6 \left(\frac{\sqrt{3}}{4} a^2\right)]$$

$$[\text{Volume} = \frac{1}{3} \times (6 \times \frac{\sqrt{3}}{4} a^2) \times h]$$



[86] A right pyramid is on a regular hexagonal base. Each side of the base is 10m and the height is 60m. The volume of the pyramid is:

- (a) 5000 m³ (b) 5100 m³ (c) 5195 m³ (d) 5196 m³

Soln.

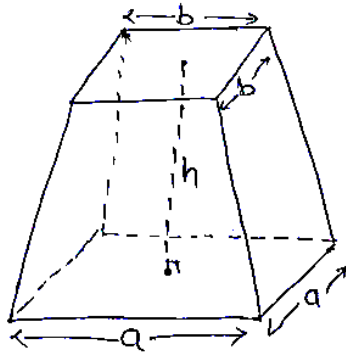
$$V = \frac{1}{3} \times \left(6 \times \frac{\sqrt{3}}{4} \times 10^2\right) \times 60$$

$$= \frac{1}{3} \times 6 \times \frac{\sqrt{3}}{4} \times 100 \times 60$$

$$= 3\sqrt{3} \times 1000 \text{ m}^3$$

$$= \boxed{5196 \text{ m}^3} \text{ Ans}$$

Frustum of Pyramid

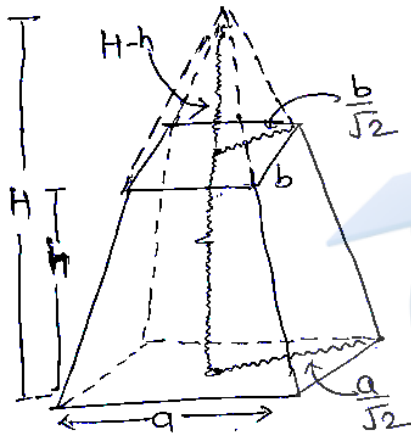


$$L.S.A = 2(a+b) \sqrt{\left(\frac{a-b}{2}\right)^2 + h^2}$$

$$T.S.A = a^2 + b^2 + L.S.A$$

$$\text{Volume} = \frac{1}{3} (a^2 + b^2 + ab) h$$

Proof:- Volume



Similarity:-

$$\frac{H-h}{H} = \frac{b/\sqrt{2}}{a/\sqrt{2}}$$

$$a(H-h) = bH$$

$$H(b-a) = -ah$$

$$H = \frac{ah}{a-b}$$

$$\begin{aligned} H-h &= \frac{ah}{a-b} - h \\ &= \frac{bh}{a-b} \end{aligned}$$

$$\text{Volume: } \frac{1}{3} \times a^2 \times H - \frac{1}{3} \times b^2 \times (H-h)$$

$$= \frac{1}{3} \left[\frac{a^3 h}{a-b} - \frac{b^3 h}{a-b} \right]$$

$$= \frac{1}{3} \times h \times \left[\frac{a^3 - b^3}{a-b} \right]$$

$$= \left[\frac{1}{3} \times h \times a^2 + b^2 + ab \right]$$

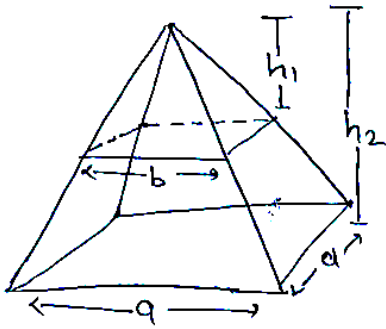
$\Rightarrow$

$$\boxed{\frac{h_1}{h_2} = \frac{b}{a}}$$

$h_1 \rightarrow$ height of small pyramid

$h_2 \rightarrow$ height of Big pyramid

1.



$$\frac{h_1}{h_2} = \frac{b}{a}$$

$$\frac{V_1}{V_2} = \frac{\frac{1}{3}a^2 \times h_2}{\frac{1}{3} \times b^2 \times h_1}$$

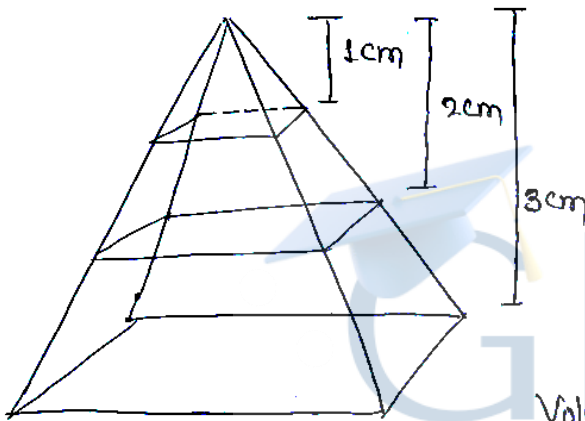
$$\frac{V_1}{V_2} = \frac{a^2}{b^2} \times \frac{h_2}{h_1}$$

$$\# \frac{V_1}{V_2} = \frac{a^3}{b^3} = \frac{h_2^3}{h_1^3}$$

$\rightarrow h_2$ height
 $V_1 \rightarrow$ Total

$V_2 \rightarrow$ half (upper)
 $\hookrightarrow h_1$

2.

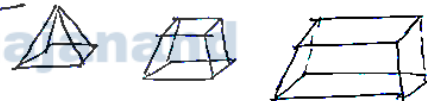


$$h_1 : h_2 : h_3 = 1 : 2 : 3$$

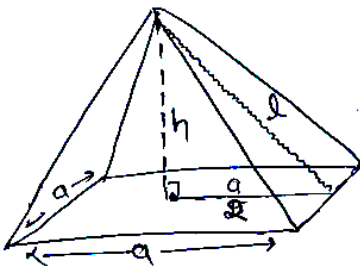
$$V_1 : V_2 : V_3 = (1)^3 : (2)^3 : (3)^3$$

$$= 1 : 8 : 27$$

$$\text{Volume } 1 : 7 : 19$$

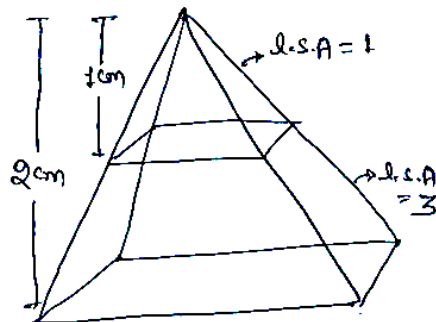


3.



$$\text{slant height (l)} = \sqrt{h^2 + \frac{a^2}{4}}$$

4.



$$\text{height : } h_1 : h_2 = 1 : 2$$

$$\text{l.s.A : } l.s.A_1 : l.s.A_2 = 1 : 4$$

$$\text{Volume : } V_1 : V_2 = 1 : 8$$

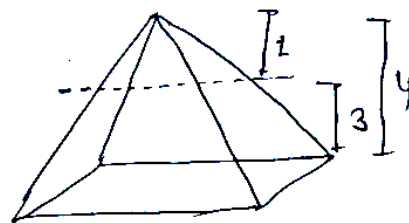
[87] Side of the square base of a pyramid is 8cm. the pyramid is cut in two parts by a plane parallel to its base such that heights of upper and lower parts are in ratio 1:3 respectively. What is the ratio of volume of lower and upper parts respectively

Soln:-

Small Big
Height $\rightarrow 1 : 4$

Volume $\rightarrow 1 : 64$

Lower : Upper $\rightarrow (64-1) : 1 = \boxed{63:1}$ Ans



[88] A pyramid has square base. The side of square is 12cm. height of pyramid is 21cm. The pyramid is cut into 3 parts by 2 cuts parallel to its base the cuts are at height of 7cm and 14cm respectively from the base. What is the difference in the volume of top most and bottom most part?

Soln:-

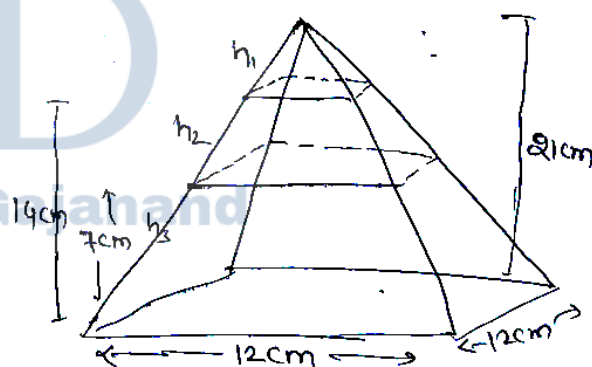
$$h_1 : h_2 : h_3 = 7 : 14 : 21$$

$$= 1 : 2 : 3$$

$$V_1 : V_2 : V_3 = 1 : 8 : 27$$

$$\text{Volume} = 1 : 7 : 19$$

upper middle bottom



$$\text{Total Volume} = \frac{1}{3} \times (12^2) \times 21$$

$$= 144 \times 7$$

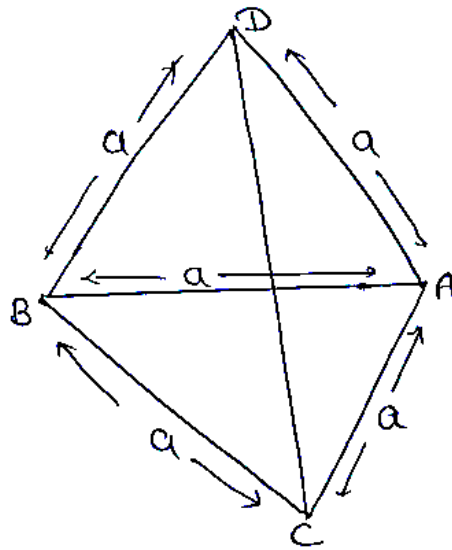
$$\text{Volume} = (\text{Top} - \text{bottom}) = (27 - 1) = 18 \text{ Unit}$$

$$27 \text{ Unit} \rightarrow 144 \times 7$$

$$\text{Volume} = 18 \text{ Unit} \rightarrow \frac{144 \times 7}{27} \times 18$$

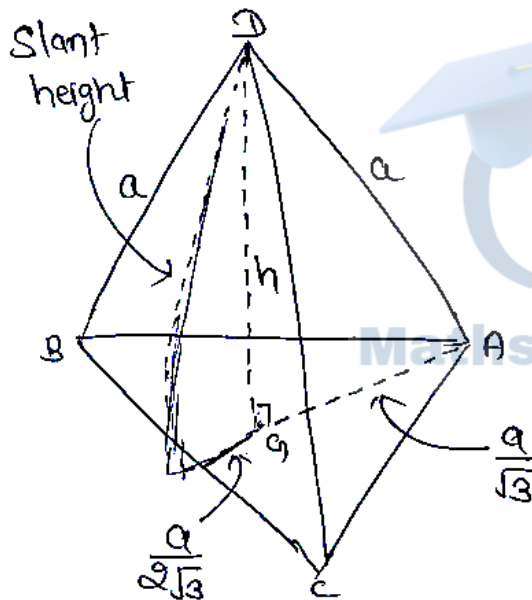
$$= \boxed{672} \text{ Ans}$$

Tetrahedron



4 $\Rightarrow$ Faces

(All are Equilateral Δ)



L.S.A = 3 $\times$ (Equilateral Δ)

$$\# \left[\text{L.S.A} = 3 \times \frac{\sqrt{3}}{4} a^2 \right]$$

T.S.A = 4 $\times$ (Equilateral Δ)

$$\# \left[\text{T.S.A} = 4 \times \frac{\sqrt{3}}{4} a^2 \right]$$

$$\text{height} = h = \sqrt{a^2 - \left(\frac{a}{\sqrt{3}}\right)^2}$$

In ΔADG

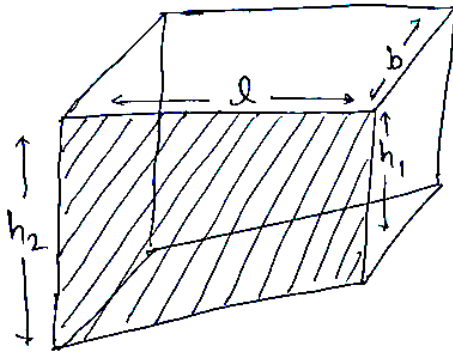
$$h = \sqrt{\frac{2}{3}} a$$

Volume = $\frac{1}{3} \times (\text{Base Area}) \times \text{height}$

$$= \frac{1}{3} \times \left(\frac{\sqrt{3}}{4} a^2\right) \times \left(\sqrt{\frac{2}{3}} a\right)$$

$$\# \left[V = \frac{a^3}{6\sqrt{2}} \right]$$

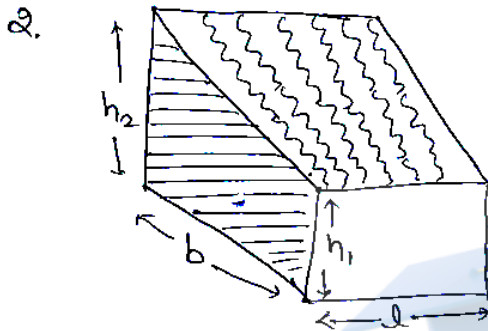
1. Swimming Pool:-



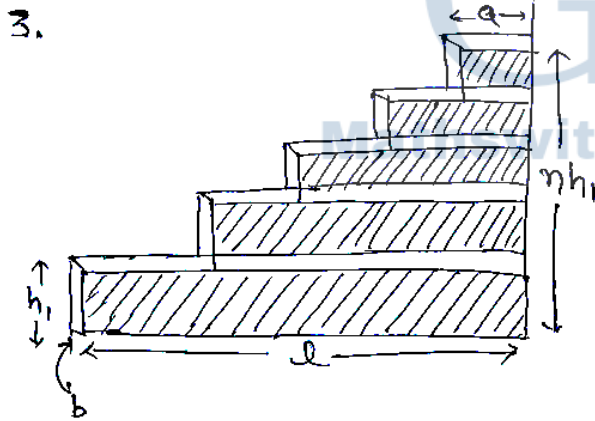
$$\text{Volume} = \text{Base} \times \text{height}$$

$$\text{Trapezium} = \frac{1}{2}(h_1 + h_2)l$$

$$\text{Volume} = \frac{1}{2}(h_1 + h_2)lb$$

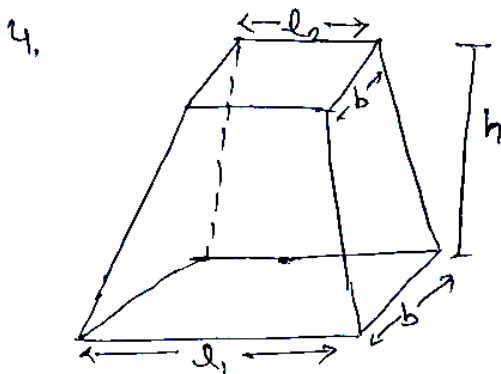


$$\text{Volume} = \frac{1}{2}(h_1 + h_2) \times b \times l$$



$$\text{Volume} = \frac{1}{2}(l + a) \times n h_1 \times b$$

$n \rightarrow$ no. of steps



$$\text{Volume} = \frac{1}{2}(l_1 + l_2) \times h \times b$$

89 A side of a Tetrahedron is 9cm then find

(i) L.S.A = ?

(ii) T.S.A = ?

(iii) Volume = ?

Soln :-

$$L.S.A = 3 \times \left(\frac{\sqrt{3}}{4} \times 9^2 \right) = \frac{243\sqrt{3}}{4}$$

$$T.S.A = 4 \times \left(\frac{\sqrt{3}}{4} \times 9^2 \right) = \frac{324\sqrt{3}}{4}$$

$$Volume = \frac{1}{3} \times \left(\frac{\sqrt{3}}{4} \times 9^2 \right) \times \sqrt{\frac{2}{3}} \times 9$$

OR

$$V = \frac{(9)^3}{6\sqrt{2}} = \frac{243\sqrt{2}}{4}$$

90 Volume of a Tetrahedron which height is $4\sqrt{3}$ cm ?

Soln :-

$$h = \sqrt{\frac{2}{3}} a = 4\sqrt{3}$$

$$a = 6\sqrt{2}$$

$$V = \frac{a^3}{6\sqrt{2}} = \frac{(6\sqrt{2})^3}{6\sqrt{2}} = \boxed{72} \text{ cm}^3$$

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